

# ANNULAR SCATTERING AND QUARTIC OPEN FJRW DESCENDENTS

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ABSTRACT. For each finite labelled collection of insertions and each deformation order, we construct a compact oriented rational piecewise-linear bordism class over the moduli spaces of open and closed quartic  $W$ -spin curves. Its cells parametrize decorated tropical forests in a cut-open integral-affine annulus together with the corresponding  $W$ -spin data. Applying a transported relative residue trace to its zero-dimensional endpoint chains defines the contact moments. After the canonical identifications of internal faces, the exterior boundary consists of the two annular ends, closed-component splittings, wall-crossing and cotangent-line faces, pairwise exchange faces, and closed factors of incompatible Euler degree. The residue trace vanishes on the wall and degree-mismatched contributions, while exchange contributions cancel by orientation. Stokes' formula gives the two Gross–Kelly–Tessler cotangent-line recursions. Their difference strictly lowers the number of labels, so the vacuum and one-marking values determine all moments and identify them with the Gross–Kelly–Tessler invariants. The two annular ends give the least-slope and greatest-slope canonical boundary conditions. As the finite data vary, the bordism classes form an inverse system equivariant for the annular core holonomy. They therefore form a holonomy-equivariant inverse system of theta-decorated relative PL bordism classes whose coefficientwise contact moments give the complete formal genus-zero descendent potential.

The affine annulus and its coefficient system are determined by the same quartic data. Applying the cofactor map to the first one-marking descendent generators produces an index-eight extension of the toric character lattice, and the minimal cover preserving this lattice is an annulus with four focus–focus singularities and hyperbolic core holonomy of trace 18. Transport of the required Kummer roots produces an inverse system of finite root extensions. The twisted volume line carries a Jacobi–Kirillov bracket; Saito's primitive form enters separately in the primary comparison. The two descendent Hamiltonians generate a consistent scattering diagram at every finite deformation order.

Resolving the annular boundary and attaching six integral-affine cylinders gives separated fine-saturated logarithmic compactifications with flat boundary divisors. A dominating system obtained by logarithmic resolution, principalization, and root constructions is log smooth with regular geometric fibers. The annular charts also transport the narrow-primary open-FJRW Frobenius structure. The construction is coefficientwise: it does not give an analytic virtual fundamental class for the infinite system or identify a coefficient in an arbitrary GKT chamber.

## 1. INTRODUCTION

Throughout,  $\mathbb{k}$  is an algebraically closed field of characteristic zero. Formal-algebraic constructions are made over  $\mathbb{k}$ ; oscillatory integrals and Saito theory are understood after base change to  $\mathbb{C}$ . The enumerative coefficient rings are defined over  $\mathbb{Q}$  and are implicitly base-changed to  $\mathbb{k}$  when used in the root-stack construction.

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**1.1. From boundary-ray continuation to an affine scattering problem.** The Gross–Siebert programme constructs mirrors by placing wall-crossing automorphisms on an integral-affine base with singularities and reading off broken lines and theta functions [GHK15, GHKK18, GS22]. What it requires as input is geometric: a base, its singular locus, a polyhedral decomposition, and a polarization. This paper asks whether the wall crossing of an open Landau–Ginzburg theory supplies that input, and settles the question for one model.

Gross, Kelly, and Tessler define genus-zero open FJRW invariants for

$$(W, G) = (x^r + y^s, \mu_r \times \mu_s)$$

as relative Euler numbers on moduli spaces of  $W$ -spin disks [GKT22]. Their invariants depend on canonical multisections, and changes of those boundary conditions act through a formal wall-crossing group closely related to the tropical vertex group. This action has the algebra of a scattering diagram but not, by itself, an integral-affine realization: there is no affine manifold, singular locus, or polyhedral decomposition on which the walls have been placed.

The companion paper [Wue26] isolates the gap precisely.

*Theorem* (Boundary-ray factorization [Wue26]). Fix a finite set of deformation monomials closed under taking divisors. After discarding every other monomial, the change between the two canonical GKT boundary conditions at the limiting boundary rays factors uniquely into wall-crossing operations ordered by slope. The slope of a factor is the ray spanned by the pair of boundary-marking counts of the corresponding critical  $W$ -spin graph. Each factor is realized by a homotopy of canonical boundary conditions. Radial motion in the punctured first quadrant of marking counts acts trivially, so the product is transport from one limiting ray to the other and not monodromy around the origin. After the degree-zero torus action is included, the two endpoint boundary conditions and their transition are unique in the finite quotient. The nonidentity factors admit successive finite homotopies in increasing-slope order.

This result supplies an ordered family of wall-crossing operations, but it does not supply the integral-affine surface on which their rays should lie or any affine monodromy. The present paper constructs that missing geometry.

We resolve this for the parabolic quartic

$$W = x^4 + y^4,$$

the parabolic singularity  $X_9$ , whose central charge is exactly one. The suspension of its marginal pencil is an anticanonical elliptic curve in  $\mathbb{P}(1, 1, 2)$ , and the toric-boundary degeneration of that pair gives the disk-shaped dual intersection complex of the Landau–Ginzburg construction of Carl–Pumperla–Siebert [CPS24]; we call it the *CPS complex*. It is the natural candidate base. The first result of this paper is that it cannot carry the descendent lattice, and that three further standard choices fail with it: a finite root stack of coefficients, a scalar Hamiltonian algebra, and a theta basis indexed by one lattice. Proposition 1.1 collects the five obstructions, each proved in the body.

**1.2. Initial Hamiltonians.** The affine and scattering constructions begin with two elements of the GKT critical Lie algebra. Write  $M = \mathbb{Z}^2$  for the toric character lattice in the chosen toric chart. A critical graph of [GKT22] is supported on a nonempty set  $J$  of boundary markings, each carrying a descendent order; we call the datum a *singleton* when  $|J| = 1$ , and *primary* when no marking carries a positive descendent order. The two first-order singleton generators of  $x^4 + y^4$  are, writing  $q^\perp = (q_2, -q_1)$ ,  $\partial_v$  for the invariant logarithmic derivation in direction  $v$ , and  $\kappa = (1, 1)$  for the shift introduced below, in the shifted notation  $X_q^\kappa = z^{q-\kappa} \partial_{q^\perp}$  of Section 3,

$$A = -\frac{1}{2}t_1 X_{(2,1)}^\kappa, \quad B = -\frac{1}{2}t_2 X_{(1,2)}^\kappa,$$

the coefficient  $-\frac{1}{2}$  being the normalization borne by the GKT invariants themselves. The integer pairs  $(2, 1)$  and  $(1, 2)$  label the generators and do lie in  $M$ . Their wall supports are obtained by the cofactor map  $F$  in (2):

$$F(2, 1) = (1/4, -1/2), \quad F(1, 2) = (-1/4, -1).$$

These support vectors do not lie in  $M$ ; their integral span with  $M$  is the index-eight overlattice  $\widetilde{M}$ . This is the lattice mismatch that produces the annular cover of Section 2. We call  $A$  and  $B$  the *singleton Hamiltonians*, and we call their deformation parameters  $t_1, t_2$  the *singleton parameters*; the ideal  $I = (t_1, t_2)$  is the *singleton ideal*. These two elements are the only enumerative input to Sections 2–4. We call  $X_{0,0} = x\partial_x - y\partial_y$  the *degree-zero* generator; the word *marginal* is reserved throughout for the charge-one deformation direction  $x^2y^2$ , the modulus of  $X_9$ . An element of  $\mu_4 \times \mu_4$  is *narrow* when its fixed locus in  $\mathbb{C}^2$  is the origin; the nine narrow elements are  $(\zeta^{a+1}, \zeta^{b+1})$  with  $0 \leq a, b \leq 2$ , matching the monomial basis  $x^a y^b$  of the Jacobian ring  $\text{Jac}(x^4 + y^4)$ .

**1.3. Coefficient system and volume line.** Two consequences of the lattice mismatch determine the coefficient system. Let a *finite transport tree* be a finite rooted tree whose vertices are chambers of the annular wall decomposition and whose oriented edges are reduced paths between adjacent chambers. For such a tree  $T$  and an integer  $N$ , adjoining the finitely many roots required to lift every slab and wall map along  $T$ , and then reducing modulo  $I^N$ , gives a finite root stack  $\mathcal{R}_{T,N}$ . Inclusion of trees, refinement of paths, and reduction of  $N$  give transition maps. We use the inverse system

$$(1) \quad \{\mathcal{R}_{T,N}\}_{(T,N)}$$

over this cofiltered category; no geometric inverse limit is asserted. It is formally analogous to the finite stages of an infinite root stack [TV18], but the indexing category here records path transport rather than divisibility alone.

The second uses a twisted volume form. In a developed torus chart put

$$\kappa = (1, 1), \quad \Omega_\kappa = z^\kappa d\log z_1 \wedge d\log z_2.$$

$\kappa$  denotes the shift in the chosen developed chart and is transported with that chart; it is not a global flat section of the character local system. A lifted slab transition preserves  $d\log z_1 \wedge d\log z_2$  but rescales  $\Omega_\kappa$  by a root-valued unit, so the local Hamiltonian brackets glue on a line bundle rather than on the structure sheaf. The resulting object is the Jacobi–Kirillov bracket on the *volume line*  $\mathcal{L}_\kappa$ ; equivalently, the complement of the zero section in  $\mathcal{L}_\kappa^\vee$  carries a homogeneous Poisson structure, the Poissonization of the Jacobi bracket [VW20]. Saito’s primitive form is a different object and enters only in Section 6.

**1.4. Annular monodromy and its consequences.** Two mechanisms produce the departures from the standard finite-type situation. The first descendent exponents force an overlattice  $\widetilde{M} \supset M$  of index eight; only one of the three affine monodromies of the CPS complex preserves it; the double cover branched at the two offending singular points is an annulus; and the holonomy  $K$  of that annulus about its core has trace 18, hence is hyperbolic. Repeated slab shear makes the transported root orbit infinite and forces the inverse system (1). Hyperbolicity makes every nonzero exponent orbit infinite and forces the theta functions to form a local system rather than a basis indexed by one lattice.

**1.5. Basic notation.** We use the standard scattering-diagram terminology of [GS22]. In particular, slabs carry the gluing functions on codimension-one cells, whereas walls are outgoing. The cofactor map below determines the supports of the two initial Hamiltonians.

Let  $M \simeq \mathbb{Z}^2$  be the character lattice of the quartic toric model,  $\varphi$  its polarization, and, in the cofactor chart of Section 2, put

$$(2) \quad F = \frac{1}{4} \begin{pmatrix} 1 & -1 \\ 0 & -2 \end{pmatrix}, \quad \widetilde{M} = F(\mathbb{Z}^2), \quad D = \widetilde{M}/M \simeq \mathbb{Z}/2 \oplus \mathbb{Z}/4.$$

Let  $\overline{\mathcal{A}}$  be the completed parity cover,  $\mathcal{A} = \text{Int}(\overline{\mathcal{A}})$ , and let  $\mathcal{A}^\circ$  be the complement of its four affine singularities. We write  $t_{a,b}$ ,  $0 \leq a, b \leq 2$ , for the primary deformation parameters dual to the monomial basis  $x^a y^b$  of  $\text{Jac}(x^4 + y^4)$ , which index the nine narrow sectors, and reserve  $t_1, t_2$  for the singleton parameters of §1.2. The linear holonomy of  $\mathcal{A}$  about its core, in the basis of  $\widetilde{M}$  fixed by  $F$  and with the orientation of Appendix A, is

$$(3) \quad K = \begin{pmatrix} -1 & 2 \\ -10 & 19 \end{pmatrix}, \quad \text{tr } K = 18.$$

## 1.6. Obstructions to the disk-shaped affine base.

**Proposition 1.1** (Obstructions to simpler models). *The following hold for the open FJRW wall crossing of  $(x^4 + y^4, \mu_4^2)$ .*

- (1) (No model on the disk.) *The first descendent Hamiltonians do not act integrally on  $M$ , which has index eight in  $\widetilde{M}$ . Of the three affine monodromies of the disk-shaped Carl–Pumperla–Siebert complex, exactly one preserves  $\widetilde{M}$ . The smallest cover on which  $\widetilde{M}$  is preserved completes to an annulus. Thus the enlarged lattice defines no model on that complex or on an ordinary root-stack refinement with the same coarse disk.*
- (2) (No finite root stack.) *No finite algebraic Kummer extension containing the initial root data is stable under the slab shear. If every transported root and its automorphisms are retained, the resulting coefficient object is not a formal scheme, a finite root stack, or an algebraic stack whose diagonal is locally of finite type.*
- (3) (No scalar bracket.) *The scalar  $\kappa$ -twisted Hamiltonian algebra containing the two singleton fields is not preserved under conjugation by the lifted slab transitions. Hence its local brackets do not glue to a bracket on the structure sheaf.*
- (4) (No global theta basis.) *No nonzero  $m \in \widetilde{M}$  has finite  $K$ -orbit, and the formal  $K$ -orbit sum of a theta section does not converge  $I$ -adically. There is no theta basis indexed by one copy of  $\widetilde{M}$ .*
- (5) (No ordinary recovery of the quartic family, and no regular minimal presentation.) *The annular cover is not the dual intersection complex of a distinguished toric degeneration of del Pezzo surfaces with simple singularities, proper generic fiber, and smooth anti-canonical divisor. Moreover some finite root-stack presentation has singular atlas and is not log regular.*

The five parts are proved, in order, by Proposition 2.2 with Theorem 2.3; Theorems 2.5 and 2.7; Proposition 3.2; Theorem 3.8; and Propositions 4.6 and 4.7.

Together these results lead to an annular parity cover, a pro-Kummer inverse system of finite root stacks, a volume line carrying a Jacobi–Kirillov bracket, a  $K$ -equivariant theta local system, and a dominating principalized presentation. On these objects the two initial GKT Hamiltonians admit an ordered scattering completion with proper, locally finite support.

### 1.7. The geometric result.

**Construction 1.2** (Flared compactification). Blow up, in the real-oriented sense, the two boundary components of  $\overline{\mathcal{A}}$  and its four focus strata, and attach to the six resulting ends outward integral-affine cylinders, here called *flares*, of widths 4, 16, 1, 1, 1, 1. Write  $\overline{\mathcal{A}}^{\text{fl}}$  for the result.

**Theorem 1.3** (Annular reconstruction). *The quartic toric degeneration and the first two descendent GKT Hamiltonians determine the following data.*

- (1) (The integral-affine surface.) *The surface  $\overline{\mathcal{A}}$  is a polarized integral-affine annulus with tangent lattice  $\widetilde{M}$ , four primitive focus–focus singularities, and boundary shear widths 4 and 16. Its polarization is the minimal integral multiple  $4p^*\varphi$  of the pullback of the quartic polarization.*
- (2) (The coefficients.) *The local slab transitions lift after adjoining roots of degrees 2, 4, 4, 2. Closing these roots under transport gives the inverse system (1); paths between chambers act on its finite stages by the lifted transition maps. For every  $N$  and every compact  $C \subset \mathcal{A}^\circ$ , the transported wall functions over  $C$  modulo  $I^N$  lie in a single finite root-stack presentation; no finite root stack contains the complete transported root orbit.*
- (3) (The wall structure.) *The inverse system (1) carries the volume line  $\mathcal{L}_\kappa$  and its Jacobi bracket. Starting with the two rays in directions  $F(2, 1)$  and  $F(1, 2)$  and the Hamiltonians  $A$  and  $B$ , the scattering algorithm gives a unique proper, locally finite wall structure on  $\mathcal{A}^\circ$ . Its path-ordered product around every sufficiently small contractible loop about a joint is the identity.*
- (4) (Theta multiplication.) *Given any finite list of theta sections based in chosen chambers, products among them, associativity comparisons, and any  $N$ , there is a finite set of exponents containing the inputs and every exponent generated in those computations modulo  $I^N$ . There is also one finite root-stack stage on which all the corresponding broken-line expansions are defined and unique. The resulting products are commutative and associative. The core holonomy  $K$  of (3) acts trivially on  $D$  and hyperbolically on  $\widetilde{M}$ . Consequently the theta functions form a  $K$ -equivariant local system; there is no single-valued basis indexed by a fixed copy of  $\widetilde{M}$ .*
- (5) (Logarithmic compactification.) *Let  $\overline{\mathcal{A}}^{\text{fl}}$  be as in Construction 1.2. At every finite root, coefficient, and deformation stage, transported toric monoid charts glued along their full boundary overlaps give a separated fine-saturated logarithmic compactification whose six boundary divisors are flat over the coefficient ring. These compactifications commute with the transition maps of (1). Logarithmic regularity fails at some finite root-stack presentation (Proposition 4.7); the system is nevertheless dominated by a cofiltered presentation by log-smooth Deligne–Mumford stacks with regular, log-regular geometric fibers (Theorem 4.8), obtained by functorial logarithmic resolution, principalization of the pulled-back branch ideals, and fourth roots of the resulting normal-crossings components.*

In the orientation of Appendix A, the two incoming coefficients in Part (3) are  $-1/2, -1/2$  and the first outgoing coefficient is  $3/4$ ; these are computed in Section 3.

Consistency in Part (3) has its standard local meaning: the path-ordered product is the identity around every sufficiently small contractible loop about a joint. It does not trivialize the prescribed affine, root, or line holonomy around a focus or around the annular core. Part (5) asserts separatedness, fine-saturated logarithmic charts, and flatness of the horizontal boundary at each finite stage. It asserts neither regularity of the underlying local rings nor logarithmic

regularity, and the inverse limit is not claimed to be an algebraic stack of finite type. The final assertion of Part (5) concerns the separate dominating presentation of Theorem 4.8; it does not retroactively make the minimal finite root-stack presentations regular.

The compactification addresses a boundary obstruction absent from the open locus. The off-diagonal walls meet the compact boundary transversely, so the usual convexity argument for tangent boundary walls does not apply. Nor can the boundary be doubled affinely: its shear is nonzero. We instead attach flaring cylinders, order the exiting supports, and transport the compactified monoid chart across each wall. Gluing adjacent transported charts along the full boundary overlap, rather than only after inverting a boundary parameter, is what gives separatedness and a horizontal flat divisor.

**1.8. The open-FJRW comparison.** The finite sector group  $D$  remembers fractional exponents but not the ordered pair of quartic twists. The latter is

$$Q_{\text{GKT}} = \widetilde{M}/4\mathbb{Z}^2 \simeq (\mathbb{Z}/4)^2.$$

Let  $\mathcal{A}_{\text{lab}}^\circ \rightarrow \mathcal{A}^\circ$  be the degree-four cover on which this quotient local system is constant. We call it the *labelled-sector cover*. A degree-two intermediate cover trivializes  $D$  and has a deck involution induced by interchanging  $x$  and  $y$ . Coordinate symmetrization below means taking invariants under this involution after multiplication, followed by the finite-cover pushforward.

The comparison has two logically distinct parts. First, the  $\kappa$ -shift identifies the GKT Lie algebra of critical-graph wall operations with a labelled subalgebra of the Jacobi algebra of wall Hamiltonians. A *GKT endpoint factor* is the ordered product that changes the GKT canonical boundary condition across one ray. Dividing the scattering correction on that ray by this embedded factor defines the *residual factor*. The first mixed GKT factor has coefficient  $1/4$ , while scattering consistency requires coefficient  $3/4$ ; the residual factor therefore begins with coefficient  $1/2$ . It cannot be omitted without destroying consistency around the joint.

The Frobenius comparison takes the following second input.

*Theorem* (Primary open mirror theorem; [GKT22, Thm. 0.3 and Cor. 4.17]). Let  $W_{\text{prim}}^{\text{GKT}}$  be the coordinate-symmetric primary open FJRW potential of  $(x^4 + y^4, \mu_4^2)$ , obtained by setting every descendent variable to zero. Then  $W_{\text{prim}}^{\text{GKT}}$  is an unfolding of  $x^4 + y^4$  with primitive form  $dX \wedge dY$  and nine good-basis cycles  $\Xi_{a,b}$ , and its nine parameters  $t_{a,b}$ ,  $0 \leq a, b \leq 2$ , are simultaneously Saito-flat coordinates for that unfolding.

What this paper adds to the primary mirror theorem above is the transport, not the Frobenius structure itself.

Second, a Frobenius comparison requires more than wall coefficients. It requires the primitive form, oscillatory periods, a residue pairing, and a Kodaira–Spencer basis. These data remove a unipotent ambiguity invisible to the unary walls. In Part (3) below, closure under dependencies means inclusion of every connected-component datum and every zero- or exchange-boundary datum required by the GKT assembling relation; Appendix B gives the formal definition. The resulting statement is as follows.

**Theorem 1.4** (Wall factorization and transported primary structure). *On the labelled-sector cover  $\mathcal{A}_{\text{lab}}^\circ$  and its finite root-stack stages:*

- (1) *Every finite labelled GKT critical-graph Lie algebra embeds without changing its coefficients in the Jacobi algebra of wall Hamiltonians, and every finite GKT endpoint operation maps to the corresponding slope-ordered product of Jacobi wall automorphisms. These embeddings commute with relabelling, restriction, and transport along a framed path.*

- (2) *Each total consistency factor has a canonical raywise factorization into a GKT endpoint factor and a residual unipotent factor. The residual factors are compatible under products and changes of framed paths. For residual factors  $e^\xi, e^\eta, e^\zeta$  of equal deformation degree the first nontrivial associator has logarithm*

$$\frac{1}{72} [[\xi, \zeta], \eta].$$

- (3) *Finite GKT endpoint homotopies can be chosen compatibly under enlargement of finite labelled systems closed under the boundary and component dependencies of Appendix B. Their union is a componentwise smooth, slope-ordered GKT homotopy whose restriction to every finite quotient realizes the corresponding endpoint factors.*
- (4) *Let  $W_{\text{prim}}^{\text{GKT}}$  be the coordinate-symmetric primary GKT potential and let  $P_\Gamma$  be transport along a framed annular path  $\Gamma$ . In the transported Jacobian algebra, the nine classes*

$$\Phi_{a,b}^\Gamma = \left[ P_\Gamma \left( \frac{\partial W_{\text{prim}}^{\text{GKT}}}{\partial t_{a,b}} \right) \right], \quad 0 \leq a, b \leq 2,$$

*form a basis. Transport along another framed path carries this basis to the corresponding basis there. Their product, unit, and normalized residue pairing have structure constants independent of  $\Gamma$  in these transported bases. Taking invariants under  $x \leftrightarrow y$  and pushing forward along the degree-two intermediate cover gives the complete narrow-primary GKT/Saito–Givental Frobenius structure in every annular chart.*

The last part uses the primary mirror theorem of [GKT22]. The contribution here is to place the potential, primitive form, Brieskorn complex, and Kodaira–Spencer basis in the annular theta local system and to prove that transport and finite Kummer refinement preserve the Frobenius tensor. Section 6 also checks the marginal period normalization and distinguishes its coefficients from the mixed Jacobi wall coefficient  $3/4$ .

**1.9. Main result: the annular cotangent correspondence.** The positive-descendent comparison is the principal result of the paper and is independent of the primary transport argument. In standard terms, its input is a finite full subdiagram of the stratified quartic  $W$ -spin moduli problem, closed under the normalization, detachment, stabilization, and exchange maps that occur on its boundary. Its output is a class in relative oriented PL bordism for a compactified tropical incidence problem. The next two definitions make this finite diagram and bordism group precise before the theorem is stated.

**Definition 1.5** (Finite labelled window). Each label  $i$  has a quartic twist  $(a_i, b_i) \in \{0, 1, 2, 3\}^2$  and a descendent degree  $d_i \in \mathbb{N}$ . An *exact open  $W$ -spin datum* is the following finite diagram:

$$(4) \quad (\Gamma, \mathcal{PM}_\Gamma, \mathbb{E}_\Gamma(\mathbf{d}), \mathcal{L}_i, t_{i,w}, F_{\Lambda\Gamma}, \phi_{\Lambda\Gamma}).$$

Here  $\Gamma$  is a stable labelled open quartic  $W$ -spin graph;  $\mathcal{PM}_\Gamma$  is its compact oriented orbifold with corners;  $\mathbb{E}_\Gamma(\mathbf{d})$  is its descendent Witten bundle;  $\mathcal{L}_i = T_{z_i}^* \mathcal{C}_\Gamma$  is the tautological line; and  $t_{i,w}$  is the standard meromorphic-differential section of  $\mathcal{L}_i$  pointing from  $z_i$  to a retained marking  $w$ . For every boundary graph  $\Lambda$ , the datum includes the normalization, detachment, stabilization, or exchange map  $F_{\Lambda\Gamma} : \mathcal{PM}_\Lambda \rightarrow \partial_\Lambda \mathcal{PM}_\Gamma$  and the corresponding oriented bundle map  $\phi_{\Lambda\Gamma}$ . These maps are the maps of the compact  $W$ -spin moduli problem, not numerical open invariants. They obey the actual composition identities on multiple boundary strata. Whenever  $i$  is retained and its component is not contracted, the datum also contains the canonical splitting

$$(5) \quad \mathbb{E}_\Gamma(\mathbf{d} + \mathbf{e}_i) \simeq \mathbb{E}_\Gamma(\mathbf{d}) \oplus \mathcal{L}_i.$$

A closed boundary entry analogously specifies its compact closed  $W$ -spin orbifold, Euler bundle and section, and its maps on all boundary strata. Thus an exact open datum is a finite piece of

the usual stratified  $W$ -spin moduli problem together with the bundles and tautological sections used in the Euler-chain construction.

A *finite labelled window*  $S = (I, P, \mathcal{E})$  consists of a finite labelled set  $I$ , a finite divisor-closed set  $P$  of deformation monomials containing 1, and a finite collection  $\mathcal{E}$  of the diagrams (4) and their closed boundary entries. It is closed under connected components, boundary graphs, and the bundle splittings (5). For each support  $J$ , write

$$(6) \quad \sum_{i \in J} a_i = r_J + 4\ell_J, \quad \sum_{i \in J} b_i = s_J + 4m_J, \quad N_J = \ell_J + m_J - |J| + 1 + \sum_{i \in J} d_i,$$

where  $0 \leq r_J, s_J \leq 3$ . The window also contains the finite sequence with exponent profiles

$$(7) \quad (r_J + 4p, s_J + 4(N_J - p)), \quad 0 \leq p \leq N_J,$$

when  $N_J \geq 0$ , together with the strata joining consecutive profiles. These are respectively the balanced and critical profiles in the terminology of [GKT22]. The window is invariant under twist-preserving permutations. Restriction means literal forgetting of labels, monomials, and their dependent strata. Equivalently, the window is the finite full subdiagram on which one fixed coefficient truncation of the open and closed degeneration formulas is defined.

Put  $\mathbf{m}_S = (t_i : i \in I)$  and

$$(8) \quad R_{S,N} = \mathbb{Q}[t_i : i \in I] / (\mathbf{m}_S^N + \langle m : m \notin P \rangle).$$

Divisor closure makes this a finite Artin ring. Under unlabelling, each  $t_i$  maps to the singleton parameter of the same twist, so the  $\mathbf{m}_S$ -filtration is the labelled pullback of the singleton filtration used in the two-generator scattering problem.

A *framed compact annular corridor* is obtained by cutting the affine annulus along a support-free transverse arc and retaining the compact polyhedral strip between stops placed before the first and after the last wall used modulo  $\mathbf{m}_S^N$ . Its two boundary arcs are the incoming and outgoing ends. Stability suppresses every unmarked bivalent wall-free vertex. A *finite root stage* is one finite presentation in (1) containing every root needed by the slab and wall maps in the corridor and the fourth root of every closed-component smoothing parameter. A *target frame* is a terminal chamber and a path from each cell to it; its transport map is the ordered composition of the wall and root-chart chain isomorphisms along that path. Transport around the core gives  $K$ ; to compare two finite corridors, both are first included in a common finite enlargement and then transported. The corridor is therefore a compact cut-open scattering chamber with its two limiting boundary conditions retained as relative boundary components; a target frame is only the standard choice needed to compare transported coefficients in one chart.

The term *broken-line chain* is used at chain level: its cells parametrize stable decorated tropical forests together with  $W$ -spin data, rather than individual Gross–Siebert broken lines. The finite chain used below is a cellular chain of stable decorated tropical trees mapping to this corridor, fibered over the orbifolds  $\mathcal{PM}_\Gamma$  in (4). Forest cuts carry elements of the transported relative twisted de Rham complex defined in Section 7; these elements retain their values in the chosen finite root extension. The *contact moment* of a zero-dimensional boundary chain is its transported relative residue trace. All endpoint and degeneration facets remain in the cellular boundary unless they are paired by an explicit mapping cylinder.

**Definition 1.6** (Relative PL bordism module). Let  $\mathfrak{X}_{S,N}$  be the finite diagram formed from the compact  $W$ -spin orbifolds, annular incidence polyhedra, and boundary maps in a finite labelled window. Two compact oriented rational PL branched chains in this diagram are *relatively PL bordant* if they are the two parameter-end restrictions of a compact oriented rational PL

branched chain over  $[0, 1]$  whose annular-end restrictions are fixed and whose other exterior faces obey the same labelled boundary maps. Coefficients are carried in the transported relative twisted de Rham complexes. We write  $\Omega_{S,N}^{\text{PL}}$  for the resulting  $R_{S,N}$ -module of bordism classes. Thus a class in  $\Omega_{S,N}^{\text{PL}}$  retains its named boundary strata; it is not obtained by discarding the degeneration faces. Section 7 gives the cellular realization of this module.

**Theorem 1.7** (Annular cotangent correspondence). *Fix a finite labelled window  $S = (I, P, \mathcal{E})$  as in Definition 1.5, a deformation order  $N$ , the finite root stage just defined, and a framed compact annular corridor. There is a finite cellular complex  $\mathcal{C}_{S,N}^{\text{dp}}$  over  $R_{S,N}$  and a canonically defined relative bordism class*

$$(9) \quad [\text{BL}_{S,N}]_{\text{PL}} \in \Omega_{S,N}^{\text{PL}}.$$

*Its contact moment  $\mathcal{A}_{\text{ann}}(I, \mathbf{d})$  satisfies the following two identities. For  $A \subseteq I$  with  $N_A = -1$ , let  $Q_A = (I \setminus A) \sqcup \{z_A\}$  be the continuing open datum obtained by replacing  $A$  by its half-node, and let  $\mathcal{C}_A^{\text{ext}}$  be the Frobenius-normalized Euler number of the complementary closed  $W$ -spin factor. If  $j_1, j_2 \in I$  are distinct, then*

$$(10) \quad \mathcal{A}_{\text{ann}}(I, \mathbf{d} + \mathbf{e}_{j_1}) = \sum_{\substack{A \subseteq I, j_1 \in A, |A| \geq 2 \\ N_A = -1}} \mathcal{C}_A^{\text{ext}} \mathcal{A}_{\text{ann}}(Q_A, \mathbf{d}|_{I \setminus A}) - \mathcal{A}_{\text{ann}}(I, \mathbf{d}),$$

and

$$(11) \quad \mathcal{A}_{\text{ann}}(I, \mathbf{d} + \mathbf{e}_{j_1}) = \sum_{\substack{A \subseteq I, j_1 \in A, j_2 \notin A, |A| \geq 2 \\ N_A = -1}} \mathcal{C}_A^{\text{ext}} \mathcal{A}_{\text{ann}}(Q_A, \mathbf{d}|_{I \setminus A}).$$

Together with

$$(12) \quad \mathcal{A}_{\text{ann}}(\emptyset, 0) = 1, \quad \mathcal{A}_{\text{ann}}(\{i\}, d) = (-1)^d,$$

these identities determine every moment by induction on  $|I|$  and identify the resulting family with the chamber-independent Gross–Kelly–Tessler moment family.

The class has the following geometric and functorial properties.

- (1) *It is represented by compact rational PL zero loci over the open and closed  $W$ -spin orbifolds in the window. The Witten-bundle splitting (5) identifies the upper cotangent face with descendent degree  $\mathbf{d} + \mathbf{e}_i$ ; the two annular ends are separate relative restrictions. Its remaining exterior boundary consists of closed-component splittings, wall-crossing homotopies, cotangent-line contraction faces, the two cyclic orders at a pair collision, and closed factors whose Euler rank does not match the dimension of their moduli space.*
- (2) *At the two annular endpoints, the coefficients equal the two GKT boundary systems obtained by retaining, recursively on support, only the least-slope or only the greatest-slope balanced graph. Divided-power pushforward gives the standard automorphism factors. No arbitrary-chamber coefficient equality is asserted.*
- (3) *The bordism classes commute with window restriction, deformation-order reduction, finite-root refinement, relabelling, transport to another target frame, coordinate transposition, and core transport after including both windows in a common finite stage. In the inverse limit they form a  $K$ -equivariant inverse system of theta-decorated relative PL bordism classes. Its coefficientwise contact moment is the complete formal genus-zero descendent potential. No map from relative PL bordism to the theta algebra is asserted.*

The geometric moduli spaces, bundles, tautological sections, boundary maps in (4), and the two normalized singleton Hamiltonians  $\mathbf{A}, \mathbf{B}$  are inputs. No higher-support numerical GKT open

*invariant, GKT choice of Euler chain, period coefficient, or target endpoint coefficient enters the construction of (9).*

Section 7 gives the construction and proof. It is finite-dimensional and coefficientwise; it does not produce an analytic infinite-window virtual fundamental class.

**1.10. Relation with reconstruction and scope.** The construction follows the usual separation between affine and wall data [GHK15, GHKK18, GS22] but does not produce a distinguished Carl–Pumperla–Siebert degeneration of the original quartic pair. Their connected anticanonical construction has a disk-shaped intersection complex; our descendent lattice forces an annular cover with different coarse topology, and Proposition 4.6 shows the two cannot be reconciled inside that class. The descendent theorem identifies the chamber-independent moment family and the coefficients in two specified extreme normal forms. It does not identify an unreexpanded pair-of-broken-lines coefficient or a raw coefficient in an arbitrary GKT chamber. Those quantities are not determined by the invariant moments.

The proof has four stages. First, the discrete Legendre transform and the descendent exponents determine the enlarged lattice, parity cover, and pathwise root stacks. Second, the shifted Hamiltonian bracket places the initial factors on the rays determined by the cofactor map; filtered scattering and broken-line re-expansion then give the framed theta algebra, while transported monoid charts compactify its ends. Third, the GKT critical algebra is embedded in the same Jacobi group. The compatible-homotopy theorem realizes its finite factors, and the GKT primary mirror theorem supplies the source Frobenius tensor that is transported through the annular charts. The fourth stage is independent of that imported primary tensor. For each compact open  $W$ -spin orbifold in the finite window, we take the fiber product with the finite incidence space of stable decorated tropical trees in the cut-open annulus. The equation bundle contains both the open Witten bundle and the annular incidence equations. The splitting  $\mathbb{E}_\Gamma(\mathbf{d} + \mathbf{e}_i) = \mathbb{E}_\Gamma(\mathbf{d}) \oplus \mathcal{L}_i$  makes the added cotangent power geometric: a PL homotopy in  $\mathcal{L}_i$  joins a generic compatible section to a signed combination of the standard pointing sections. The zeros of those pointing sections are precisely the open–closed boundary divisors. Compatible equivariant PL multisections regularize the remaining equations. Explicit mapping cylinders fill zero-length tree edges, collisions of directions, wall homotopies, and the fourth-root normal to an open–closed divisor. The two choices of pointing combination give the two standard one- and two-marking cotangent-line relations. A relative Fermat residue trace turns their cellular boundary formulas into scalar recursions, which prove uniqueness before comparison with the GKT invariants.

Section 2 constructs the affine annulus and its coefficient system. Section 3 proves Jacobi consistency and theta multiplication. Section 4 constructs the logarithmic caps. Section 5 gives the labelled GKT factorization and residual coherence, and Section 6 proves the primary comparison. Section 7 constructs the virtual broken-line chain and proves the descendent correspondence. Appendix A records the nonstandard affine and logarithmic calculations. Appendix B proves the compatible GKT completion theorem.

The manuscript sources for this paper and its two companions are available at <https://github.com/bwuebben/open-fjrw-scattering>.

## 2. THE ANNULAR AFFINE MANIFOLD AND ITS COEFFICIENT SYSTEM

**2.1. The quartic degeneration.** Let  $N_{\text{tor}} = \mathbb{Z}^2$  and take the primitive fan rays of  $\mathbb{P}(1, 1, 2)$  to be

$$v_X = (1, 0), \quad v_Y = (-1, -2), \quad v_Z = (0, 1).$$

The anticanonical polygon and its polar are

$$(13) \quad \Delta = \text{conv}\{(-1, -1), (-1, 1), (3, -1)\}, \quad \Delta^\vee = \text{conv}\{v_X, v_Z, v_Y\}.$$

The edge  $[v_Y, v_X]$  has lattice length two and midpoint  $m_* = (0, -1)$ . This is the unique non-basic edge of  $\Delta^\vee$ .

Consider the marginal quartic pencil

$$(14) \quad E_s = \{X^4 + Y^4 + Z^2 + sX^2Y^2 = 0\} \subset \mathbb{P}(1, 1, 2)$$

and the toric-boundary degeneration

$$(15) \quad XYZ + q(X^4 + Y^4 + Z^2 + sX^2Y^2) = 0.$$

For  $q \neq 0$ , division by  $q$  and the substitution  $Z' = Z + XY/(2q)$  give the same pencil with effective parameter  $s - 1/(4q^2)$ . At  $q = 0$  the fiber is the toric boundary.

**Proposition 2.1.** *The family (15) is a relative anticanonical section of the coherent Mumford degeneration associated with the star subdivision of  $\Delta$  at the origin. Its logarithmic smoothing charts are semistable on the canonical stack. At the coarse  $A_1$  point the smoothing is Kummer of index two.*

*Proof.* The lattice points of  $\Delta$  index the anticanonical Cox monomials by

$$m \mapsto X^{\langle m, v_X \rangle + 1} Y^{\langle m, v_Y \rangle + 1} Z^{\langle m, v_Z \rangle + 1}.$$

The five terms in (15) correspond to the origin and the four boundary points indexing  $X^4, Y^4, Z^2, X^2Y^2$ . Assigning height zero to the origin and height one to the boundary gives the star subdivision, hence the stated coherent degeneration.

On  $Y \neq 0$ , with  $x = X/Y$  and  $z = Z/Y^2$ , a smooth toric node has equation

$$xz + q(z^2 + x^4 + sx^2 + 1) = 0.$$

The parenthesized term is a unit at the node, so an étale change of coordinates gives  $xz = q$ . At  $[0 : 0 : 1]$ , pass to the canonical stack chart with residual involution  $(x, y) \mapsto (-x, -y)$ . The equation is again  $xy = q$  up to a unit. On the coarse chart set  $a = x^2, b = xy, c = y^2$ ; since  $ac = b^2$ , eliminating  $b$  gives  $AC = q^2$ .  $\square$

Let  $u$  denote the primitive character along the length-two slab. Its free coefficient may be written  $f_\rho = 1 + az^u + z^{2u}$ . The associated curve is compared with (14) by

$$(16) \quad s^2 = \frac{4a^2}{a^2 - 4}, \quad a^2 = \frac{4s^2}{s^2 - 4}.$$

Indeed, eliminating  $U$  from  $U + V + aV^{-1} + U^{-1}V^{-2} = 0$  gives  $Y^2 = (V^2 + a)^2 - 4$ , and rescaling by a square root of  $a^2 - 4$  gives the quartic. Equation (16) compares the anticanonical curves; it does not identify the slab automorphism with an open-FJRW wall.

**2.2. The cofactor lattice.** Let  $N_\partial = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2$  be the boundary-charge lattice. The coordinate divisors give

$$(17) \quad \iota : N_\partial \longrightarrow N_{\text{tor}}, \quad \iota = \begin{pmatrix} 1 & -1 \\ 0 & -2 \end{pmatrix}, \quad [N_{\text{tor}} : \iota(N_\partial)] = 2.$$

In particular,  $\iota(e_1 + e_2) = 2m_*$ . The discrete-Legendre support map is the cofactor

$$(18) \quad L = \det(\iota)\iota^{-T} = \begin{pmatrix} -2 & 0 \\ 1 & 1 \end{pmatrix}.$$

If

$$H_{\partial} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

then

$$(19) \quad J\iota H_{\partial} = L, \quad L^T \iota = \det(\iota) I.$$

Thus  $L$  is the oriented conormal obtained by transporting a Hamiltonian tangent through  $\iota$ .

The first descendent exponents force the lattice

$$(20) \quad F = \frac{1}{4}\iota, \quad \widetilde{M} = F(\mathbb{Z}^2) \supset M = \mathbb{Z}^2, \quad [\widetilde{M} : M] = 8.$$

For a shifted Hamiltonian exponent  $q = k + (1, 1)$  put

$$(21) \quad \widetilde{n}_q = F^{-T}(q_2, -q_1) \in \widetilde{M}^{\vee}.$$

**Proposition 2.2.** *Conjugation by  $F$  carries every logarithmic Hamiltonian derivation in the boundary-charge lattice to an integral monomial derivation of  $\mathbb{C}[\widetilde{M}]$ . More precisely,*

$$(22) \quad F_* X_k(z^m) = \langle \widetilde{n}_q, m \rangle z^{m+F(k)}, \quad m \in \widetilde{M}.$$

The wall character is  $z^{F(q)}$ , its derivation shifts exponent by  $F(k)$ , and its support direction is  $Lq$ .

*Proof.* For  $n_q = (q_2, -q_1)$  and  $\ell \in \mathbb{Z}^2$ ,

$$\langle F^{-T}n_q, F\ell \rangle = \langle n_q, \ell \rangle.$$

This proves integrality and (22). The identity  $\langle \widetilde{n}_q, F(q) \rangle = 0$  makes  $F(q)$  tangent to the wall character, while (19) identifies  $Lq$  with the oriented conormal of the transported Hamiltonian tangent.  $\square$

In the  $F$ -basis, the old lattice is generated by  $(4, 0)$  and  $(-2, -2)$ . The determinant is eight, so the quotient is

$$(23) \quad D = \widetilde{M}/M \simeq \mathbb{Z}/2 \oplus \mathbb{Z}/4.$$

The two singleton classes  $\alpha = [e_1]$  and  $\beta = [e_2]$  have order four, whereas  $\rho = \alpha + \beta$  has order two. The homomorphism

$$(24) \quad \sigma : D \longrightarrow \mathbb{Z}/4, \quad \sigma([xe_1 + ye_2]) = x + y \pmod{4}$$

has kernel generated by  $\delta = \alpha - \beta$ .

**2.3. The parity cover.** The straight-boundary monodromies of the disk-shaped complex, on its Legendre-dual character lattice, are

$$(25) \quad M_A = \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix}, \quad M_B = \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}, \quad M_C = \begin{pmatrix} 4 & -1 \\ 9 & -2 \end{pmatrix}.$$

Only  $M_B$  preserves  $\widetilde{M}$ . Indeed, conjugating the other two by  $F$  gives half-integral matrices; see Appendix A. Let  $\Delta_{\text{aff}}^{\circ}$  be the affine base disk with the three singular points  $A, B, C$  removed, and define

$$(26) \quad \chi(\gamma_A) = \chi(\gamma_C) = 1, \quad \chi(\gamma_B) = 0 \quad \text{in } \mathbb{Z}/2.$$

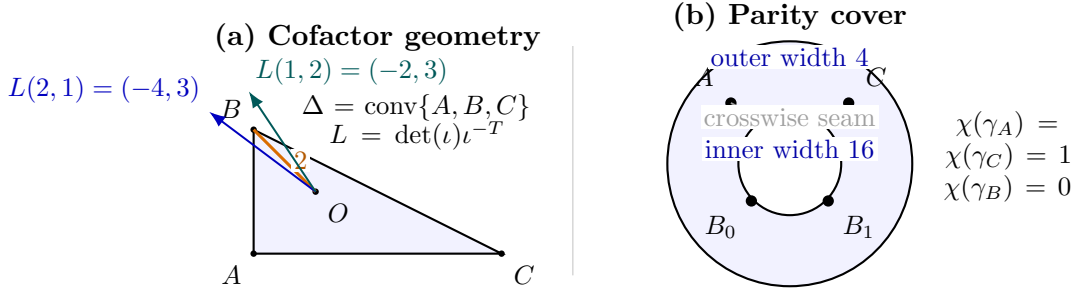


FIGURE 1. The quartic polygon and its annular parity cover. The first two singleton supports are the cofactor directions  $L(2, 1)$  and  $L(1, 2)$ . The branch points are  $A, C$ ; the width-two point  $B$  has two unramified lifts.

**Theorem 2.3** (Annular parity cover). *The connected double cover of  $\Delta_{\text{aff}}^\circ$  classified by  $\chi$  extends over the compact disk as a cover ramified at  $A$  and  $C$ . Its completion  $\overline{\mathcal{A}}$  is an annulus. The point  $B$  has two unramified lifts  $B_0, B_1$ , and all four points  $A, B_0, B_1, C$  are primitive focus–focus singularities for the  $\widetilde{M}$ -affine structure. The pullback polarization becomes integral after multiplication by four, and  $4p^*\varphi$  is the minimal integral multiple.*

*Proof.* Cut the disk along an arc from  $A$  to  $C$  disjoint from  $B$  and glue two copies crosswise. The local model at each endpoint is  $z \mapsto z^2$ . Since the boundary loop has parity  $1 + 0 + 1 = 0$ , it has two lifts. Riemann–Hurwitz gives  $\chi(\overline{\mathcal{A}}) = 2\chi(D^2) - 2 = 0$ ; a connected oriented surface with Euler characteristic zero and two boundary components is an annulus.

In the  $F$ -basis the four lifted monodromies are

$$(27) \quad \begin{aligned} H_A &= \begin{pmatrix} -2 & 9 \\ -1 & 4 \end{pmatrix}, & H_{B,0} &= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}, \\ H_{B,1} &= \begin{pmatrix} -14 & 25 \\ -9 & 16 \end{pmatrix}, & H_C &= \begin{pmatrix} -2 & 1 \\ -9 & 4 \end{pmatrix}. \end{aligned}$$

Their rank-one parts factor as

$$(28) \quad \begin{aligned} H_A - I &= \begin{pmatrix} 3 \\ 1 \end{pmatrix} (-1, 3), & H_{B,0} - I &= \begin{pmatrix} 1 \\ -1 \end{pmatrix} (1, 1), \\ H_{B,1} - I &= \begin{pmatrix} 5 \\ 3 \end{pmatrix} (-3, 5), & H_C - I &= \begin{pmatrix} 1 \\ 3 \end{pmatrix} (-3, 1). \end{aligned}$$

Both factors are primitive in each row, proving the focus–focus assertion. The old facet conormals have  $F$ -dual coordinates with denominator four, and one contains a coefficient  $1/4$ . Multiplication by four clears all three and is minimal.  $\square$

The two boundary holonomies have normal forms

$$(29) \quad \begin{pmatrix} 1 & -4 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & -16 \\ 0 & 1 \end{pmatrix}.$$

Their widths are therefore 4 and 16. The core holonomy, with the orientation fixed in Appendix A, is the hyperbolic matrix  $K$  of (3).

The affine annulus is consequently an abstract polarized affine manifold forced by the descendent lattice. It is not the intersection complex of an ordinary distinguished degeneration of the connected quartic pair. The next step determines the coefficient object on which its slab maps act.

**2.4. Local Kummer charts.** Each matrix in (27) exchanges  $\alpha$  and  $\beta$  in  $D$ , while the core matrix  $K$  acts trivially. Hence  $\sigma$  in (24) is the maximal constant quotient of the sector local system. The finite sectors do not, however, account for the roots required by the slab maps.

For  $V \in \{A, B_0, B_1, C\}$  let  $f_V$  be the pulled-back slab function and  $N_V$  its primitive conormal in  $\widetilde{M}^\vee$ . With compatible orientations,

$$(30) \quad \begin{array}{c|cccc} V & A & B_0 & B_1 & C \\ \hline d_V & 2 & 4 & 4 & 2 \\ N_V & (1, -3) & (-1, -1) & (-3, 5) & (-3, 1). \end{array}$$

On the complement of the slab divisor put

$$(31) \quad R_V^{\text{root}} = \mathbb{k}[\widetilde{M}][f_V^{-1}, g_V]/(g_V^{d_V} - f_V), \quad \tilde{\theta}_V(z^m) = g_V^{\langle N_V, m \rangle} z^m.$$

**Proposition 2.4.** *The formula (31) defines a slab automorphism extending the old transition. The root degrees 2, 4, 4, 2 are minimal.*

*Proof.* The root  $g_V$  is a unit on the localized chart and  $\langle N_V, m \rangle$  is integral. Moreover,  $N_V = 4n_V$  for the old rational conormal. At the ramified lifts  $A, C$  the path traverses the old loop twice, so  $g_V^{\langle N_V, m \rangle} = f_V^{2\langle n_V, m \rangle}$ ; at  $B_0, B_1$  it traverses it once, giving  $f_V^{\langle n_V, m \rangle}$ . Primitivity of  $N_V$  shows that exponent  $1/d_V$  occurs on some character, so no proper divisor of  $d_V$  suffices.  $\square$

At the two cusps the degree-four class drops to order two. If  $f_B(a, z) = 1 + az + z^2$ , then

$$(32) \quad \begin{aligned} a = 2: & \quad g^4 - f_B = (g^2 - (1 + z))(g^2 + (1 + z)), \\ a = -2: & \quad g^4 - f_B = (g^2 - (1 - z))(g^2 + (1 - z)). \end{aligned}$$

Away from the reduced zero divisor these are two connected quadratic torsors; their closures meet over the divisor. We retain this fourth-root stack rather than replacing it by one connected square-root cover.

**2.5. The infinite root orbit.** The failure of finite Kummer closure already occurs on the overlap of the  $A$  and  $B_0$  charts. Write

$$x_A = z^{2u_A}, \quad x_B = z^{2u_{B,0}}, \quad f_A = 1 + x_A.$$

The  $A$ -slab shear fixes  $f_A$  and sends

$$(33) \quad x_B \mapsto f_A^4 x_B.$$

**Theorem 2.5** (Infinite Kummer orbit). *Closure under all powers of (33) requires roots*

$$(34) \quad h_n^2 = 1 + f_A^{4n} x_B, \quad n \in \mathbb{Z}.$$

*Their square classes are linearly independent in  $\mathbb{k}(x_A, x_B)^\times / \mathbb{k}(x_A, x_B)^{\times 2}$ . Consequently no finite algebraic Kummer extension containing the initial  $A$ - and  $B_0$ -roots is stable under the slab shear.*

*Proof.* Let  $E_n$  be the prime divisor  $1 + f_A^{4n} x_B = 0$ . It occurs with odd valuation in the  $n$ th radicand and with valuation zero in every other radicand: if  $m \neq n$ , substitution  $x_B = -f_A^{-4n}$  leaves  $1 - f_A^{4(m-n)}$ , nonzero at the generic point of  $E_n$ . No nonempty finite product of distinct radicands is therefore a square. Kummer theory gives degree at least  $2^r$  after adjoining any  $r$  of the roots. Since the shear takes  $h_n$  to  $h_{n+1}$ , a stable extension containing  $h_0$  contains them all.  $\square$

Let  $\mathfrak{C}$  be the chamber adjacency graph and fix a chamber  $u_0$ . Its universal reduced-path tree  $\mathcal{T}$  has vertices the reduced edge paths beginning at  $u_0$ . For a finite rooted subtree  $T \in \mathcal{T}$ , construct  $\mathfrak{X}_T$  recursively: at an edge labelled by  $V$ , pull back the current  $V$ -slab divisor along the unique preceding path, take its  $d_V$ th root stack, and base-change the roots already constructed. If  $T \subset T'$ , forgetting the new roots gives  $\mathfrak{X}_{T'} \rightarrow \mathfrak{X}_T$ .

**Definition 2.6.** Let  $R$  be the deformation coefficient algebra and  $I$  the singleton ideal. The pathwise root object and its adic completion are

$$(35) \quad \mathfrak{X}_{\text{path}} = \varprojlim_{T \in \mathcal{T}} \mathfrak{X}_T, \quad \widehat{\mathfrak{X}}_{\text{path}} = \varprojlim_N \varprojlim_{T \in \mathcal{T}} (\mathfrak{X}_T \times \text{Spec } R/I^N).$$

If  $\Pi$  is the deck group of the path tree over the annular chamber groupoid, the *annular pro-Kummer coefficient object* is

$$(36) \quad \widehat{\mathfrak{X}}_{\text{ann}}^{\text{root}} = [\widehat{\mathfrak{X}}_{\text{path}}/\Pi].$$

This inverse system is of the same formal kind as the infinite root stacks of [TV18]; the additional feature here is that the root divisors are indexed by reduced transport paths and are subsequently identified by the annular deck group.

Path concatenation permutes the complete set of pulled-back root divisors, preserves their degrees, and gives the action in (36). The quotient is descent along the chamber groupoid. It does not yet assert the consistency of the wall automorphisms, which will be proved in Section 3.

**Theorem 2.7** (Finite descent and nonrepresentability). *Every compact finite transport diagram modulo  $I^N$ , including finitely presented coefficient sheaves and their morphisms, descends to some finite root-stack presentation  $\mathfrak{X}_T \times \text{Spec } R/I^N$ . Any two such presentations have a common finite refinement. On the other hand, if all transported roots and their automorphisms are retained, (36) is not represented by a formal scheme, a finite root stack, or an algebraic stack whose diagonal is locally of finite type.*

*Proof.* A finite transport diagram involves finitely many paths and, modulo  $I^N$ , finitely many deformation bidegrees. The rooted union of their path indices is a finite subtree. Finite presentation then descends the finitely many generators, relations, arrows, and cocycle identities after a further finite enlargement. Rooted union gives a common refinement.

For the negative assertion, specialize the tower (34) at a geometric point with  $f_A^4 = 1$  and  $x_B = -1$ . Rooting the divisors indexed by a finite set  $S \subset \mathbb{Z}$  gives stabilizer  $\prod_{n \in S} \mu_2$ ; the inverse system contains

$$\prod_{n \in \mathbb{Z}} \mu_2.$$

This group scheme is not of finite type. A formal scheme has trivial stabilizers, a finite root stack has finite-type diagonal, and an algebraic stack with diagonal locally of finite type cannot have the displayed geometric stabilizer.  $\square$

This is the only inverse-limit principle used later. Every stated product, period, or logarithmic chart is first constructed on an ordinary finite root stage. Passage to the pro-object records compatibility; it does not replace an infinite calculation by notation.

### 3. JACOBI SCATTERING AND THETA FUNCTIONS

**3.1. The twisted volume line.** Fix a root chamber and an integral basis  $e_1, e_2$  of  $\widetilde{M}$ . Put

$$(37) \quad \kappa = e_1 + e_2, \quad \Omega_0 = d \log z_1 \wedge d \log z_2, \quad \Omega_\kappa = z^\kappa \Omega_0.$$

Let  $J(q_1, q_2) = (q_2, -q_1)$  and denote the invariant logarithmic derivation in direction  $v$  by  $\partial_v$ .

**Proposition 3.1** (Shifted Hamiltonian bracket). *The inverse bivector  $\pi_\kappa = z^{-\kappa}\pi_0$  of  $\Omega_\kappa$  has Hamiltonian vector fields*

$$(38) \quad X_q^\kappa = z^{q-\kappa}\partial_{Jq},$$

and

$$(39) \quad [X_q^\kappa, X_r^\kappa] = -\det(q, r)X_{q+r-\kappa}^\kappa.$$

Thus, after writing  $q = k + \kappa$  and  $r = \ell + \kappa$ , the output is indexed by  $k + \ell$ .

*Proof.* The Hamiltonian field of  $z^q$  for  $z^{-\kappa}\pi_0$  is (38). The bracket of two monomial derivations has direction

$$\langle Jq, r - \kappa \rangle Jr - \langle Jr, q - \kappa \rangle Jq = -\det(q, r)J(q + r - \kappa),$$

which gives (39). □

The lifted slab map at  $V$  is symplectic for  $\Omega_0$ , but

$$(40) \quad \tilde{\theta}_V^* \Omega_\kappa = g_V^{\langle N_V, \kappa \rangle} \Omega_\kappa, \quad (\langle N_V, \kappa \rangle)_V = (-2, -2, 2, -2).$$

Choose a local line frame  $e$  and identify frames across the slab by

$$(41) \quad \tilde{\theta}_V^* e' = g_V^{-\langle N_V, \kappa \rangle} e.$$

The products  $\Omega_\kappa \otimes e$  then glue. We denote the resulting line by  $\mathcal{L}_\kappa$ .

The transition law therefore requires a line bundle.

**Proposition 3.2** (No scalar frame). *The scalar  $\kappa$ -twisted Hamiltonian algebra containing the two singleton fields is not preserved under conjugation by any lifted slab transition. Consequently its local brackets do not glue to a bracket on the structure sheaf.*

*Proof.* If  $p_V$  is the slab exponent and

$$S_V = \frac{1}{d_V} \log(1 + z^{p_V}) \partial_{N_V},$$

then for  $k = q - \kappa$  one computes

$$(42) \quad [S_V, X_q^\kappa] = \sum_{r \geq 1} \frac{(-1)^{r+1}}{d_V} z^{k+r p_V} \left( \frac{\langle N_V, k \rangle}{r} \partial_{Jq} - \langle Jq, p_V \rangle \partial_{N_V} \right).$$

With respect to  $\Omega_\kappa$ , the divergence of the  $r$ th summand has coefficient

$$\frac{(-1)^r}{d_V} \langle Jq, p_V \rangle \langle N_V, \kappa \rangle.$$

For  $r = 1$  and the singleton exponents  $q_1 = (2, 1)$  and  $q_2 = (1, 2)$ , the eight coefficients are

|       | $A$ | $B_0$ | $B_1$ | $C$ |
|-------|-----|-------|-------|-----|
| $q_1$ | 2   | 3     | 1     | -10 |
| $q_2$ | 10  | 3     | -7    | -2. |

Thus every singleton-slab interaction has nonzero divergence. □

**Proposition 3.3** (Jacobi transport). *The line  $\mathcal{L}_\kappa$  on the pathwise pro-root object carries a canonical Jacobi bracket whose expression in the distinguished chamber is the  $\kappa$ -twisted Poisson bracket. Every lifted slab transition is a Jacobi isomorphism. The construction is compatible with finite root restriction and descends with the pro-Kummer coefficient object.*

*Proof.* In a frame  $e$ , write a Jacobi bracket as

$$\{fe, he\} = (\Lambda(df, dh) + fE(h) - hE(f))e.$$

Under  $e' = ae$  the local pair transforms by

$$(43) \quad \Lambda' = a\Lambda, \quad E' = aE + \Lambda^\sharp(da).$$

Start at the root chamber with  $(\Lambda, E) = (\pi_\kappa, 0)$  and transport the pair across each edge using (41) and (43). Every vertex of the path tree has a unique route from the root, so the construction is unambiguous and compatible with restriction to finite subtrees. Change of frame and pullback preserve the Jacobi identity. The deck action transports the same data, giving descent.  $\square$

Equivalently,  $\mathcal{L}_\kappa^\vee \setminus 0$  carries the Poissonization of this Jacobi structure. The line-bundle formulation displays the slab multiplier in (41) explicitly.

**Definition 3.4** (Jacobi walls and consistency). Fix a finite root-stack presentation and reduce the coefficient ring modulo  $I^N$ . A *Jacobi wall* is a pair  $(\mathfrak{d}, \Theta_\mathfrak{d})$ , where  $\mathfrak{d}$  is an oriented rational affine segment, ray, or full chord in the smooth locus and  $\Theta_\mathfrak{d}$  is a locally constant unipotent Jacobi automorphism along  $\mathfrak{d}$ , congruent to the identity modulo  $I$ . The automorphism is read in the local frame of  $\mathcal{L}_\kappa$  and is transported between wall charts by Proposition 3.3.

For a transverse path  $\gamma$ , its *path-ordered product*  $\Theta_\gamma$  is the product of the crossed wall automorphisms in crossing order, using the inverse when the coorientation is reversed. A locally finite collection of Jacobi walls is *consistent at a joint* if the path-ordered product around a sufficiently small positively oriented loop about that joint is the identity. It is *contractibly consistent* if  $\Theta_\gamma$  depends only on the homotopy class of  $\gamma$  relative to its endpoints for homotopies through compact transverse paths.

All these notions are finite-stage notions. A wall structure on the pro-Kummer object means a compatible system of such structures as the root stage and  $N$  vary. This convention prevents inverse-limit notation from hiding an infinite path-ordered product.

**3.2. The singleton joint.** Let  $t_1, t_2$  be the two singleton parameters,  $I = (t_1, t_2)$ , and put

$$(44) \quad E_{a,b} = t_1^a t_2^b X_{\kappa+(a,b)}^\kappa, \quad (a,b) \in \mathbb{N}^2 \setminus \{(0,0)\}.$$

Then

$$(45) \quad [E_{a,b}, E_{c,d}] = -\det(\kappa + (a,b), \kappa + (c,d)) E_{a+c, b+d}.$$

The two incoming GKT normalizations are

$$(46) \quad A = -\frac{1}{2}E_{1,0}, \quad B = -\frac{1}{2}E_{0,1}.$$

Since the shifted exponents are  $(2,1)$  and  $(1,2)$ ,

$$(47) \quad [A, B] = -\frac{3}{4}E_{1,1}.$$

**Proposition 3.5** (Ordered singleton completion). *Let  $\widehat{\mathfrak{g}}_{\text{sing}}$  be the product of the lines spanned by the  $E_{a,b}$ , completed in total deformation degree. The inverse commutator*

$$(48) \quad (e^A e^B e^{-A} e^{-B})^{-1}$$

*has a unique factorization into ray groups in increasing shifted slope. The factorization is finite modulo  $I^N$ , is compatible under reduction of  $N$ , and begins with  $\exp(\frac{3}{4}E_{1,1})$  in the chosen outgoing orientation.*

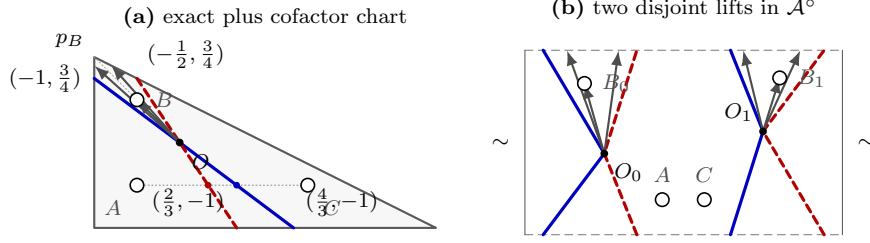


FIGURE 2. Cofactor placement. The two full chords carry the singleton factors. Off-diagonal corrections exit the deleted boundary; the diagonal support ends at a lift of  $B$ . Crosswise gluing of the branch cut produces two copies of the joint without adding intersections.

*Proof.* Formula (45) gives an  $\mathbb{N}^2$ -grading with finitely many components modulo  $I^N$ . Generators whose shifted exponents lie on one ray commute. At each total degree, take the logarithm of the residual word, decompose its homogeneous part into ray summands, and insert the summands in slope order. This is the standard filtered factorization for two-dimensional scattering diagrams [GPS10]. Directness of the ray decomposition gives uniqueness. Equation (47) gives the initial coefficient and the inverse-loop sign.  $\square$

**3.3. Cofactor supports and consistency.** In the plus cofactor chart, the incoming shifted exponents have support directions

$$(49) \quad L(2, 1) = (-4, 3), \quad L(1, 2) = (-2, 3).$$

Each is placed on the full chord through the central point  $O$ . Full chords are forced: a small loop around the endpoint of a truncated degree-one wall would cross that wall once, while every correction begins in degree two.

If  $q = \lambda(2, 1) + \mu(1, 2)$  with  $\lambda, \mu \geq 0$ , then

$$(50) \quad Lq = (-4\lambda - 2\mu, 3\lambda + 3\mu).$$

For  $\lambda > \mu$  the ray exits one upper facet, for  $\mu > \lambda$  it exits the other, and for  $\lambda = \mu$  it runs to the  $B$  singularity. A correction parallel to an incoming ray is multiplied into the positive half of the same chord. Every other correction is placed on its own outgoing radial support.

**Proposition 3.6.** *Modulo  $I^N$ , the cofactor placement is a proper locally finite wall support system on  $\mathcal{A}^\circ$ . Its only finite joints are the two lifts  $O_0, O_1$  of  $O$ . Distinct winding lifts do not meet on the annulus, even when their developed images intersect in an affine plane.*

*Proof.* Modulo  $I^N$  there are finitely many deformation bidegrees. After resonant corrections have been grouped with the incoming chords, the compact polygon contains a finite embedded chord-and-ray skeleton whose members meet only at  $O$ . In the normalization of Appendix A, the branch segment joins  $(-1/2, -1/2)$  to  $(3/2, -1/2)$  and crosses the two negative singleton chords at  $(2/3, -1/2)$  and  $(1/3, -1/2)$ , away from  $O$ . Crosswise gluing lifts each chord smoothly and produces two copies of the skeleton, without creating another intersection. Deleting the outer and focus endpoints makes each support relatively closed. A compact subset of  $\mathcal{A}^\circ$  stays a positive distance from these ends and therefore meets finitely many supports. Developed intersections of different deck lifts do not descend to intersections of the supports themselves.  $\square$

Transport the ordered factorization of Proposition 3.5 through the Jacobi path atlas.

**Theorem 3.7** (Open annular consistency). *The transported factors on the supports of Proposition 3.6 define a deck-equivariant Jacobi wall structure on  $\mathcal{A}^\circ$ . It is consistent at every joint and descends to  $\widehat{\mathfrak{X}}_{\text{ann}}^{\text{root}}$ . Every path-ordered transport along a compact transverse path modulo  $I^N$  is defined on one finite root-stack presentation and depends only on the contractible homotopy class of the path relative to its endpoints.*

*Proof.* At either joint  $O_i$ , the two full chords contribute the commutator in (48), while the outgoing walls carry its unique ordered inverse factorization. Their product is the identity modulo  $I^N$  for every  $N$ . Changing a path chart conjugates this complete identity by a Jacobi isomorphism. Deck transport carries one wall skeleton to the next, and Proposition 3.6 excludes new joints between different lifts. This proves descent and contractible consistency.

A compact path crosses finitely many walls modulo  $I^N$ . Theorem 2.7 places all of its root and wall transitions on one finite stage. Subdivision of a contractible homotopy into small joint loops then gives homotopy invariance.  $\square$

The qualification “contractible” retains the global annular holonomy absent from the boundary-charge cone.

**3.4. Hyperbolic holonomy and finite covers.** The matrix  $K$  in (3) satisfies

$$(51) \quad \chi_K(T) = T^2 - 18T + 1, \quad \lambda_{\pm} = 9 \pm 4\sqrt{5}.$$

For  $q_1 = (2, 1)$  and  $q_2 = (1, 2)$ , put  $q_{i,n} = K^n q_i$ . Then

$$(52) \quad q_{i,n+2} = 18q_{i,n+1} - q_{i,n}, \quad \det(q_{1,n}, q_{2,n}) = 3.$$

The core fixes their classes in  $D$  but moves every nonzero exponent through an infinite orbit.

More precisely, if a path  $\gamma$  has linear holonomy  $L_\gamma$ , its coefficient action in a developed chart has the form

$$(53) \quad \Psi_\gamma(z^m) = c_\gamma(m) z^{L_\gamma m},$$

where  $c_\gamma$  is a multiplicative character with values in finite products of path-indexed root units. Its unique lift to the volume line has multiplier

$$(54) \quad c_\gamma(\kappa)^{-1} z^{\kappa - L_\gamma \kappa}.$$

Thus the annular holonomy is a semidirect action on exponents, roots, and the volume line; a residual finite root phase alone does not describe it.

**Theorem 3.8** (Theta local system). *No nonzero  $m \in \widetilde{M}$  has finite  $K$ -orbit. A single-valued theta function on the annular quotient therefore cannot have a unique nonzero leading monomial  $z^m$ . Moreover, the formal orbit sum*

$$\sum_{n \in \mathbb{Z}} \text{Hol}(K)^n(\vartheta_m)$$

*does not converge in the  $I$ -adic completion. The framed theta modules, with the action (53)–(54), form a  $K$ -equivariant local system rather than a globally lattice-indexed basis.*

*Proof.* Neither eigenvalue in (51) is a root of unity, so  $K^r - 1$  is invertible over  $\mathbb{R}$  for every  $r \neq 0$ . Thus no nonzero lattice vector is periodic. Holonomy takes a theta section with leading term  $z^m$  to one with leading term  $z^{K^r m}$ , proving the first assertion. The terms of the orbit sum have distinct leading monomials, all of deformation order zero. Their orders do not tend to infinity, so the  $I$ -adic topology does not sum them.  $\square$

Two finite covers retain the finite grading data. Let  $p_{\text{sec}} : \mathcal{A}_{\text{sec}}^\circ \rightarrow \mathcal{A}^\circ$  be the degree-two cover on which  $D$  is constant. The residual involution exchanges  $\alpha$  and  $\beta$ . Put

$$(55) \quad M_{\text{GKT}} = 4\mathbb{Z}^2 \subset M \subset \widetilde{M}, \quad Q_{\text{GKT}} = \widetilde{M}/M_{\text{GKT}} \simeq (\mathbb{Z}/4)^2.$$

Every peripheral monodromy acts on  $Q_{\text{GKT}}$  by the order-four matrix

$$(56) \quad P_{\text{res}} = \begin{pmatrix} 2 & 1 \\ 3 & 0 \end{pmatrix} \pmod{4}.$$

The corresponding degree-four ordered-residue cover factors through  $p_{\text{sec}}$ . Both covers preserve the pro-Kummer roots, volume line, and wall structure. Neither kills the hyperbolic exponent holonomy: the first closed core lift to the ordered-residue cover acts by  $K^2$ .

On the sector cover, let  $\delta_{\text{deck}}$  denote the deck involution. For a  $D$ -graded deck-linearized object  $\mathcal{F}$ , define

$$(57) \quad \text{Corr}(\mathcal{F}) = p_*\mathcal{F}, \quad \text{Sym}(\mathcal{F})_j = \left( \bigoplus_{\sigma(d)=j} p_*\mathcal{F}_d \right)^{\delta_{\text{deck}}}.$$

The first functor retains the two singleton labels and the deck operator; the second takes the coordinate-symmetric summand. Since the ground field has characteristic zero, the idempotents

$$e_+ = \frac{1 + \delta_{\text{deck}}}{2}, \quad e_- = \frac{1 - \delta_{\text{deck}}}{2}$$

split the labelled object into symmetric and Prym parts. Products are first constructed equivariantly and only then restricted to  $e_+$ .

**3.5. Framed theta multiplication.** A *path frame* consists of a lifted source chamber, an initial exponent and generic endpoint, a compact region containing the broken lines under consideration, a deformation order  $N$ , and a finite root-stack presentation supporting the resulting finite diagram. For such a frame define

$$(58) \quad \vartheta_p = \sum_{\gamma \in \text{BL}(p, Q)} a_\gamma z^{m(\gamma)} = z^p + O(I).$$

Modulo  $I^N$  the sum is finite: only finitely many walls meet the compact region, and a contributing broken line has at most  $N - 1$  nontrivial bends.

**Theorem 3.9** (Finite-window theta multiplication). *Fix finitely many framed theta sections, products, and associativity diagrams modulo  $I^N$ . After a finite enlargement of the exponent set and refinement of the root-stack presentation, every requested product has a unique theta re-expansion. These products are commutative, associative, and equivariant under path and deck transport.*

If  $B(z^m) = \vartheta_m$  is the finite change-of-basis matrix and  $b_{p,q}$  is the vector of monomial coefficients in  $\vartheta_p \vartheta_q$ , then

$$(59) \quad B = 1 + U, \quad B^{-1} = \sum_{j=0}^{N-1} (-U)^j \pmod{I^N}, \quad c_{p,q} = B^{-1} b_{p,q}.$$

*Proof.* Proper local finiteness and the bend bound make every chosen theta sum and ordinary product finite. If re-expansion requires a new terminal exponent, adjoin its theta sum and repeat. Every correction raises  $I$ -order, so the process stops after at most  $N - 1$  rounds. Theorem 2.7 gives one finite root-stack presentation for the resulting diagram.

The no-bend term makes  $B$  unipotent, and the finite geometric series in (59) is its two-sided inverse. Re-expansion is therefore unique. Commutativity and associativity follow from ordinary multiplication in the source chart. Path and deck transports are algebra isomorphisms and carry the complete change-of-basis diagram with them.  $\square$

The raw monomial vector  $b_{p,q}$  and the theta structure constants  $c_{p,q}$  are different quantities. This distinction will be essential in Section 6, where Jacobian reduction and period normalization supply two further changes of basis.

#### 4. LOGARITHMIC COMPACTIFICATION

The wall structure of Theorem 3.7 is proper on the open annulus, but its supports have genuine exits at the deleted boundary and focus ends. Restoring the compact boundary does not turn these exits into ordinary harmless boundary joints. We replace them by logarithmic caps. The construction is first made for a finite GKT window  $\mathcal{W} = (L, P, \mathcal{E})$  in the sense of Definition 1.5, a deformation order  $N$ , and a finite root subtree  $T$ . Its coefficient ring is  $R_{\mathcal{W},N}$  from (8), with  $L$  in place of  $I$ . The symbol  $\mathfrak{X}_{T,\mathcal{W},N}$  denotes the finite logarithmic cap obtained from the root subtree  $T$  over this ring.

**4.1. Why the ends must be modified.** Near a smooth boundary point choose affine coordinates in which the tangent wedge is the upper half-plane and the boundary tangent is  $(1, 0)$ . If a wall terminates transversely with primitive inward direction  $m = (u, v)$ ,  $v > 0$ , and outgoing multiplicity  $d > 0$ , the extremal bend changes the boundary tangent to

$$(60) \quad (1 + dvu, dv^2).$$

Its normal component is positive. The extremal broken line remains inside the wedge, so the boundary-convexity argument for tangent canonical walls does not apply [GHS22]. The endpoint retains nontrivial one-sided transport.

Nor can one remove the boundary by an affine reflection. In a boundary tangent–normal basis, write its nontrivial holonomy and a putative pointwise-fixing reflection as

$$U = \begin{pmatrix} 1 & w \\ 0 & 1 \end{pmatrix}, \quad R = \begin{pmatrix} 1 & c \\ 0 & -1 \end{pmatrix}.$$

The relation  $RU = UR$  forces  $2w = 0$ , whereas the two annular boundary widths are 4 and 16.

At a finite stage, close the wall skeleton in  $\overline{\mathcal{A}}$  and retain the two one-sided germs at every transverse endpoint. A path may terminate at such a germ but is not continued across it. Let

$$(61) \quad E_{\mathcal{W},N} = \Theta_m \cdots \Theta_1$$

be the complete ordered product along one attaching circle, after coincident abelian supports have been combined. At  $A, C$  this word is empty; at  $B_0, B_1$  all diagonal exits lie in one abelian group. This stopped object is a useful intermediate form, but it is not yet a logarithmic compactification.

The focus germs already have standard Kummer nearby charts. If  $s$  is the slab divisor and  $xy = ts$  is the node chart, the degree- $d$  root is the fs monoid map

$$(62) \quad \langle x, y, t, s \mid x + y = t + s \rangle \longrightarrow \langle x, y, t, r \mid x + y = t + dr \rangle, \quad s \longmapsto dr.$$

On the atlas this is  $xy = tr^d$ ; inverting  $r$  recovers the root-slab chart of Section 2.

**Lemma 4.1** (The focus root chart). *For every  $d \geq 1$ , the monoid*

$$P_d = \langle x, y, t, r \mid x + y = t + dr \rangle$$

is fine and saturated. Its monoid algebra is

$$\mathbb{k}[P_d] \simeq \mathbb{k}[x, y, t, r]/(xy - tr^d).$$

The map in (62) is Kummer with group cokernel  $\mathbb{Z}/d$ , and its atlas is finite free, given by the monic equation  $r^d = s$ . After any coefficient base change, both the chart and its boundary  $r = 0$  are flat over the coefficient ring.

*Proof.* The defining binomial is prime because its exponent relation is primitive. The hypersurface  $xy = tr^d$  is Cohen–Macaulay. For  $d = 1$  its singular locus is the origin; for  $d > 1$  it is contained in  $x = y = r = 0$ , of codimension two. Serre’s criterion therefore makes  $\mathbb{k}[P_d]$  normal, equivalently  $P_d$  is saturated. The group cokernel and the monic presentation follow from  $s \mapsto dr$ . Finally,

$$\mathbb{k}[P_d]/(r) \simeq \mathbb{k}[x, y, t]/(xy),$$

so tensoring either ring with a coefficient algebra over  $\mathbb{k}$  preserves flatness.  $\square$

**4.2. Affine cylinder ends and peripheral holonomy.** Real-oriented blowup replaces the two boundary components and four focus points by six attaching circles. Their shear widths are

$$(63) \quad (w_0, w_1, w_A, w_{B,0}, w_{B,1}, w_C) = (4, 16, 1, 1, 1, 1).$$

**Proposition 4.2** (Outward affine flares). *Let an attaching circle have affine length  $\ell > 0$  and holonomy  $T^{-w}$ . The quotient of  $r \geq 0$  by*

$$(64) \quad (x, r) \mapsto (x + \ell + wr, r)$$

*is a flaring integral-affine cylinder. The action is free and proper, the circle at height  $r$  has length  $\ell + wr$ , and  $r$  is a proper integral convex exhaustion. Every finite-stage wall exit extends to a proper ray and cannot return to the annular core.*

*Proof.* The  $n$ th power of (64) translates  $x$  by  $n(\ell + wr)$ . This quantity is nonzero for  $n \neq 0$ , which proves freeness and properness. The normal component of a tangent vector is preserved by the shear. An exiting wall with positive normal component therefore crosses each level of  $r$  at most once.  $\square$

The affine cylinder has the correct affine holonomy, but its peripheral loop also crosses the product (61). Let  $\rho(H)$  denote the prescribed affine, root, sector, and volume-line transport around the attaching circle. Choose a radial seam disjoint from the exit supports and glue its two sides by

$$(65) \quad G_{\mathcal{W},N} = \rho(H)E_{\mathcal{W},N}^{-1}.$$

A positive peripheral loop crosses  $E_{\mathcal{W},N}$  and then the seam, so its total transport remains

$$(66) \quad G_{\mathcal{W},N}E_{\mathcal{W},N} = \rho(H).$$

The stopped operation has not been killed; it is the correction between the bare peripheral gluing and the cap seam.

The polarization extends over every flare with positive integral kink seven. Indeed, if one branch on the core is  $ax + br + c$ , use  $ax + (b + 7)r + c$  on the flare. The value seven is a convenient uniform choice: the directions and pole orders of the four basic wall characters at their negative ends are

| $q$        | (2, 1)  | (4, 3)  | (3, 4)  | (1, 2)  |
|------------|---------|---------|---------|---------|
| $Lq$       | (−4, 3) | (−8, 7) | (−6, 7) | (−2, 3) |
| pole order | 3       | 7       | 7       | 3.      |

Additivity gives a bound of seven per deformation degree. Equivalently, the auxiliary Rees relation

$$(67) \quad \lambda = u^7 s$$

regularizes every finite ordered product of exiting-wall automorphisms in one dominating expansion. Equation (67) does not define the boundary: its divisor  $u = 0$  is vertical over  $\lambda = 0$ . Horizontal flatness will instead come from transporting the monoid chart itself.

**Proposition 4.3** (Separation of the exits). *At each finite stage, the positive and negative support families on either outer flare are pairwise disjoint, apart from coincident supports combined in one abelian ray group. A support-free seam can therefore be chosen, and the chamber nerve of the cut flare is an interval. The focus flares  $A, C$  carry no support, while  $B_0, B_1$  carry one abelian radial support.*

*Proof.* Develop an outer attaching circle in invariant tangent–normal coordinates. A support of tangential speed  $\sigma$  has equation  $x(r) = \phi_i - \sigma(r + h_i)$ , while the deck period at height  $r$  is  $w_i(r + h_i)$ . Thus two supports meet modulo deck translation exactly when their slope difference lies in  $w_i\mathbb{Z}$ . The exact slope intervals, derived in Appendix A, have width smaller than  $w_i$ ; every positive-minus-negative difference lies strictly between 0 and  $w_i$ . Hence no two distinct supports meet. The focus assertions follow from the cofactor placement: only the diagonal ray reaches a  $B$  lift.  $\square$

**4.3. Transported monoids.** Let  $R = R_{\mathcal{W}, N}$  be the finite Artin coefficient ring just defined. Choose a base chamber  $C_0$ , an fs toric cap monoid  $P$ , and a finite Kummer algebra  $S_0$  over  $R[P]$ . Locally,  $S_0$  is obtained by tangent localizations and monic root equations

$$(68) \quad S_i = S_{i-1}[r_i]/(r_i^{d_i} - f_i).$$

Let  $\nu : P \rightarrow \mathbb{N}$  be the outward valuation and

$$(69) \quad J_0 = (z^p : \nu(p) > 0)S_0.$$

If  $\Phi_C$  is the ordered Jacobi product from  $C_0$  to a chamber  $C$ , set inside the punctured Kummer algebra

$$(70) \quad S_C = \Phi_C(S_0), \quad J_C = \Phi_C(J_0), \quad P \longrightarrow S_C, \quad p \longmapsto \Phi_C(z^p).$$

The transport acts on the monoid chart itself. A character with a pole in  $S_0$  is regular in the appropriate  $S_C$  by construction.

Around a wall endpoint choose a small toric open star  $U$ . If  $\Phi_-$  and  $\Phi_+$  are the products in the adjacent chambers, identify the complete transported charts, including their boundary strata, by

$$(71) \quad \Phi_+ \Phi_-^{-1} : \Phi_-(S_0(U)) \xrightarrow{\sim} \Phi_+(S_0(U)).$$

The identification includes the full boundary overlap. Gluing only after inverting the outward uniformizer would leave two copies of the boundary point.

**Theorem 4.4** (Finite-stage logarithmic cap). *The charts (70), glued by the full boundary identifications (71), form a separated fine-saturated formal logarithmic cap over  $R$ . Its asymptotic divisor is defined chamberwise by  $J_C$  and is flat over  $R$ . The complement of this divisor is the corrected mapping torus (65); reduction modulo the deformation ideal is the ordinary toric/Kummer cap with peripheral transport  $\rho(H)$ . The construction commutes with restriction of the root tree and labelled coefficient system and with reduction of  $N$ .*

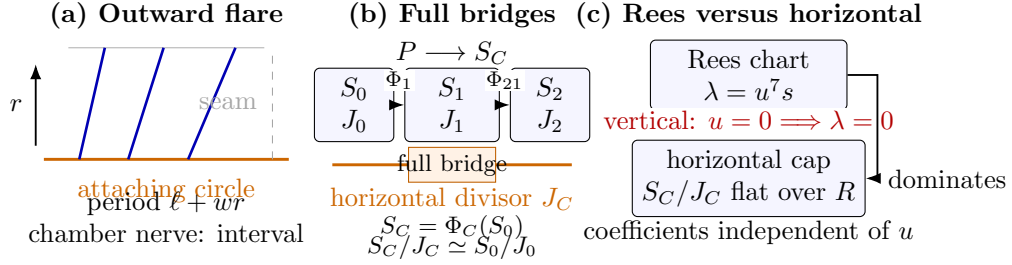


FIGURE 3. A logarithmic cap. Wall rays on the outward flare give a linear chamber order. Each chamber carries the transported monoid chart  $P \rightarrow S_C$ , and adjacent charts are identified across a full bridge containing the horizontal boundary. The Rees relation controls pole order but does not define that boundary.

*Proof.* Every equation in (68) is monic. Before quotienting by the finite diagonalizable root group,  $S_0$  is therefore finite free over the preceding toric ring; after restriction to the toric boundary the same monic equations remain. Since  $R$  is a characteristic-zero  $\mathbb{k}$ -algebra, the Reynolds operator splits invariants under this finite group. Thus the invariant affine charts, as well as the root-stack atlases themselves, have  $R$ -flat boundary, and in particular  $S_0/J_0$  is  $R$ -flat.

Transport by  $\Phi_C$  is an  $R$ -algebra isomorphism. The prelog chart in (70) is strictly isomorphic to the original toric chart, so its characteristic monoid is fs, and

$$(72) \quad S_C/J_C \simeq S_0/J_0$$

is  $R$ -flat.

By Proposition 4.3, the wall closures meet the asymptotic divisor at distinct points and the chamber nerve is an interval. The bridge (71) is an isomorphism of the entire chart rings. On endpoint bridges the overlap ring is either chart ring; on chamber-band overlaps, conjugation by  $\Phi_C$  reduces to an ordinary intersection of toric opens. In each case the affine-intersection multiplication map is surjective, proving separatedness. Since the endpoint stars are disjoint, there are no further intersections or cap joints.

If  $C_m$  is the last chamber,  $\Phi_{C_m} = E_{\mathcal{W},N}$ . At the seam,

$$G_{\mathcal{W},N}(S_{C_m}) = \rho(H)E_{\mathcal{W},N}^{-1}E_{\mathcal{W},N}(S_0) = \rho(H)(S_0),$$

which closes the cap with total transport (66). Modulo the deformation ideal every wall product is the identity. All operations are functorial under the stated restrictions.  $\square$

There are three distinct restrictions of a finite cap. Removing its asymptotic divisor gives the punctured corrected mapping torus. Reducing the coefficient ring modulo its deformation ideal gives the toric/Kummer scattering-special fiber. Restricting to the asymptotic divisor gives  $\mathrm{Spf}(S_C/J_C)$ , which remains flat over the entire coefficient ring by (72). No coefficient variable is identified with a spatial boundary uniformizer.

**4.4. The compactification theorem.** Truncate each flare at an integral level. The extension of the polarization has positive kink seven, and the product subdivision gives locally convex boundary. The original focus points have become peripheral end loops; their primitive monodromies remain as holonomy rather than codimension-two singularities of the truncation. The

finite tropical spaces are therefore positive, locally rigid, and pre-polarized in the sense used in [CPS24].

**Theorem 4.5** (Separated fs logarithmic compactification). *The finite caps of Theorem 4.4, including the focus charts (62), form an inverse system indexed by finite root subtrees, finite labelled coefficient systems closed under the dependencies of Appendix B, and deformation orders. Each stage is separated and fs logarithmic, and its six horizontal boundary divisors are flat over its finite Artin coefficient ring. Its affine base has flare widths 4, 16, 1, 1, 1, 1 and a positive integral polarization. Its punctured restriction carries the original annular Jacobi wall structure, with all root, sector, exponent, and volume-line holonomies unchanged.*

*The inverse limit and chamber-groupoid quotient give a logarithmic compactification in the adic proalgebraic pro-Kummer category. No regularity or log-regularity assertion is made, and the limit is not claimed to be an algebraic stack of finite type.*

*Proof.* Proposition 4.2 supplies the affine ends and their proper convex exhaustion. Proposition 4.3 supplies a support-free seam and linearly ordered chamber cover, so every end satisfies Theorem 4.4. Tensoring the Kummer monoid charts (62) with the cap monoids extends the root data. Lemma 4.1 gives fs charts and coefficient-flat boundary at these four ends; root stacks commute with the base change. Proposition 3.3 then extends the volume line and its bracket.

The full bridges introduce neither doubled boundary points nor new joints. All new charts restrict to the original ones before the exits, so contractible consistency and every framed annular product remain unchanged. Monic root equations and (72) give flatness at each stage. Root construction, ordered wall products, and bridge maps commute with every finite restriction. Passing to the compatible inverse system and then to the groupoid quotient proves the theorem.  $\square$

The special fiber in this theorem is the toric/Kummer cap of the annular model. It is not identified with the central fiber of the original disk-shaped quartic degeneration. This distinction is topological as well as categorical, and the topological half of it is an obstruction.

**Proposition 4.6** (No ordinary CPS recovery). *Neither the annular cover  $\overline{\mathcal{A}}$  nor its noncompact Legendre-dual cylindrical form is the dual intersection complex of a distinguished toric degeneration of del Pezzo surfaces with simple singularities, proper generic fiber, and smooth anticanonical divisor.*

*Proof.* For such a degeneration the dual intersection complex is homeomorphic to  $\mathbb{R}^2$  [CPS24, Prop. 7.13], so its first cohomology vanishes. An annulus and an open cylinder both have  $H^1(-, \mathbb{k}) \simeq \mathbb{k}$ .  $\square$

The conclusion is limited to that class of degenerations. It does not exclude logarithmic or orbifold degenerations, non-proper models, a boundary of different topology, or a comparison which agrees only on part of the sector grading. It does exclude reading the annulus as an unnoticed substitute for the ordinary Carl–Pumperla–Siebert affine base.

**4.5. A regular dominating presentation.** The finite stages of Theorem 4.5 are not themselves log regular in general. This failure comes from the transported root divisors, not from the asymptotic boundary.

**Proposition 4.7** (A multiple-root obstruction). *There is a finite root-stack presentation whose atlas is singular and not log regular. More precisely, three divisors in the orbit (34) meet with distinct tangent directions at one point of the two-dimensional torus.*

*Proof.* Write  $f_A = 1 + x_A$  and

$$F_n = 1 + f_A^{4n} x_B.$$

At  $p = (x_A, x_B) = (-2, -1)$  one has  $f_A(p) = -1$  and  $F_n(p) = 0$  for every  $n$ . For  $n = 0, 1, 2$  the gradients are

$$\nabla F_0(p) = (0, 1), \quad \nabla F_1(p) = (4, 1), \quad \nabla F_2(p) = (8, 1),$$

so the three reduced divisors are smooth and pairwise transverse at  $p$ . The simultaneous root atlas has equations

$$r_i^2 = F_i, \quad 0 \leq i \leq 2.$$

It is finite over a surface. At the point above  $p$  with all  $r_i = 0$ , its three-by-five Jacobian has rank two, so its tangent space has dimension three. The atlas is therefore singular. If the three root divisors are put in the log structure, the log ideal cuts out the closed point but the sharp characteristic group has rank three. Kato's dimension equality would read  $2 = 0 + 3$ , so the chart is not log regular. Such a finite stage occurs by taking a rooted subtree containing the paths which give  $F_0, F_1, F_2$ .  $\square$

Regularity is recovered by a modification of the finite presentations. The distinction is useful: it keeps the minimal root object of Theorem 4.5, while providing a toroidal model when regular geometry is required.

**Theorem 4.8** (Regular dominating presentation). *For every finite stage  $\mathfrak{X}_{T, \mathcal{W}, N}$  of Theorem 4.5, over its Artin coefficient ring  $R = R_{\mathcal{W}, N}$ , there is a cofiltered system of proper morphisms*

$$\mathfrak{X}_\lambda^{\text{reg}} \longrightarrow \mathfrak{X}_{T, \mathcal{W}, N}$$

*with the following properties.*

- (1) *Each morphism is an isomorphism away from the spatial boundary and the transported slab-zero divisors.*
- (2) *Each  $\mathfrak{X}_\lambda^{\text{reg}}$  is a separated logarithmic Deligne–Mumford stack, log smooth over  $(\text{Spec } R, 0)$ , and has a smooth atlas over  $R$ . Every geometric fiber has a regular atlas and is log regular.*
- (3) *Its total logarithmic boundary is simple normal crossings on an étale atlas. In particular, the six asymptotic divisors remain flat over  $R$ .*
- (4) *All finite root phases, the volume line, the Jacobi bracket, the wall maps, and the corrected peripheral transports pull back. Full bridges and seams lift to strict logarithmic isomorphisms.*
- (5) *Common refinements are compatible with enlargement of the root tree, restriction of  $\mathcal{W}$ , and reduction of  $N$ . Their inverse system carries the chamber-groupoid action and gives a regularized adic pro-Kummer compactification dominating that of Theorem 4.5.*

*For a fixed resolved boundary, its component-root maps are Kummer log étale. A transition which introduces a previously absent root divisor is not asserted to be Kummer log étale.*

*Proof.* Fix the finite stage and form the reduced union  $\Delta_T$  of its toric boundary and its finitely many transported root divisors. The root radicands have deformation degree zero, so this pair is defined over  $\mathbb{k}$  and then base-changed to  $R$ . First apply functorial logarithmic resolution in characteristic zero to the root-free toroidal chart:

$$\rho : Z_T^{\text{sm}} \longrightarrow Z_T.$$

This is an isomorphism off  $\Delta_T$ , and  $Z_T^{\text{sm}}$  is a smooth toroidal orbifold. On it, functorially principalize the product of the pulled-back radicand ideals, obtaining

$$\pi : Y_T \longrightarrow Z_T^{\text{sm}}, \quad \tilde{\pi} = \rho \circ \pi : Y_T \longrightarrow Z_T.$$

Then  $Y_T$  is smooth and the reduced total transform  $D_T$  is simple normal crossings [ATW20]. Both steps are functorial for toroidal isomorphisms, hence for the chamber maps, full bridges, finite diagonal actions, and seams.

Etale locally, let  $x_1, \dots, x_s$  cut out the components of  $D_T$ . For every radicand  $f_\alpha$  of degree  $d_\alpha \in \{2, 4\}$ , principalization gives

$$\tilde{\pi}^* f_\alpha = u_\alpha \prod_{j=1}^s x_j^{a_{\alpha j}}, \quad u_\alpha \in \mathcal{O}_{Y_T}^\times.$$

Take the fourth root stack of every component, writing  $x_j = y_j^4$ , and etale locally retain all solutions of  $\eta_\alpha^{d_\alpha} = u_\alpha$ . Then

$$(73) \quad g_\alpha = \eta_\alpha \prod_{j=1}^s y_j^{4a_{\alpha j}/d_\alpha} \quad \text{satisfies} \quad g_\alpha^{d_\alpha} = \tilde{\pi}^* f_\alpha.$$

All exponents are integral. Thus the component-root stack maps to the original root-stack presentation and retains its full root-choice orbit. At a  $B$  cusp,  $f_B = h^2$ , formula (73) reads  $h = y^4$  and  $g = \eta y^2$ . The ambient  $\mu_4$  action exchanges the two normalized quadratic branches, while the effective class on either branch has order two, as in (32).

Before the finite diagonal quotient, an etale local atlas is smooth over  $R$  with boundary  $y_1 \cdots y_s = 0$ . Its chart is  $0 \rightarrow \mathbb{N}^s$ , so Kato's chart criterion proves log smoothness [Kat89]. Finite diagonal groups are etale in characteristic zero; their quotients are therefore log-smooth Deligne–Mumford stacks. Geometric fibers have regular atlases with simple-normal-crossings boundary and hence are log regular. The fourth-root map  $\mathbb{N}^s \rightarrow \frac{1}{4}\mathbb{N}^s$  is Kummer log etale. Smoothness over  $R$  and the normal-crossings description also give flatness of the asymptotic divisors.

The resolution–principalization composite is an isomorphism on the punctured transport locus, so all wall, line, and theta data pull back unchanged. Given two finite regularizations, pass to the rooted union of their stages, take the closure of their common graph, and apply the same resolution–principalization procedure. This gives a regular model dominating both. The resulting index category is cofiltered. Path concatenation reindexes it, while functoriality under isomorphisms lifts the groupoid action. This proves the pro-system assertion. Finally, adding a new branch divisor raises the characteristic rank at its generic point, so such an enlargement map need not be Kummer; this explains the qualifier in the statement.  $\square$

## 5. GKT FACTORS INSIDE ANNULAR SCATTERING

The consistent wall structure is determined by the two singleton factors and the requirement that the path-ordered product about every joint be the identity. A GKT endpoint operation has a different origin: it is the transition between two systems of open invariants defined by canonical multisections. The two constructions use the same Hamiltonian Lie algebra after the  $\kappa$ -shift, but their coefficients need not agree. This section identifies their exact relation.

**5.1. Labelled critical algebras.** Let  $Q = Q_{\text{GKT}} = (\mathbb{Z}/4)^2$  be the ordered residue group of (55). An object  $\mathcal{I} = (I, \tau)$  consists of a finite set and a twist map  $\tau : I \rightarrow Q$ ; morphisms are twist-preserving bijections. Put

$$(74) \quad \mathcal{A}_{\mathcal{I}} = \mathbb{Q}[\{u_{i,d} : i \in I, d \geq 0\}] / \langle u_{i,d} u_{i,d'} : i \in I \rangle.$$

Thus two variables belonging to the same marking label have zero product.

Write  $\tau(i) = (a_i, b_i)$  with representatives in  $\{0, 1, 2, 3\}^2$ . The quartic GKT critical Lie algebra is spanned over  $A_{\mathcal{I}}$  by  $u_{J,\mathbf{d}}X_k$ , where  $J \subset I$  is nonempty,  $u_{J,\mathbf{d}} = \prod_{i \in J} u_{i,d_i}$ , and

$$(75) \quad k \equiv \sum_{i \in J} \tau(i) \pmod{4}, \quad k_1 + k_2 = \sum_{i \in J} (a_i + b_i + 4(d_i - 1)).$$

The vector field is

$$(76) \quad X_k = x^{k_1} y^{k_2} ((k_2 + 1)x\partial_x - (k_1 + 1)y\partial_y).$$

The marginal generator  $X_{0,0} = x\partial_x - y\partial_y$  is adjoined when it occurs.

**Theorem 5.1** (Labelled critical-algebra embedding). *For every  $\mathcal{I}$ , the assignment*

$$(77) \quad j_{\mathcal{I}} : u_{J,\mathbf{d}}X_k \mapsto u_{J,\mathbf{d}}X_{k+\kappa}^{\kappa}$$

*is a natural coefficient-preserving injective Lie map from the GKT critical algebra to the Jacobi wall algebra in the distinguished root chamber. It sends  $X_{0,0}$  to  $X_{\kappa}^{\kappa}$ . Jacobi transport carries its image to a strictly path-natural local system on the ordered-residue cover. The construction commutes with relabelling, finite restriction, and deformation truncation.*

*Proof.* In GKT notation,

$$[X_k, X_{\ell}] = -\det(k + \kappa, \ell + \kappa)X_{k+\ell}.$$

Equation (39) gives the same formula after applying (77). If two coefficient monomials share a label, their product vanishes in (74); if their supports are disjoint, both conditions in (75) add. Hence the assignment is a Lie map, and it is injective on the displayed monomial basis. The marginal bracket is preserved because

$$[X_{0,0}, X_{a,b}] = (a - b)X_{a,b} = -\det(\kappa, (a, b) + \kappa)X_{a,b}.$$

Jacobi path transport is a Lie isomorphism by Proposition 3.3. On the ordered-residue cover the  $Q$ -grading is constant; affine transport moves the Hamiltonian exponent and the  $\kappa$ -shift together. The remaining naturality statements are immediate from the labelled formula.  $\square$

In a finite nilpotent quotient, exponential integrates (77) to an injective group homomorphism. The GKT jump theorem identifies an isolated critical zero of a canonical homotopy with the exponential of its critical-graph generator, while the boundary-ray theorem of [Wue26] gives its unique increasing-slope factorization, giving the following comparison.

**Proposition 5.2** (Finite endpoint operations). *Every finite GKT endpoint transition maps to a framed Jacobi automorphism, and its increasing-slope factors map factor by factor to the shifted rays  $\mathbb{R}_{\geq 0}(k + \kappa)$ . The image depends only on the realizable endpoints, not on the chosen canonical homotopy. It is natural under labelled restriction and path transport.*

*Proof.* Exponentiation and Baker–Campbell–Hausdorff are finite in the chosen quotient and are functorial for the map of Theorem 5.1. GKT prove that their finite action on realizable chamber indices is faithful and transitive [GKT22, Thm. 4.27 and Cor. 5.14]. This alone would not give uniqueness of a transition element. The enlarged-torsor statement in Theorem 1.1, which includes the marginal generator, proves that the enlarged action is free and transitive. Its freeness makes the element carrying one endpoint to the other unique. The slope assertion follows because  $k$  maps to shifted exponent  $k + \kappa$ .  $\square$

Appendix B strengthens this formal statement: the finite endpoint homotopies can be chosen compatibly as the labelled system grows. Thus every GKT factor distinguished below has an actual finite multisection realization.

The ordered-residue cover does not identify annular transport with standard retwisting of one fixed polynomial critical algebra. Indeed, the peripheral residue action completes the singleton twists into

$$\{(1, 0), (2, 3), (3, 2), (0, 1)\}.$$

Two distinct labels of twist  $(1, 0)$  and descendent degree one have a joint critical block of rank one, with exponent  $(2, 0)$ . Retwisting both by the matrix  $P_{\text{res}}$  of (56) produces three exponents  $(0, 10), (4, 6), (8, 2)$ . Hence no coefficient-preserving linearization can fix labels and descendent degrees while retwisting every marking by  $P_{\text{res}}$ . Annular transport instead moves the exponent and volume frame and may produce root-valued Laurent expressions in the target chart.

**5.2. Enumerative and residual factors.** Take two distinct labels with

$$t_1 = u_{i,1}, \quad \tau(i) = (1, 0), \quad t_2 = u_{j,1}, \quad \tau(j) = (0, 1),$$

and work modulo  $(t_1^2, t_2^2)$ . The two extreme GKT potentials are

$$(78) \quad \begin{aligned} W^{\min} &= x^4 + y^4 - 4t_1xy^4 - 2t_2y^5 + 12t_1t_2xy^5, \\ W^{\max} &= x^4 + y^4 - 2t_1x^5 - 4t_2x^4y + 12t_1t_2x^5y. \end{aligned}$$

Their increasing-slope endpoint transition is

$$(79) \quad \exp\left(-\frac{1}{2}t_2X_{0,1}\right) \exp\left(\frac{1}{4}t_1t_2X_{1,1}\right) \exp\left(-\frac{1}{2}t_1X_{1,0}\right).$$

By contrast, the inverse commutator of the two singleton walls has exact mixed factor

$$(80) \quad T = \exp\left(\frac{3}{4}t_1t_2X_{1,1}\right).$$

Define

$$(81) \quad G_{\text{mix}} = \exp\left(\frac{1}{4}t_1t_2X_{1,1}\right), \quad R_{\text{mix}} = \exp\left(\frac{1}{2}t_1t_2X_{1,1}\right).$$

**Proposition 5.3** (First GKT–residual factorization). *The three factors lie in one abelian shifted-ray group and*

$$(82) \quad T = G_{\text{mix}}R_{\text{mix}}, \quad \frac{3}{4} = \frac{1}{4} + \frac{1}{2}.$$

*The first factor on the right is the unique mixed GKT endpoint factor; the second is the unique residual factor. Replacing the total wall by the GKT factor leaves contractible-loop holonomy  $R_{\text{mix}}^{\pm 1}$ .*

*Proof.* The balanced two-label profiles are  $(1, 5)$  and  $(5, 1)$ . The GKT chamber equations give endpoint coefficient  $-12$  on each extreme. After applying the first singleton factor, the remaining difference in the coefficient of  $xy^5$  is two. Since

$$X_{1,1}(x^4 + y^4) = 8x^5y - 8xy^5,$$

the mixed endpoint generator has coefficient  $2/8 = 1/4$ , proving (79). Equation (47) gives the scattering coefficient  $3/4$ . The square-zero mixed ray is central, so subtraction of logarithms gives the residual coefficient  $1/2$  exactly. The total factor is the one which cancels the joint product.  $\square$

There is no group projection which fixes the two incoming factors and sends  $T$  to  $G_{\text{mix}}$ : a homomorphism fixing the inputs fixes their commutator. Nor can a rank-one character remove  $R_{\text{mix}}$ , since a character is trivial on the commutator subgroup. The GKT factor is therefore a distinguished subfactor of the total wall, not a quotient wall of a second consistent diagram.

**5.3. All-order factorization.** Let  $\mathcal{I}$  be a finite set of distinct descendent-one labels, each of twist  $\alpha = (1, 0)$  or  $\beta = (0, 1)$ , in the square-zero labelled ring. For  $\emptyset \neq J \subseteq \mathcal{I}$  put

$$p_J = |J_\alpha|, \quad q_J = |J_\beta|,$$

and write  $p_J = r_J + 4\ell_J$ ,  $q_J = s_J + 4m_J$  with  $0 \leq r_J, s_J \leq 3$ . The balanced profiles and critical exponents in the full-support GKT block are

$$(83) \quad b_{J,h} = (r_J + 4h, s_J + 4(\ell_J + m_J + 1 - h)), \quad 0 \leq h \leq \ell_J + m_J + 1,$$

$$(84) \quad k_{J,h} = (r_J + 4h, s_J + 4(\ell_J + m_J - h)), \quad 0 \leq h \leq \ell_J + m_J.$$

Write

$$b_J^{\min} = b_{J,0}, \quad b_J^{\max} = b_{J,\ell_J+m_J+1},$$

and let  $\nu_J^\epsilon$  be the full-support endpoint invariant in the chamber  $\epsilon \in \{\min, \max\}$ . For any exponent  $(u, v)$  in this residue class, set

$$(85) \quad w_J(u, v) = \frac{\Gamma((u+1)/4)\Gamma((v+1)/4)}{\Gamma((r_J+1)/4)\Gamma((s_J+1)/4)}.$$

This is the quartic Gamma weight in the GKT chamber equation; it is nonzero for all balanced profiles occurring here.

Let

$$(86) \quad E_J = t_J \sum_h g_{J,h} X_{k_{J,h}}$$

be the unique slope-ordered GKT endpoint logarithm. If  $a_{p,q}$  is the coefficient of  $E_{p,q}$  in the unlabelled singleton scattering factorization, set  $a_{p,0} = a_{0,q} = 0$  for  $p, q > 0$  and define the full-support scattering logarithm

$$(87) \quad S_J = p_J! q_J! a_{p_J, q_J} t_J X_{p_J, q_J}.$$

For a shifted ray  $\mathfrak{r}$ , let  $E_{\mathfrak{r}}$  and  $S_{\mathfrak{r}}$  be the sums of the corresponding terms and put

$$(88) \quad R_{\mathfrak{r}} = S_{\mathfrak{r}} - E_{\mathfrak{r}}.$$

**Theorem 5.4** (Canonical all-order GKT factorization). *The endpoint recursions determine every coefficient in (86) canonically, and*

$$(89) \quad \prod_{\mathfrak{r}} \exp(S_{\mathfrak{r}}) = \prod_{\mathfrak{r}} (\exp(E_{\mathfrak{r}}) \exp(R_{\mathfrak{r}})).$$

*The equality is raywise; no factor is commuted past a factor on another slope. It is natural under relabelling, deletion of labels, restriction of finite labelled systems closed under the dependencies of Appendix B, reduction of deformation order, and framed Jacobi transport. It adds no new geometric support to the annular diagram.*

*Proof.* The GKT dimension formula gives (83)–(84). For an endpoint  $\epsilon \in \{\min, \max\}$ , the full-support invariant is determined by the partition recursion

$$(90) \quad w_J(b_J^\epsilon) \nu_J^\epsilon + \sum_{\substack{\pi \in \text{Part}(J) \\ |\pi| \geq 2}} w_J \left( \sum_{C \in \pi} b_C^\epsilon \right) \prod_{C \in \pi} \nu_C^\epsilon = \begin{cases} -1, & |J| = 1, \\ 0, & |J| \geq 2. \end{cases}$$

The coefficient of  $\nu_J^\epsilon$  is a nonzero Gamma weight, so induction on  $|J|$  determines the endpoints. The critical generators form the weighted incidence matrix of the path of balanced profiles. Indeed,

$$(91) \quad X_{u,v}(x^4 + y^4) = 4(v+1)x^{u+4}y^v - 4(u+1)x^u y^{v+4}.$$

For  $(u, v) = k_{J,h}$  the two monomials in (91) are precisely the adjacent vertices  $b_{J,h+1}$  and  $b_{J,h}$ . The resulting path incidence matrix has full column rank, and its one left-kernel relation is the chamber equation. Hence the coefficients  $g_{J,h}$  exist and are unique.

For the scattering term, substitute

$$t_\alpha = \sum_{\tau(i)=\alpha} t_i, \quad t_\beta = \sum_{\tau(i)=\beta} t_i$$

in the unlabelled factorization. The coefficient of  $t_J$  in  $t_\alpha^{p_J} t_\beta^{q_J}$  is  $p_J! q_J!$ , which gives (87). Both constructions are triangular in proper subsets, proving naturality under restriction. Generators on one shifted ray commute, so (88) exponentiates and yields (89). A nonzero  $E_\tau$  with  $S_\tau = 0$  is recorded as the identity pair  $\exp(E_\tau) \exp(-E_\tau)$ , not as a geometric wall.  $\square$

**5.4. Compatible geometric realization.** The preceding factorization has a geometric realization. The relative extension arguments in Appendix B give the following geometric realization theorem.

For the theorem, an *exact GKT datum* is a graded open  $(4, 4)$ -graph together with its descendent vector. A *finite partial system* records a finite twist-labelled set, a finite set of labelled monomials closed under divisors, and finitely many exact data closed under the connected-component and boundary dependencies of the GKT assembling relation. The precise definition is given in Appendix B.

**Theorem 5.5** (Compatible completed GKT realization). *There are nested finite partial systems  $\mathbb{D}_1 \subset \mathbb{D}_2 \subset \dots$  exhausting all exact GKT data, compatible canonical families realizing the two extremal chamber indices, and a componentwise smooth GKT homotopy between them such that:*

- (1) *the countably many ray factors occur in pairwise disjoint open time intervals, ordered by increasing slope;*
- (2) *each fixed component is nonconstant on only finitely many such intervals and has constant endpoint collars;*
- (3) *every finite divisor-closed quotient has precisely its prescribed finite collection of nonzero ray jumps; and*
- (4) *the finite restrictions form an inverse limit in the category of sets.*

*Simplicity is asserted for the finite elementary homotopies, not for one global assignment of jump times to all label subsets.*

Thus every GKT factor in Theorem 5.4 is the image of an actual finite restriction of a compatible homotopy. The residual factors are not enumerative jumps; they remain comparison automorphisms.

**5.5. Residual coherence.** The residual factors can be retained coherently in multiplication. Fix a finite labelled system and two framed input paths. Let  $U_i$  be the total consistent transport,  $V_i$  its GKT subword, and  $Q_i = U_i V_i^{-1}$  the residual transport. In a finite nilpotent quotient define the midpoint

$$(92) \quad H = Q_1(Q_1^{-1}Q_2)^{1/2}, \quad C_i = H Q_i^{-1}.$$

Rational powers are the finite polynomials  $g^c = \exp(c \log g)$ .

**Lemma 5.6.** *The midpoint is symmetric in  $Q_1, Q_2$  and*

$$(93) \quad C_i U_i = H V_i, \quad \mu(C_1 U_1 f_1, C_2 U_2 f_2) = H \mu(V_1 f_1, V_2 f_2).$$

*It is natural under finite restriction and bi-equivariant under common terminal total and GKT transports.*

*Proof.* With  $A = Q_1^{-1}Q_2$ ,  $Q_2(Q_2^{-1}Q_1)^{1/2} = Q_1A(A^{-1})^{1/2} = Q_1A^{1/2}$ , proving symmetry. The first identity in (93) is immediate; the second uses that  $H$  is an algebra automorphism. Conjugation-equivariance of rational powers gives the claimed bi-equivariance.  $\square$

For positive degrees  $r, s$ , set

$$(94) \quad M_{r,s}(P, Q) = P(P^{-1}Q)^{s/(r+s)}.$$

For a rooted binary product tree, recursively define a residual interpolant, with weights equal to the sums of the leaf degrees.

**Proposition 5.7** (Degree-weighted coherence). *If  $H_\tau$  is the interpolant associated with a product tree  $\tau$ , its corrected product is*

$$P_\tau = H_\tau \prod_i V_i f_i.$$

*For two trees  $\tau, \sigma$ , the automorphism  $\mathcal{A}_{\tau \leftarrow \sigma} = H_\tau H_\sigma^{-1}$  carries one product to the other and satisfies*

$$\mathcal{A}_{\tau \leftarrow \sigma} \mathcal{A}_{\sigma \leftarrow \rho} = \mathcal{A}_{\tau \leftarrow \rho}.$$

*All associahedral coherence identities therefore hold. For three equal-degree residuals  $e^x, e^y, e^z$  in the free labelled square-zero group,*

$$(95) \quad \log \mathcal{A}_{L \leftarrow R} = \frac{1}{72} [[x, z], y].$$

*The comparison is coherent but not generally strict monoidal.*

*Proof.* At an internal vertex, the corrections carry both child products to the same interpolant  $H_\tau$ . Since this interpolant is an algebra automorphism, induction gives the displayed form of  $P_\tau$ . Ratios of interpolants carry one tree product to another, and cancellation proves the cocycle identity. The weighted interpolation is symmetric under exchange of the two weighted children and is equivariant under the left–right bitorsor action. Baker–Campbell–Hausdorff expansion for three equal weights gives the common linear term  $(x + y + z)/3$ , no quadratic difference, and cubic difference

$$\frac{1}{72}(xzy - zxy - yxz + yzx) = \frac{1}{72} [[x, z], y],$$

which is (95). By comparison, unweighted iteration assigns leaf weights  $(1/4, 1/4, 1/2)$  and  $(1/2, 1/4, 1/4)$  in the two parenthesizations already in the abelian quotient.  $\square$

Changes of framed input paths act on the individual legs as well as on the output interpolant. The precise naturality statement is the following.

Let  $\Gamma = (\gamma_i)$  and  $\Gamma' = (\gamma'_i)$  be two tuples of framed input paths with the same endpoints. For a fixed product tree, write  $V_i^\Gamma, V_i^{\Gamma'}$  for the corresponding GKT transports and  $H_\Gamma, H_{\Gamma'}$  for their residual interpolants. Put

$$(96) \quad W_i^{\Gamma' \leftarrow \Gamma} = V_i^{\Gamma'} (V_i^\Gamma)^{-1}, \quad B^{\Gamma' \leftarrow \Gamma} = H_{\Gamma'} H_\Gamma^{-1}.$$

**Lemma 5.8** (Change of framed paths). *For  $a_i = V_i^\Gamma f_i$  one has the commutative comparison square*

$$(97) \quad H_{\Gamma'} \prod_i V_i^{\Gamma'} f_i = B^{\Gamma' \leftarrow \Gamma} H_\Gamma \prod_i (W_i^{\Gamma' \leftarrow \Gamma} a_i).$$

*The leg maps and output maps compose under successive changes of framed paths, and the resulting squares commute with the tree-transition cocycle of Proposition 5.7. Thus path and tree changes form a coherent factorization groupoid.*

*There is no output-only descent for arbitrary independent changes of the input paths. In particular, if one leg changes by a nontrivial unital automorphism  $W$  and another leg is fixed, no output automorphism  $Z$  can satisfy  $Z(ab) = W(a)b$  for all  $a, b$ .*

*Proof.* The identities  $V_i^{\Gamma'} f_i = W_i^{\Gamma' \leftarrow \Gamma} a_i$  and  $B^{\Gamma' \leftarrow \Gamma} H_\Gamma = H_{\Gamma'}$  give (97). For a third tuple  $\Gamma''$ , cancellation gives

$$W_i^{\Gamma'' \leftarrow \Gamma'} W_i^{\Gamma' \leftarrow \Gamma} = W_i^{\Gamma'' \leftarrow \Gamma}, \quad B^{\Gamma'' \leftarrow \Gamma'} B^{\Gamma' \leftarrow \Gamma} = B^{\Gamma'' \leftarrow \Gamma}.$$

Tree transitions are likewise ratios of the corresponding centers, so the mixed squares commute by the same cancellation. Finally, if  $Z(ab) = W(a)b$ , then  $b = 1$  gives  $Z = W$ , whereas  $a = 1$  gives  $Z = \text{id}$ . Hence an output-only comparison exists only when the leg change is trivial.  $\square$

The result of this section is therefore a coherent factorization comparison. The total factors remain the walls of the consistent diagram. The GKT endpoint factors are marked inside them, and the residual automorphisms are retained as comparison data. No equality between these wall coefficients and commutative theta structure constants has been asserted.

## 6. THE PRIMARY FROBENIUS COMPARISON

The theta multiplication of Theorem 3.9 is defined before enumerative normalization. The factorization of Theorem 5.4 identifies the enumerative subfactors in a noncommutative wall group. Neither construction alone determines a Frobenius structure. The comparison also includes the primitive form, oscillatory periods, residue pairing, and Kodaira–Spencer basis.

Work first on the ordered-residue cover, retaining labels and the two coordinates. General annular transport carries the primitive form in  $\mathcal{L}_\kappa$ , while every enumerative or residual GKT factor is a volume-preserving right equivalence. Both act on Brieskorn complexes. We construct the product and period-normalized basis in this labelled setting, restrict to the narrow primary object, and only then apply the symmetric idempotent of (57).

Suppress the descendent index in the primary variables and put

$$(98) \quad A_{\text{prim}} = \mathbb{Q}[[t_{a,b} : 0 \leq a, b \leq 2]], \quad \mathfrak{m}_{\text{prim}} = (t_{a,b}).$$

Statements over this complete ring mean the compatible systems over finite quotients closed under the dependencies of Appendix B.

**6.1. The classical vertex and coefficient layers.** At the Fermat point the primary state space is

$$(99) \quad H_{4,4} = \mathbb{Q}[X, Y]/(X^3, Y^3), \quad e_{a,b} = [X^a Y^b], \quad 0 \leq a, b \leq 2.$$

Normalize the Grothendieck residue trace by

$$(100) \quad \epsilon_0(f) = 16 \text{Res}_0 \frac{f dX \wedge dY}{(4X^3)(4Y^3)}.$$

**Proposition 6.1** (Classical primary vertex). *The pairing  $\eta_0(f, g) = \epsilon_0(fg)$  is nondegenerate and*

$$(101) \quad \eta_0(e_{a,b}, e_{c,d}) = \delta_{a+c, 2} \delta_{b+d, 2}.$$

*Thus the ordinary GKT node involution is the socle channel  $e_{a,b} e_{2-a, 2-b} = e_{2,2}$  with coefficient one. In a fixed ordered-residue chart, its leading monomials give the no-bend theta vertex.*

*Proof.* The residue in (100) extracts the coefficient of  $X^2 Y^2$ . This proves (101), nondegeneracy, and the socle identity. Additivity of  $F$  carries the identity to leading theta monomials.  $\square$

Five coefficient layers occur in the comparison. A Jacobi wall coefficient belongs to the logarithm of a crossing automorphism. A raw terminal coefficient is obtained by summing bend contributions in an ambient monomial product. A theta structure constant appears only after the re-expansion (59). A potential coefficient depends on a right-equivalence and primitive-form gauge. A Frobenius structure constant is obtained only after Jacobian reduction in the period-normalized Kodaira–Spencer basis. Numerically equal rationals in different layers are not identified without an intertwining theorem.

There is already a unipotent ambiguity in the first mixed theta window. If the deck involution exchanges  $\vartheta_\alpha$  and  $\vartheta_\beta$  and fixes  $\vartheta_\rho$ , then

$$(102) \quad \vartheta_\rho^{(\lambda)} = \vartheta_\rho + \lambda t_\alpha t_\beta \vartheta_0$$

preserves the leading term, grading, deck equivariance, and framed transport. Unary scattering therefore does not select the invariant product basis. The commutator lies in the Prym summand, while  $\vartheta_\alpha \vartheta_\beta$  lies in the symmetric summand. Periods remove the ambiguity.

**6.2. Marginal calibration.** Set  $t = t_{2,2}$ . The marginal quartics are

$$(103) \quad W_{A,B,C} = AX^4 + BX^2Y^2 + CY^4, \quad \Omega = dX \wedge dY.$$

The volume-preserving coordinate change  $(X, Y) \mapsto (qX, q^{-1}Y)$  acts by

$$(104) \quad (A, B, C) \mapsto (q^4A, B, q^{-4}C).$$

This is the quartic GHS zero-cochain action; its invariants are  $B$  and  $AC$  [GHS22]. Each orbit with  $AC$  a unit has a unique reflection-fixed representative  $A = C = \sqrt{AC}$  after choosing the branch at the Fermat point.

Let  $\Xi_{0,0}$  and  $\Xi_{2,2}$  be the two relevant product good-basis cycles. For  $\alpha = A - 1$  and  $\beta = B$ , define

$$(105) \quad F_0(\alpha, \beta) = \sum_{i,j,\ell \geq 0} \frac{(-1)^{i+j} \left(\frac{1}{4}\right)_{i+\ell} \left(\frac{1}{4}\right)_{j+\ell}}{i!j!(2\ell)!} \alpha^{i+j} \beta^{2\ell},$$

$$(106) \quad F_2(\alpha, \beta) = \sum_{i,j,\ell \geq 0} \frac{(-1)^{i+j} \left(\frac{3}{4}\right)_{i+\ell} \left(\frac{3}{4}\right)_{j+\ell}}{i!j!(2\ell+1)!} \alpha^{i+j} \beta^{2\ell+1}.$$

**Theorem 6.2** (Marginal period calibration). *There is a unique pair  $A(t), B(t) \in \mathbb{Q}[[t]]$ , with  $A(0) = 1$  and  $B(t) = t + O(t^2)$ , such that*

$$(107) \quad F_0(A(t) - 1, B(t)) = 1, \quad F_2(A(t) - 1, B(t)) = t.$$

*It is the reflection-fixed GKT/Saito-flat marginal potential*

$$W_t^{\text{flat}} = A(t)(X^4 + Y^4) + B(t)X^2Y^2,$$

where

$$(108) \quad A(t) = 1 + \frac{t^2}{16} + \frac{t^4}{768} + \frac{t^6}{61440} + O(t^8),$$

$$(109) \quad B(t) = t + \frac{t^5}{1280} + O(t^9).$$

*The two quadratic endpoint invariants satisfy*

$$(110) \quad \nu_{(4,0)} = \nu_{(0,4)} = -\frac{1}{8}.$$

The cubic marginal invariant vanishes, while the first product-visible correction is

$$(111) \quad X^3 = \left(-\frac{t}{2} + \frac{t^3}{32}\right) XY^2, \quad Y^3 = \left(-\frac{t}{2} + \frac{t^3}{32}\right) X^2Y \pmod{t^4}.$$

*Proof.* The quartic good-basis moments are

$$P_0(4k) = (-\hbar)^k \left(\frac{1}{4}\right)_k, \quad P_2(4k+2) = (-\hbar)^k \left(\frac{3}{4}\right)_k,$$

and vanish in the complementary congruence classes. Expanding

$$\exp\left(\frac{\alpha(X^4 + Y^4) + \beta X^2Y^2}{\hbar}\right)$$

against  $\Xi_{0,0}$  and  $\Xi_{2,2}$  gives precisely (105) and (106): an even number of central insertions contributes to the first period and an odd number to the second. The primitive-form normalization is (107).

At the origin the Jacobian of  $(F_0, F_2)$  with respect to  $(\alpha, \beta)$  is  $\text{diag}(-1/2, 1)$ . The formal inverse function theorem gives existence and uniqueness; coefficient extraction gives (108)–(109).

At order  $t^2$ ,

$$F_0 = 1 - \frac{1}{2}(A - 1) + \frac{B^2}{32} + O(t^3),$$

so  $A - 1 = t^2/16 + O(t^3)$ . The sign and order-two automorphism factor in the GKT potential multiply either reflection-related endpoint invariant by  $-1/2$ . Hence (110). The coefficient of  $t^3$  in the second period equation gives the cubic vanishing. Finally the Jacobian relation is  $X^3 = -B(2A)^{-1}XY^2$ , and expansion of  $-B/(2A)$  gives (111).  $\square$

The coefficient  $1/16$  in  $A(t)$  is a potential coefficient. The coefficient  $1/32$  in (111) is a Jacobian product coefficient. The latter is path independent when read on transported source sections, because framed transport is an algebra isomorphism and carries the Jacobian ideal by the chain rule.

The residue pairing detects the flat gauge already at quadratic order. For  $W_c = (1 + ct^2)(X^4 + Y^4) + tX^2Y^2$ ,

$$(112) \quad 16 \text{Res}_0 \frac{X^2Y^2 dX \wedge dY}{\partial_X W_c \partial_Y W_c} = \frac{1}{(1 + ct^2)^2 - t^2/4}.$$

Thus the primitive form distinguishes  $c = 1/16$  before ordinary multiplication does.

**6.3. Brieskorn transport.** Let  $R$  be a finite labelled coefficient quotient and  $\mathcal{P}$  the polynomial or finite root-valued theta-coordinate algebra in a path chart. For  $W \in \mathcal{P}$  define

$$(113) \quad \text{GM}_h(W) = H^2\left(\Omega_{\mathcal{P}/R}^\bullet[[\hbar]], \hbar d + dW \wedge\right).$$

This is a scalar complex in a chosen path chart. The complexes in different charts are glued by coordinate pullback. The distinguished primitive-form classes glue at the same time with the transition of  $\mathcal{L}_\kappa$ ; we do not identify the scalar de Rham complexes by tensoring their differentials with a nontrivial line bundle. The quartic annular path group is generated infinitesimally by

$$(114) \quad \mathcal{X}_{a,b} = X^a Y^b ((b+1)X \partial_X - (a+1)Y \partial_Y).$$

Its divergence with respect to  $dX \wedge dY$  is zero.

**Proposition 6.3** (Framed Brieskorn transport). *Let  $P_\Gamma$  be the total coordinate transport along a framed annular path represented at a finite stage. Pullback gives an isomorphism*

$$(115) \quad \mathrm{GM}_\hbar(W) \xrightarrow{\sim} \mathrm{GM}_\hbar(P_\Gamma W),$$

*carrying the primitive form to  $dP_\Gamma(X) \wedge dP_\Gamma(Y)$ , with the line multiplier across slabs. These maps compose strictly and commute with coefficient and root restriction, residual corrections, tree associators, and changes of framed paths. Modulo  $\hbar$  they preserve the Jacobian algebra and normalized residue pairing; dually they preserve oscillatory periods after inverse transport of cycles.*

*Proof.* For every coordinate automorphism  $\phi$  and differential form  $\xi$ ,

$$(\hbar d + d(\phi^* W) \wedge) \phi^* \xi = \phi^* ((\hbar d + dW \wedge) \xi).$$

Thus pullback is a chain isomorphism. The flows of (114) preserve  $dX \wedge dY$ . Across a slab, Equation (41) supplies the inverse conformal multiplier and preserves the line-valued primitive form. Composition and finite-stage compatibility follow from functoriality. Reduction modulo  $\hbar$  gives the Koszul complex of  $dW$ , and change of variables with the primitive form preserves its Grothendieck residue.  $\square$

If a total wall factor is written as a GKT factor followed by its residual factor, the latter therefore acts as a volume-preserving right equivalence of potential-with-form pairs. It is retained as a coordinate change, not replaced by a scalar. The same observation turns the residual associativity and framed-path cocycles into ordinary composition identities of Brieskorn chain maps.

**6.4. The simultaneous primary unfolding.** Let  $W_{\mathrm{prim}}^{\mathrm{GKT}} \in A_{\mathrm{prim}}[[X, Y]]$  be the coordinate-symmetric primary GKT potential after all descendent variables are set to zero. It comes with the nine good-basis cycles  $\Xi_{a,b}$  and the primitive form  $dX \wedge dY$ . The GKT mirror theorem states simultaneously that the nine parameters  $t_{a,b}$  are Saito-flat coordinates for this unfolding [GKT22, Thm. 0.3 and Cor. 4.17]; see also [LLSS17, HLSW22].

**Theorem 6.4** (Transport of the GKT primary Saito structure). *Transport  $W_{\mathrm{prim}}^{\mathrm{GKT}}$ , its primitive form, and its nine good-basis cycles along any finite annular path frame. The transported primitive-form class remains primitive, and all nine  $t_{a,b}$  remain simultaneous Saito-flat coordinates. These transports compose under path concatenation and commute with all finite coefficient, root, residual, tree, and framed-path maps. After narrow restriction, the degree-two symmetric pushforward gives a framed narrow Saito local system.*

*Proof.* Proposition 6.3 preserves every oscillatory period after inverse transport of the cycles. Hence the primitive-form and flat-coordinate identities in the source frame hold in every transported frame, simultaneously in all nine variables. Strict composition gives the cocycle statements. The primary narrow variables are stable under the degree-two deck action; exactness of the Reynolds idempotent in characteristic zero then gives the symmetric local system.  $\square$

The first two-dimensional slice already tests simultaneity. Set  $u = t_{1,1}$  and  $t = t_{2,2}$ . Through total degree three,

$$(116) \quad W_{u,t} = \left(1 + \frac{t^2}{16}\right) (X^4 + Y^4) + tX^2Y^2 + \left(u - \frac{ut^2}{16}\right) XY - \frac{u^2t}{32} + O((u,t)^4).$$

The two new labelled block invariants are

$$\nu_{ut^2;(1,1)} = -\frac{1}{8}, \quad \nu_{u^2t;(0,0)} = -\frac{1}{16}.$$

The repeated-marking automorphisms divide these by two, giving the last two corrections in (116). For the  $\Xi_{1,1}$  period, the direct correction, its quadratic cross term, and the cubic singleton term contribute

$$-\frac{ut^2}{16\hbar}, \quad -\frac{ut^2}{16\hbar}, \quad \frac{ut^2}{8\hbar};$$

for  $\Xi_{0,0}$  the two contributions are  $-u^2t/(32\hbar)$  and  $u^2t/(32\hbar)$ . Both sums vanish. Thus the constant term  $-u^2t/32$  is fixed by the oscillatory integral although it is invisible to the critical locus.

### 6.5. Kodaira–Spencer classes in theta coordinates. Put

$$(117) \quad \mathcal{P}_0 = A_{\text{prim}}[[X, Y]], \quad \mathcal{R}_0 = \mathcal{P}_0/(\partial_X W_{\text{prim}}^{\text{GKT}}, \partial_Y W_{\text{prim}}^{\text{GKT}}).$$

For a framed path  $\Gamma$ , write

$$U_\Gamma = P_\Gamma(X), \quad V_\Gamma = P_\Gamma(Y), \quad W_\Gamma = P_\Gamma(W_{\text{prim}}^{\text{GKT}})$$

and define the Jacobian quotient in the transported coordinate algebra

$$(118) \quad \mathcal{R}_\Gamma = P_\Gamma(\mathcal{P}_0)/(\partial_{U_\Gamma} W_\Gamma, \partial_{V_\Gamma} W_\Gamma).$$

The transported period-normalized Kodaira–Spencer classes are

$$(119) \quad \Phi_{a,b}^\Gamma = P_\Gamma\left(\left[\frac{\partial W_{\text{prim}}^{\text{GKT}}}{\partial t_{a,b}}\right]\right) \in \mathcal{R}_\Gamma, \quad 0 \leq a, b \leq 2.$$

Here “in theta coordinates” means that the classes are expressed in the framed theta-coordinate algebra; they need not be individual canonical theta functions.

**Theorem 6.5** (Transported narrow-primary Frobenius structure). *The nine sections (119) form an  $A_{\text{prim}}$ -basis of  $\mathcal{R}_\Gamma$  and satisfy*

$$(120) \quad \Phi_{a,b}^\Gamma = P_\Gamma(\Phi_{a,b}^0), \quad \Phi_{a,b}^0 = X^a Y^b + O(\mathfrak{m}_{\text{prim}}).$$

*Their multiplication, unit  $\Phi_{0,0} = [1]$ , and normalized residue pairing have coefficient matrices independent of  $\Gamma$ . They are the complete primary Saito–Givental Frobenius structure transported from  $W_{\text{prim}}^{\text{GKT}}$ . By the GKT primary mirror theorem, the source tensor is the narrow-primary closed FJRW Frobenius structure.*

*Constructing the product equivariantly on the labelled cover, restricting to the narrow primary subobject, and then applying  $e_+$  gives an isomorphism of framed narrow-primary Frobenius algebras. This construction is compatible with every finite coefficient and Kummer presentation.*

*Proof.* The chain rule gives

$$P_\Gamma(\partial_X W_{\text{prim}}^{\text{GKT}}) = \partial_{U_\Gamma} W_\Gamma, \quad P_\Gamma(\partial_Y W_{\text{prim}}^{\text{GKT}}) = \partial_{V_\Gamma} W_\Gamma.$$

Hence  $P_\Gamma$  identifies  $\mathcal{R}_0$  with  $\mathcal{R}_\Gamma$ . Although  $P_\Gamma$  fixes the coefficient ring, its coordinate expressions may depend on the parameters. For  $\partial_i = \partial/\partial t_{a,b}$ , put

$$\xi_i = (\partial_i \circ P_\Gamma - P_\Gamma \circ \partial_i) P_\Gamma^{-1}.$$

Thus  $\xi_i$  is the coefficient-linear infinitesimal variation of the coordinate map. It is a derivation of the transported coordinate algebra, and

$$\partial_i W_\Gamma = P_\Gamma(\partial_i W_{\text{prim}}^{\text{GKT}}) + \xi_i(W_\Gamma).$$

Writing  $\xi_i = A_i \partial_{U_\Gamma} + B_i \partial_{V_\Gamma}$  shows that the second term lies in the Jacobian ideal. Consequently

$$[\partial_i W_\Gamma] = P_\Gamma([\partial_i W_{\text{prim}}^{\text{GKT}}]) \quad \text{in } \mathcal{R}_\Gamma.$$

Equivalently, (119) uses the connection obtained by transporting the trivial coefficient connection from the source frame. This proves naturality even for parameter-dependent path maps.

The primary GKT potential is a formal miniversal unfolding, so its critical algebra is finite free of rank nine. At the Fermat point the derivatives in (119) are the nine monomials in (99); their change-of-basis determinant is therefore a unit. This proves the basis assertion.

Since  $t_{0,0}$  is the constant deformation in the source potential,  $\Phi_{0,0}^0 = [1]$  and hence  $\Phi_{0,0}^\Gamma = [1]$  in every frame.

The normalized residue pairing is nondegenerate at the Fermat point by Proposition 6.1 and remains so over the complete local base. Its trace identity

$$\eta(fg, h) = \epsilon(fgh) = \eta(f, gh)$$

makes the transported critical algebra Frobenius. Theorem 6.4 identifies the  $t_{a,b}$  as simultaneous flat coordinates, so the Kodaira–Spencer product is the Saito Frobenius product. The GKT primary mirror theorem [GKT22, Thm. 0.3] identifies the source tensor with the narrow-primary closed FJRW tensor. Path independence follows by applying the algebra isomorphism  $P_\Gamma$  to the source product and comparing coefficients in the transported basis.

The deck involution preserves the potential and primitive form. Its symmetric and Prym summands are therefore orthogonal for the residue metric; nondegeneracy of the full metric implies nondegeneracy on each summand. Deck equivariance makes the symmetric summand a unital subalgebra. The idempotent  $e_+$  commutes with path transport and all finite restrictions, completing the descent statement.  $\square$

The theorem compares the complete primary structure. It includes all nine deformation directions, flat coordinates, multiplication, unit, and residue metric. It does not identify an unquotiented pair-of-broken-lines coefficient with an individual open descendent invariant. On the primary slice there are no positive-weight critical walls: the source theta product is ordinary multiplication, and nontrivial coefficients such as  $1/32$  appear only after reduction of an out-of-range monomial in the period-normalized Jacobian quotient.

Thus the annular geometry changes the presentation of the Frobenius algebra but not its tensor. Hyperbolic holonomy remains, so the algebras form a path-equivariant local system rather than an unframed global theta basis.

## 7. THE DESCENDENT BROKEN-LINE CORRESPONDENCE

We now construct the class in Theorem 1.7 and the cellular model of Definition 1.6. Fix a finite labelled window  $S = (I, P, \mathcal{E})$  as in Definition 1.5, a deformation order  $N$ , the finite root stage, and a framed compact annular corridor defined there. All coefficients are taken in the finite Artin ring  $R_{S,N}$  of (8). The closure conditions on  $\mathcal{E}$  retain every proper support, every open boundary graph and bundle map in (4), and every closed-component splitting, including those whose Euler rank does not equal the dimension of its compact moduli factor. All spaces below are therefore finite-dimensional and compact. The virtual chains are rational PL branched zero loci of finite orbibundle sections; no analytic infinite-window virtual class is used.

The construction has four parts. First, a relative twisted de Rham complex for the Fermat polynomial supplies a residue cocycle compatible with the two coordinate divisors. Second, a deformation retract of this complex gives coherent primitives for the factorization faces of tropical forests. Third, the Witten-bundle splitting realizes insertion of a cotangent class by a parameterized Euler chain. Finally, these data are assembled on a compactified incidence space of tropical forests and  $W$ -spin curves. The boundary stratification of that compactification gives the two cotangent recursions by Stokes' formula.

The chain-level argument has the following conceptual order:

$$\begin{array}{ccccc} \text{relative residue complex} & \longrightarrow & \text{forest primitives} & \longrightarrow & \text{incidence zero chain} \\ & & & & \downarrow \\ \text{GKT moments} & \longleftarrow & \text{cotangent recursions} & \longleftarrow & \text{boundary strata} \end{array}$$

Here a broken-line chain is a cellular chain whose strata parametrize decorated tropical forests together with their  $W$ -spin data; it is not a single Gross–Siebert broken line.

**7.1. Relative contact Stokes.** Put  $W_0 = X^4 + Y^4$ ,  $\Omega = dX \wedge dY$ , and  $D_0 = \hbar d + dW_0 \wedge$ . Let  $D_x = \{X = 0\}$ ,  $D_y = \{Y = 0\}$ , and  $D_{xy} = D_x \cap D_y$ . On the two divisors and their intersection put

$$D_x = \hbar d + d(Y^4) \wedge, \quad D_y = \hbar d + d(X^4) \wedge, \quad D_{xy} = 0.$$

Let  $\mathcal{B}_A, \mathcal{B}_x, \mathcal{B}_y$ , and  $\mathcal{B}_{xy} = \mathbb{Q}[\hbar]$  be the corresponding twisted polynomial de Rham complexes, and let  $r_x, r_y, r_{xy}^x, r_{xy}^y$  be their ordinary restriction maps. Define

$$(121) \quad \mathcal{B}_{\text{con}}^k = \mathcal{B}_A^k \oplus \mathcal{B}_x^{k-1} \oplus \mathcal{B}_y^{k-1} \oplus \mathcal{B}_{xy}^{k-2}$$

with differential

$$(122) \quad \begin{aligned} \delta(\omega, \eta_x, \eta_y, \xi) = & (D_0\omega, r_x\omega - D_x\eta_x, r_y\omega - D_y\eta_y, \\ & r_{xy}^x\eta_x - r_{xy}^y\eta_y + D_{xy}\xi). \end{aligned}$$

The restriction square commutes, so  $\delta^2 = 0$ .

For  $0 \leq r \leq 3$ , set

$$(123) \quad G_r(p) = \prod_{j=0}^{p-1} \frac{r+1+4j}{4}, \quad G_r(0) = 1.$$

The bulk, divisor, and corner traces are

$$(124) \quad \begin{aligned} \tau_A(X^{r+4p}Y^{s+4q}\Omega) &= (-\hbar)^{p+q}G_r(p)G_s(q), \\ \tau_x(Y^{s+4q}dY) &= 4(-\hbar)^qG_s(q), \\ \tau_y(X^{r+4p}dX) &= -4(-\hbar)^pG_r(p), \quad \tau_{xy}(c) = 16c. \end{aligned}$$

The factors 4 and 16 are the inverse traces of  $B\mu_4$  and  $B(\mu_4^2)$ . Set

$$(125) \quad \text{Tr}_{\text{con}}(\omega, \eta_x, \eta_y, \xi) = \tau_A(\omega) - \tau_x(\eta_x) - \tau_y(\eta_y) - \tau_{xy}(\xi).$$

Let  $\tilde{R}_{S,N}$  be the finite  $R_{S,N}$ -algebra obtained by adjoining the roots occurring at the chosen root stage. Fix a chamber  $\Gamma_0$  in which the developed coordinates are  $(X, Y)$  and the reference polynomial is  $W_0 = X^4 + Y^4$ ; this is the *reference Fermat chamber*. For any chamber frame  $\Gamma$ , let  $\mathsf{T}_{\Gamma\Gamma_0}$  be the ordered composition of the wall and root-chart maps along its framed path, and define

$$(126) \quad \mathcal{B}_{\Gamma, \text{con}} := \mathsf{T}_{\Gamma\Gamma_0}(\mathcal{B}_{\text{con}} \otimes_{\mathbb{Q}} \tilde{R}_{S,N}), \quad \delta_{\Gamma} := \mathsf{T}_{\Gamma\Gamma_0} \delta \mathsf{T}_{\Gamma\Gamma_0}^{-1}, \quad \text{Tr}_{\Gamma} := \text{Tr}_{\text{con}} \mathsf{T}_{\Gamma\Gamma_0}^{-1}.$$

Thus Laurent and root-valued expressions live in the transported complex, not in the original polynomial presentation. If  $\Gamma'$  is another frame, then  $\mathsf{T}_{\Gamma'\Gamma} := \mathsf{T}_{\Gamma'\Gamma_0} \mathsf{T}_{\Gamma\Gamma_0}^{-1}$  is a chain isomorphism and

$$(127) \quad \text{Tr}_{\Gamma'} \mathsf{T}_{\Gamma'\Gamma} = \text{Tr}_{\Gamma}.$$

The transported coordinate divisors are part of  $\mathcal{B}_{\Gamma, \text{con}}$ ; no wall map is required to preserve the original pair  $\{X = 0\} \cup \{Y = 0\}$ .

**Proposition 7.1** (Relative contact Stokes formula). *One has  $\text{Tr}_{\text{con}} \delta = 0$ . Moreover, (121) has a coordinate-equivariant deformation retract onto*

$$(128) \quad \bigoplus_{0 \leq r, s \leq 3} \mathbb{Q}[\hbar] X^r Y^s \Omega.$$

*Proof.* The recurrence  $G_r(p+1) = \frac{r+1+4p}{4} G_r(p)$  gives

$$(129) \quad \tau_A(D_0 H) = \tau_x(r_x H) + \tau_y(r_y H), \quad \tau_x(D_x f) = 16f(0), \quad \tau_y(D_y g) = -16g(0).$$

Substitution in (122) proves the Stokes identity. In one variable, let  $R_4$  send  $t^{r+4p} dt$  to  $(-\hbar)^p G_r(p) t^r dt$ , and put  $R_{\text{rel}}(gdt, c\epsilon) = R_4(gdt) - 4ct^3 dt$ . The recursive primitive

$$(130) \quad s_4(t^a) = \frac{1}{4} t^{a-3} - \hbar \frac{a-3}{4} s_4(t^{a-4}) \quad (a \geq 4),$$

with  $s_4(t^a) = 0$  for  $a \leq 3$ , satisfies the contraction identity. The full relative homotopy sends  $(gdt, c\epsilon)$  to  $s_4(g) + c$ ; together with the displayed reduction it satisfies the degreewise contraction identity  $\delta h_{\text{rel}} + h_{\text{rel}} \delta = 1 - \iota R_{\text{rel}}$ . The oriented tensor product of the two contractions gives (128); averaging its two orders makes it coordinate equivariant.  $\square$

This relative complex is necessary. Although  $D_0(\frac{1}{4} dY) = X^3 \Omega$  in the plane complex, in (121) the complete boundary is

$$(131) \quad \delta(\tfrac{1}{4} dY, 0, 0, 0) = (X^3 \Omega, \tfrac{1}{4} dY, 0, 0).$$

The bulk and divisor traces both equal one and cancel. Thus residue-three contact data survive without assigning a nonzero trace to an ordinary exact form.

**7.2. Factorization chains for tropical forests.** For a finite nonempty set  $A$ , put  $V_A = \ker(\mathbb{Q}^A \xrightarrow{\Sigma} \mathbb{Q})$ . If  $F$  is a rooted forest with leaves  $I$ , write  $R(F)$  for its root clusters and  $\text{In}(v)$  for the child clusters entering an internal vertex. The block-sum exact sequences

$$(132) \quad 0 \longrightarrow \bigoplus_{B \in \text{In}(v)} V_B \longrightarrow V_{\coprod B} \longrightarrow V_{\text{In}(v)} \longrightarrow 0$$

give a determinant isomorphism

$$(133) \quad \mathfrak{o}_F = \det \left( V_{R(F)} \oplus \bigoplus_v V_{\text{In}(v)} \right) \xrightarrow{\Phi_F} \det V_I.$$

Children, vertices, and roots are ordered by least label. Cutting two edges in opposite orders changes  $\Phi_F$  by the exchange of the two outward metric normals, exactly the cubical boundary sign.

For a nonempty cluster  $C \subseteq I$ , use the twist representatives fixed in Definition 1.5 and write

$$(134) \quad \sum_{i \in C} a_i = r_C + 4\ell_C, \quad \sum_{i \in C} b_i = s_C + 4m_C, \quad 0 \leq r_C, s_C \leq 3.$$

Its balanced exponent at position  $p$  is

$$(135) \quad \mathbf{b}_{C,p} = (r_C + 4p, s_C + 4(N_C - p)), \quad 0 \leq p \leq N_C.$$

Every vertex  $v$  of a decorated forest records a cluster and either one of these profiles or the exponent of a specified wall Hamiltonian. Denote that explicit exponent by  $\mathbf{b}_v = (b_{v,x}, b_{v,y}) \in \mathbb{N}^2$ . Its scalar  $c_v \in \widetilde{R}_{S,N}$  is the displayed coefficient of that Hamiltonian or of the normalized fourth-root collar. A closed  $W$ -spin factor is not put into  $c_v$ : it remains a geometric Euler factor in the incidence space below. Thus, beyond the two specified singleton Hamiltonian normalizations,

no higher-support GKT open invariant, endpoint coefficient, or cotangent-recursion value enters a vertex scalar.

A *decorated tropical forest*  $h$  is a stable labelled forest together with these choices at all of its vertices and the ordered wall word on each edge. The coefficient associated with  $h$  is the single bulk form

$$(136) \quad \beta_h = c_h X^{A_h} Y^{B_h} \Omega, \quad c_h = \prod_v c_v, \quad (A_h, B_h) = \sum_v \mathbf{b}_v.$$

All wall, contact, and homotopy operators thereafter act linearly on this one element of  $\mathcal{B}_{\text{con}}$ ; forms from distinct vertices are never multiplied.

For  $\mathbf{b} = (r + 4p, s + 4q)$  with  $0 \leq r, s \leq 3$ , put

$$(137) \quad w(\mathbf{b}) = G_r(p) G_s(q).$$

Let  $\pi$  be a partition of a label set  $J$ , with  $a = |\pi|$ . Choose one profile (135) on each block and set  $\mathbf{b}_\pi = \sum_{C \in \pi} \mathbf{b}_C$ . The parent profile  $\mathbf{b}_J$  has the same two residues and the factorization degree one lower at each new forest edge; hence uniquely  $\mathbf{b}_\pi = \mathbf{b}_J + 4(c_x, c_y)$  with  $c_x + c_y = a - 1$ . Let  $c$  be the product of all vertex scalars outside this contracted corolla and of the corolla's displayed wall coefficients. Define

$$(138) \quad \begin{aligned} \lambda_{J,\pi} &= -\frac{w(\mathbf{b}_\pi)}{w(\mathbf{b}_J)}, \\ \beta_{J,\pi}^{\text{sp}} &= c X^{(\mathbf{b}_\pi)_x} Y^{(\mathbf{b}_\pi)_y} \Omega, \\ \beta_{J,\pi}^{\text{par}} &= c(-\hbar)^{a-1} \lambda_{J,\pi} X^{(\mathbf{b}_J)_x} Y^{(\mathbf{b}_J)_y} \Omega. \end{aligned}$$

If  $\mathbf{b}_J = (r + 4p, s + 4q)$ , the Gamma recurrence gives the explicit cancellation

$$(139) \quad \text{Tr}_{\text{con}}(\beta_{J,\pi}^{\text{sp}} + \beta_{J,\pi}^{\text{par}}, 0, 0, 0) = c(-\hbar)^{p+q+a-1} (w(\mathbf{b}_\pi) + \lambda_{J,\pi} w(\mathbf{b}_J)) = 0.$$

Let  $R_{\text{con}}$  and  $h_{\text{con}}$  denote the projection and homotopy in the tensor-product deformation retract of Proposition 7.1. The same calculation takes place before applying the trace: both monomials reduce to the same basis vector in (128), with coefficients in the parenthesis in (139). Consequently

$$(140) \quad R_{\text{con}}(\beta_{J,\pi}^{\text{sp}} + \beta_{J,\pi}^{\text{par}}, 0, 0, 0) = 0.$$

This includes residue-three sectors because the relative normal form retains all sixteen residues.

Let

$$(141) \quad G_{J,\pi}^{\text{con}} = h_{\text{con}}(\beta_{J,\pi}^{\text{sp}} + \beta_{J,\pi}^{\text{par}}), \quad \delta G_{J,\pi}^{\text{con}} = \beta_{J,\pi}^{\text{sp}} + \beta_{J,\pi}^{\text{par}}.$$

For the oriented interval  $e = [v_0, v_1]$ , define

$$(142) \quad \kappa_{J,\pi} = e \otimes \beta_{J,\pi}^{\text{sp}} + v_0 \otimes G_{J,\pi}^{\text{con}}.$$

For the tensor differential  $d(c \otimes \alpha) = \partial c \otimes \alpha + (-1)^{\dim c} c \otimes \delta \alpha$ , one has

$$(143) \quad d\kappa_{J,\pi} = v_1 \otimes \beta_{J,\pi}^{\text{sp}} + v_0 \otimes \beta_{J,\pi}^{\text{par}}.$$

Let  $Q_F = [0, 1]^{E_{\text{int}}(F)}$ . The standard cellular contraction of  $Q_F$  to its all-zero vertex, tensored with the two-variable contraction  $(R_{\text{con}}, \iota, h_{\text{con}})$  of Proposition 7.1, gives an operator  $H_F$  with

$$(144) \quad dH_F + H_F d = 1 - \iota_F \Pi_F.$$

We now give the coherent extension, rather than choosing a primitive on each square independently. Order the nonempty faces of  $Q_F$  first by dimension and then lexicographically by their sets of free coordinates. On vertices and edges use the coefficients (138) and the corolla chain

(142). Suppose chains have been assigned to all proper faces of a  $k$ -face  $G$  and agree on their intersections. Their oriented sum is a cycle  $Z_G$ . At the all-zero vertex, every refinement order has the same block-sum exponent and product of vertex scalars. On all other terms (140) gives zero normal-form projection. Thus

$$(145) \quad dZ_G = 0, \quad \Pi_F Z_G = 0, \quad d(H_F Z_G) = Z_G.$$

Assign  $H_F Z_G$  to  $G$ . This recursion extends the chain across every face and automatically uses the already fixed primitive on every proper face. Because the order,  $H_F$ , and the block-sum maps are fixed and equivariant, the construction commutes with relabelling, coordinate transposition, restriction to a subforest, and transport of the coefficient complex. In particular, a refinement cube receives one coherent extension, rather than unrelated primitives on its squares; all higher compatibility follows from the induction in (145).

Define  $C_*^{\text{fac}}(F)$  to be the smallest differential submodule of  $C_*^{\text{cell}}(Q_F) \otimes \mathcal{B}_{\text{con}} \otimes \mathfrak{o}_F$  containing every vertex coefficient, every corolla chain (142), and every recursively assigned chain  $H_F Z_G$ , and closed under restriction to faces. Identify a length-one face with its cut forest through (133). Its differential is

$$(146) \quad d_{\text{fac}}(c \otimes \alpha) = \partial_{\text{cell}} c \otimes \alpha + (-1)^{\dim c} c \otimes \delta \alpha.$$

For a target frame  $\Gamma$ , transport the assembled coefficient into  $\mathcal{B}_{\Gamma, \text{con}}$ . Define  $\mu_\Gamma$  to be zero off total degree zero and, on a zero-dimensional boundary cell, by

$$(147) \quad \mu_\Gamma(c \otimes \alpha) = \text{Tr}_\Gamma(\alpha).$$

Equations (139), (143), and (144) prove

$$(148) \quad \mu_\Gamma d_{\text{fac}} = 0.$$

**7.3. The cotangent-line zero cobordism.** Fix an exact open datum  $\delta \in \mathcal{E}$ , a distinguished label  $j_1 \in I(\delta)$ , and write

$$(149) \quad \mathcal{M}_\delta = \mathcal{P}\mathcal{M}_{\Gamma_\delta}, \quad E_\delta^- = \mathbb{E}_{\Gamma_\delta}(\mathbf{d}), \quad E_\delta^+ = \mathbb{E}_{\Gamma_\delta}(\mathbf{d} + \mathbf{e}_{j_1}) \simeq E_\delta^- \oplus \mathcal{L}_{j_1}.$$

The last isomorphism is the Witten-bundle splitting (5). Choose, simultaneously on the finite boundary diagram of  $\delta$ , compatible equivariant rational PL multisections  $\sigma_\delta^-$  of  $E_\delta^-$  and  $s_\delta^+$  of  $\mathcal{L}_{j_1}$ . They are chosen by induction from the deepest boundary strata, using relative PL extension on each finite proper-etale atlas. Thus their restrictions are the exterior sums prescribed by the bundle maps  $\phi_{\Lambda\Gamma}$ , and no numerical Euler count is used in their choice.

For every retained marking or boundary flag  $w$ , let  $t_{j_1, w}$  be the standard section of  $\mathcal{L}_{j_1}$  obtained by evaluating at  $z_{j_1}$  the normalized meromorphic differential with residues  $+1$  at  $w$  and  $-1$  at the conjugate of  $z_{j_1}$  [GKT22, Notation 6.1]. Relative PL extension gives a transverse homotopy

$$(150) \quad H_{\delta, w} : [0, 1] \times \mathcal{M}_\delta \longrightarrow E_\delta^- \oplus \mathcal{L}_{j_1}, \quad H_{\delta, w}(0) = \sigma_\delta^- \oplus t_{j_1, w}, \quad H_{\delta, w}(1) = \sigma_\delta^- \oplus s_\delta^+.$$

The construction is relative to every face where the sections are already fixed. On a boundary map that contracts the component carrying  $j_1$ , the tautological morphism need not be invertible; that collar is retained as a relative stop and is disjoint from the zero set by the rank and positivity condition included in the exact datum. On every other face, (150) pulls back under  $F_{\Lambda\Gamma}$  and  $\phi_{\Lambda\Gamma}$ .

Let  $\mathcal{Z}_{\delta, w}$  be the determinant-oriented rational PL zero chain of (150). Its ordinary parameterized Euler boundary is

$$(151) \quad \partial \mathcal{Z}_{\delta, w} = [Z(\sigma_\delta^- \oplus s_\delta^+)] - [Z(\sigma_\delta^- \oplus t_{j_1, w})] + [Z(H_{\delta, w}|_{[0, 1] \times \partial \mathcal{M}_\delta})].$$

The first term is the Euler chain for descendent degree  $\mathbf{d} + \mathbf{e}_{j_1}$ , by (149). The divisor of  $t_{j_1, w}$  is the sum of open–closed boundary divisors on which  $j_1$  lies on the closed component and, when  $w$  is an interior marking,  $w$  remains on the open component [GKT22, Lem. 6.4]. The local equation is  $uz^4$  in the quartic root coordinate, with  $u$  a unit. Intersecting with  $\sigma_\delta^-$  leaves exactly the rank-matched splittings  $N_A = -1$ ; the others remain as residual-degree faces with zero degree-zero pushforward. Thus the degree shift and the open–closed term in the recursion are both geometric terms of (151).

The signed coefficients introduced below do not weight branches of a multisection. They weight separate chains  $\mathcal{Z}_{\delta, w}$ . Because their sum is one, all copies of the high endpoint in (151) combine to the single canonical Euler chain  $Z(\sigma_\delta^- \oplus s_\delta^+)$ . Negative coefficients are therefore legitimate rational-chain coefficients and never PL-multisection weights.

#### 7.4. Compactified annular incidence spaces.

**7.4.1. Compactified incidence types.** The geometric source of the correspondence is a compactification of a fiber product. Its open stratum parametrizes a stable labelled tropical forest in the annular corridor, together with an open quartic  $W$ -spin curve at each root and a closed quartic  $W$ -spin curve at each detached component. The evaluation positions satisfy the affine edge and wall-incidence equations, while the Witten sections vanish on the corresponding moduli factors. The coefficient of a forest is the relative twisted de Rham class constructed above.

The compactification records each geometric degeneration rather than identifying it numerically. Zero and infinite forest lengths give contractions and cuts; wall crossings acquire their Cartan homotopies; colliding directions are resolved by Fulton–MacPherson spaces; the universal boundary circle records the pointing direction; and fourth-root collars record the normal direction to an open–closed splitting. The usual compact open and closed  $W$ -spin spaces supply their own boundary strata. Auxiliary length caps are placed beyond the incidence locus. These constructions give the following local model.

**Definition 7.2** (Stable resolved type and incidence section). *A stable resolved labelled type  $\tau$  consists of the following finite data.*

- (1) It has a rooted forest with leaf set contained in  $I$ . Every internal vertex is either at least trivalent or carries one of the following specified geometric events: crossing a retained wall; identifying two affine-chart copies on an overlap; contracting or cutting an internal forest edge; colliding two tangent directions; moving the pointing flag on the compactified boundary circle; or splitting off a closed  $W$ -spin component. In particular, unmarked bivalent wall-free vertices are suppressed. Each open root component names an exact datum  $\delta$  of (4), including its boundary maps and Witten bundles.
- (2) Every oriented segment  $e$  has a fixed integral slope  $v_e$ , a nonnegative length  $\lambda_e$ , and an ordered word in the finitely many walls retained modulo  $\mathfrak{m}_S^N$ . Each nonidentity bend has positive deformation order, and the sum of the orders on a path is less than  $N$ .
- (3) Every detached component names its compact quartic  $W$ -spin base  $\mathcal{PM}_\delta$  and Euler bundle  $\mathbb{E}_\delta$  from  $\mathcal{E}$ . Every coincident set of directions names the corresponding Fulton–MacPherson factor. At a vertex carrying  $j_1$ , the oriented real blowup of its tangent-direction configuration is the compactified universal boundary circle. Its named vertices are the root flag, retained markings, the two Fermat phase flags, wall directions, collision directions, and closed-component directions. A type chooses one closed cell  $M_\tau$  of this circle and one cotangent-homotopy interval  $[0, 1]_{\psi, \delta, w}$  from (150); without a distinguished label these factors are points. Adjacent cells are identified at their common endpoints in the face category below.

- (4) The type contains a nondegenerate composable chain of  $q$  boundary or overlap maps from (4), and an ordered word of  $b$  retained wall maps. Boundary arrows strictly lower the graph-complexity order of Appendix B. For overlaps, fix a finite ordered corridor cover and retain only the strictly increasing Čech chains of distinct chart indices. The inverse transition is the coefficient map on the same overlap, not a second nerve arrow; identities and repeated indices are omitted. Mixed chains use the product normalization: a boundary arrow retains the ordered overlap multi-index, and an overlap arrow retains the graph type. The corresponding simplices  $\Delta_{\text{nerve}}^q$  and  $\Delta_{\text{bar}}^b$  record composition of the boundary maps and the ordered Cartan homotopy, respectively. Every affine overlap constraint records as part of  $\tau$  the restriction  $(A_j, c_j)$  of the fixed integral-affine transition  $x^+ = A_j x^- + c_j$  between its two corridor cells.

Let  $\mathcal{V}_\tau$  be the position vertices,  $\mathcal{S}_\tau$  the segments,  $\mathcal{H}_\tau$  the affine-overlap parameters,  $\mathcal{E}_\tau$  the internal forest edges,  $\mathcal{W}_\tau$  the prescribed wall incidences,  $\mathcal{M}_\tau$  the overlap and common-vertex matching constraints, and  $\mathcal{D}_\tau$  the closed-component smoothing roots. Let  $\mathcal{E}_\tau^{\text{op}}$  and  $\mathcal{E}_\tau^{\text{cl}}$  be the named open and closed moduli data.  $\mathcal{Q}_\tau$  is the set of chosen pairs  $(\delta, w)$  carrying a cotangent homotopy. A constraint  $j \in \mathcal{M}_\tau$  has dimension  $d_j \in \{1, 2\}$ . In a chamber  $\mathfrak{c}$ , put

$$(152) \quad A_{\tau, \mathfrak{c}} = \left\{ (x_v, \lambda_e) : \begin{array}{ll} x_v \in \mathbb{R}^2 & (v \in \mathcal{V}_\tau), \\ 0 \leq \lambda_e \leq L_e & (e \in \mathcal{S}_\tau), \\ \epsilon_{ch} \ell_h(x_{v(h)}) \geq 0 & (h \text{ a corridor side}) \end{array} \right\}.$$

The cap  $L_e$  lies beyond the incidence zero set. For the specified chains of lengths  $q$  and  $b$ , define

$$(153) \quad \begin{aligned} P_\tau &= A_{\tau, \mathfrak{c}} \times [0, 1]^{\mathcal{H}_\tau} \times [0, 1]^{\mathcal{E}_\tau} \\ &\times \Delta_{\text{nerve}}^q \times \Delta_{\text{bar}}^b \times \prod_C \text{FM}(C) \\ &\times M_\tau \times \prod_{(\delta, w) \in \mathcal{Q}_\tau} [0, 1]_{\psi, \delta, w} \times \prod_{D \in \mathcal{D}_\tau} ([\mathbb{D}_R / \mu_4] \times [0, 1]_D) \times \prod_{\delta \in \mathcal{E}_\tau^{\text{op}}} \mathcal{M}_\delta \times \prod_{\delta \in \mathcal{E}_\tau^{\text{cl}}} \mathcal{PM}_\delta. \end{aligned}$$

Each factor in (153) occurs once.

The oriented equation orbibundle  $E_\tau \rightarrow P_\tau$  and its section  $s_\tau$  are the following ordered direct sums:

| index                                     | bundle summand                        | component of $s_\tau$                 |
|-------------------------------------------|---------------------------------------|---------------------------------------|
| $e \in \mathcal{S}_\tau$                  | $\mathbb{R}^2$                        | $x_{h(e)} - x_{t(e)} - \lambda_e v_e$ |
| $h \in \mathcal{W}_\tau$                  | $\mathbb{R}$                          | $\ell_h(x_{v(h)})$                    |
| $j \in \mathcal{M}_\tau$                  | $\mathbb{R}^{d_j}$                    | $x_j^+ - A_j x_j^- - c_j$             |
| $D \in \mathcal{D}_\tau$                  | $\mathbb{C}$                          | $z_D^4 - t_D$                         |
| $\delta \in \mathcal{E}_\tau^{\text{op}}$ | $E_\delta^- \oplus \mathcal{L}_{j_1}$ | $H_{\delta, w}(u_\psi)$               |
| $\delta \in \mathcal{E}_\tau^{\text{cl}}$ | $\mathbb{E}_\delta$                   | $s_\delta^{\text{Euler}}$ .           |

(154)

For a common-vertex constraint  $(A_j, c_j) = (1, 0)$ ; for an overlap constraint it is the integral-affine transition recorded in Item (4). The open-bundle component is (150), so its  $u_\psi = 1$  restriction is the degree- $\mathbf{d} + \mathbf{e}_{j_1}$  Euler section and its  $u_\psi = 0$  restriction contains the pointing divisor. Thus

every component map and its real rank are specified. Put

$$\begin{aligned}
 d_\tau^{\text{op}} &= \sum_{\delta \in \mathcal{E}_\tau^{\text{op}}} \dim_{\mathbb{R}} \mathcal{M}_\delta, \\
 d_\tau^{\text{cl}} &= \sum_{\delta \in \mathcal{E}_\tau^{\text{cl}}} \dim_{\mathbb{R}} \mathcal{PM}_\delta, \\
 r_\tau^{\text{mod}} &= \sum_{\delta \in \mathcal{E}_\tau^{\text{op}}} \text{rank}_{\mathbb{R}}(E_\delta^- \oplus \mathcal{L}_{j_1}) + \sum_{\delta \in \mathcal{E}_\tau^{\text{cl}}} \text{rank}_{\mathbb{R}} \mathbb{E}_\delta.
 \end{aligned}
 \tag{155}$$

In particular,

$$\begin{aligned}
 \text{vdim}(\tau) &= 2|\mathcal{V}_\tau| + |\mathcal{S}_\tau| + |\mathcal{H}_\tau| + |\mathcal{E}_\tau| + q + b + \sum_C \dim \text{FM}(C) + \dim M_\tau + |\mathcal{Q}_\tau| \\
 &\quad + 3|\mathcal{D}_\tau| + d_\tau^{\text{op}} + d_\tau^{\text{cl}} - 2|\mathcal{S}_\tau| - |\mathcal{W}_\tau| - \sum_j d_j - 2|\mathcal{D}_\tau| - r_\tau^{\text{mod}},
 \end{aligned}
 \tag{156}$$

and its orientation line is

$$\mathfrak{o}_\tau = \det TP_\tau \otimes (\det E_\tau)^\vee.
 \tag{157}$$

All subsequent chain identities are degree-wise in this virtual grading.

There are only finitely many stable types. Indeed, a stable forest on a fixed finite leaf set has finitely many unmarked vertices; every remaining vertex carries one of the finite named events, and a path has at most  $N - 1$  positive-order bends. The finite wall list, finite dependency closure, finite profile sequence (7), and finite subdivisions then leave a finite set of choices. Moreover, the length  $q$  is bounded by the sum of the number of graph-complexity levels and the number of charts in the ordered corridor cover, because every nonidentity step strictly advances one of these two finite ranks. Hence the normalized Čech-boundary chains also leave only finitely many choices.

**Proposition 7.3** (Boundary stratification). *Every real codimension-one stratum of the compactified incidence space is one of the following: an annular endpoint; a contraction or cut of a tropical edge; a wall, joint, or affine-overlap stratum; a collision of tangent directions; an endpoint of the cotangent homotopy or the fourth-root normal collar; or a boundary stratum of an open or closed  $W$ -spin moduli factor. The auxiliary caps and disk boundaries are disjoint from the perturbed zero locus. After the canonical identifications on paired faces, the exterior boundary consists precisely of the canonical cotangent face, open-closed splittings, wall crossings, cotangent contractions, pairwise exchanges, and closed factors of nonzero Euler degree, together with the two annular restrictions.*

*Proof.* Every facet  $D \subset P_\tau$  is obtained by placing one factor in (153) on a codimension-one face. It has a resolved type  $\tau_D$  and a map  $f_D : P_{\tau_D} \rightarrow D$ . The following table records the local chart map and its normal-bundle comparison. A “trivial summand” is a coordinate together with a constraint on which the displayed elimination is an oriented isomorphism.

| Facet of $P_\tau$                          | Target type $\tau_D$                                                               | Map and normal-bundle comparison                                                                             |
|--------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| $\lambda_e = 0$ or an overlap-cylinder end | contracted or stabilized tree                                                      | substitute $x_{h(e)} = x_{t(e)}$ or $x^+ = A_j x^- + c_j$ and delete the corresponding trivial summand       |
| $\lambda_e = L_e$ or a corridor end        | empty, respectively $\partial_{\text{ann}}^\pm$                                    | the auxiliary cap is disjoint from the zero set; corridor ends are retained for the restriction maps $e^\pm$ |
| forest metric $t_e = 0, 1$                 | corolla, respectively cut forest                                                   | at 0 use (142); at 1 use $\Phi_F$ and delete the matching trivial summand                                    |
| wall-side equality                         | wall, joint, tangency, or bend-collision type                                      | conjugate the affine constraints by the labelled transition; use the bracket square at a joint               |
| boundary-map simplex face                  | first/last map or composite of adjacent maps                                       | restrict, delete, or compose the corresponding geometric maps                                                |
| ordered wall-homotopy face                 | shorter wall word or bracket joint                                                 | apply the chain map $T_H$ or the homotopy $P_H$ defined below                                                |
| collision scale 0                          | exchanged pair or collision block                                                  | restrict to the real blowup face and split off its oriented normal                                           |
| moving-circle vertex                       | root, pointing, contraction, wall, or closed-splitting direction                   | restrict the same open bundle and pointing section to the named vertex of $M_\tau$                           |
| cotangent parameter $u_\psi = 0, 1$        | pointing divisor, respectively degree- $\mathbf{d} + \mathbf{e}_{j_1}$ Euler chain | use the two endpoint sections in (150)                                                                       |
| root parameter $t_D = 0, 1$                | closed splitting, respectively pointing face                                       | restrict $z_D^4 - t_D$ to the indicated endpoint                                                             |
| open-moduli boundary                       | the named boundary graph $\Lambda$                                                 | apply $(F_{\Lambda\Gamma}, \phi_{\Lambda\Gamma})$ from (4)                                                   |
| closed degeneration                        | the named dependent closed datum in $\mathcal{E}$                                  | apply its stated boundary map and restrict the single Euler bundle already present                           |

Auxiliary caps and disk boundaries are disjoint from the zero set. The closed orbifold bases and moving circles have no other real codimension-one faces.

The stable cancellations used in these local comparisons are

|       | facet           | substitution          | eliminated coordinate and constraint |
|-------|-----------------|-----------------------|--------------------------------------|
| (158) | $\lambda_e = 0$ | $x_{h(e)} = x_{t(e)}$ | $(x_{h(e)}, x_{h(e)} - x_{t(e)})$    |
|       | overlap end     | $x^+ = A_j x^- + c_j$ | $(x^+, x^+ - A_j x^- - c_j)$         |
|       | $t_e = 1$       | cut the forest at $e$ | $(t_e, \text{matching constraint})$  |
|       | $z_D^4 = t_D$   | $t_D = z_D^4$         | $(t_D, z_D^4 - t_D)$                 |

The determinant of each displayed coordinate-to-constraint derivative is positive in the listed order. At  $t_e = 0$ , the coefficient constraint is not removed: it is the boundary of (142). At an open or closed moduli face, the oriented bundle comparison is the explicitly included map  $\phi_{\Lambda\Gamma}$ .

Restricting the ordered components of (154) along a facet gives a stable oriented identification

$$(159) \quad f_D^*(E_\tau|_D) \oplus \mathbb{R}^{a_D} \simeq E_{\tau_D} \oplus \mathbb{R}^{b_D},$$

where the trivial summands are precisely the stable cancellations in (158). The maps around a codimension-two face are equal because affine substitution is associative, the exact-data maps obey their actual composition identities, and the coherent forest primitives are defined on the whole refinement cube by (145). Taking determinants removes one outward normal. Interchanging two facet normals reverses their exterior product and proves the two-face boundary sign even when the rank of  $ds_\tau$  changes. At a partition face, permutation of incoming affine blocks acts by the sign character on (157); the same character acts on  $\mathfrak{o}_F$  in (133). Their tensor is therefore relabelling invariant. The displayed product has no further kinds of codimension-one face, which proves the asserted exhaustion.  $\square$

**7.4.2. Virtual PL zero loci.** Choose simultaneous equivariant triangulations of the finite proper étale atlases, the bundles, the sections, and all face maps. After relative simplicial approximation,  $s_\tau$  is PL in each stable bundle chart. A rational PL multisection on a chart is a finite set of PL branches with positive rational weights summing to one. On overlaps, replace two branch sets by their finite weighted common refinement; at isotropy include the complete finite orbit with equal subdivided weights, and at product corners take Cartesian products of branch sets.

These positive numbers are used only to perturb the incidence section. They are distinct from the signed rational coefficients of the moving-circle chains used in the cotangent recursions below.

Induct on face depth and simplex dimension. Prescribed branches on a downward-closed family of faces extend across each new simplex, and their new vertex values can be chosen outside the finite union of affine subspaces for which a branch fails PL general position. The stable identifications (159) make the choices agree across every face and orbifold chart. The same induction on  $[0, 1] \times P_\tau$  joins any two systems relative to the prescribed faces. Thus compatible arbitrarily small multisections  $\nu_\tau$  exist and

$$(160) \quad [Z_\tau]^{\text{vir}} = [(s_\tau + \nu_\tau)^{-1}(0)]$$

is a compact determinant-oriented rational branched PL chain of dimension  $\text{vdim}(\tau)$  whose boundary is its intersection with the ambient facets.

**7.4.3. Wall, joint, and gluing homotopies.** For completeness, the coefficient homotopies used on wall facets are fixed as follows. If  $H$  is a retained wall Hamiltonian, contraction with its Hamiltonian vector field gives a degree-one operator  $K_H$  on the transported contact complex. Put

$$(161) \quad \Delta_H = [\delta, K_H], \quad T_H = \exp(\Delta_H), \quad P_H = \sum_{n=0}^{N-1} \frac{K_H \Delta_H^n}{(n+1)!}.$$

All sums are finite modulo  $\mathfrak{m}_S^N$ , and the Cartan commutator identity gives

$$(162) \quad \delta P_H + P_H \delta = T_H - \text{id}.$$

For an ordered word  $H_1 \cdots H_b$ , use the telescoping homotopy

$$(163) \quad P_{H_b \cdots H_1} = \sum_{i=1}^b T_{H_b} \cdots T_{H_{i+1}} P_{H_i}.$$

At a two-wall joint the two boundary words have equal transport by scattering consistency. Let  $Z$  be the difference of their two instances of (163). It is a degree  $-1$  cycle in the endomorphism complex of the contracted contact complex. The retract (128) is concentrated in cohomological degree 2, while the contact complex has no terms above degree 2; hence  $R_{\text{con}} Z = 0$ . The contraction identity now gives  $[\delta, h_{\text{con}} Z] = Z$ , so  $h_{\text{con}} Z$  is the specified square primitive. At a

joint of codimension  $r \geq 3$ , the oriented sum of the already assigned proper faces is a cycle of degree  $1 - r$ . Its projection to the normal form again vanishes by degree, and the same contraction fills it. Induction over the ordered faces, exactly as in (145), therefore supplies a compatible cubical primitive. This defines the source, target, and signs of every wall and joint map used below.

7.4.4. *The cellular coend.* Choose the simultaneous subdivisions so that every map in the boundary-stratification table is cellular. Let  $\mathcal{K}_{I,d,N}$  be their finite face category. Its objects are pairs  $(\tau, \sigma)$ , and a generating arrow

$$(164) \quad f : (\tau_D, \sigma_D) \longrightarrow (\tau, \sigma)$$

is either the inclusion of an ordinary codimension-one subdivision face  $\sigma_D \subset \sigma$  with  $\tau_D = \tau$ , or the inclusion of a cell lying in one of the ambient facets  $D$  in the table followed by its stated map. The coefficient map for the first kind is the identity. Composites are the actual composites of restrictions, exact-data maps, and the fixed primitives above. Hence the two arrows obtained from a codimension-two corner are equal, rather than imposed by an unnamed relation. Dimension strictly increases along a nonidentity arrow, so the category is direct; its finiteness follows from the finiteness proved after Definition 7.2.

Let  $\mathcal{G}_{\tau,\sigma}$  be the rational PL branched-chain group on  $Z_\tau^{\text{vir}} \cap \sigma$ , relative to its intersections with the two corridor-end facets  $\partial_{\text{ann}}^- \cup \partial_{\text{ann}}^+$ . Collar restriction to either relative facet is retained as a separate map  $e^\pm$  below. In the transported contact complex, let  $\mathcal{B}_{\tau,\sigma}^{\text{sel}}$  be the smallest differential submodule containing the selected coefficient  $\alpha_\tau$ , all restrictions of the coherent forest primitives, all operators (161)–(163) applied to those elements, and all their root, open-boundary, and closed-boundary images. It is finite at the chosen deformation order. If  $\Gamma(\tau, \sigma)$  is the chamber frame, set

$$(165) \quad \mathcal{C}_{\tau,\sigma} = \mathcal{B}_{\tau,\sigma}^{\text{sel}} \otimes \mathfrak{o}_{F_\tau}.$$

This is the differential subcomplex generated by every coefficient and primitive used in the construction, not the line spanned by one selected coefficient. The arrow (164) induces  $f_* : \mathcal{G}_{\tau_D, \sigma_D} \rightarrow \mathcal{G}_{\tau, \sigma}$  and a chain map

$$(166) \quad f^\# : \mathcal{C}_{\tau, \sigma} \longrightarrow \mathcal{C}_{\tau_D, \sigma_D}.$$

For an ordinary subdivision face it is the identity. For an ambient facet it is restriction followed, according to the table, by chamber transport, the determinant map  $\Phi_F$ , the exact-data bundle map  $\phi_{\Lambda\Gamma}$ , or the stated root or closed-boundary map. On a metric-zero generator it uses  $G^{\text{con}}$  from (141); on a metric-one generator it uses the cut-forest element. The source module was closed under exactly these images, so the map is defined on every generator and extends linearly. Equations (127), (143), (162), and the exact-data composition identities prove  $f^\# \delta = \delta f^\#$  and equality on every composite.

Define

$$(167) \quad \mathcal{C}_{I,d,N}^{\text{tot}} = \left( \bigoplus_{(\tau,\sigma) \in \mathcal{K}_{I,d,N}} \mathcal{G}_{\tau,\sigma} \otimes \mathcal{C}_{\tau,\sigma} \right) / \langle [f_* z, \alpha] - [z, f^\# \alpha] \rangle_f.$$

The closed factor, metric cube, boundary-map simplex, and ordered wall-homotopy simplex occur in the branch polyhedron once; they are not repeated in the coefficient. The differential is

$$(168) \quad d[z, \alpha] = [\partial_{\mathcal{K}} z, \alpha] + (-1)^{\dim z} [z, \delta \alpha],$$

where  $\partial_{\mathcal{K}}$  is the ordinary outward-normal-first cellular boundary followed by the corresponding subdivision or ambient-facet arrow (164). This is an endomorphism because (165) is closed

under  $\delta$  and (166) is a chain map. The ordinary two-face cancellation gives  $\partial_K^2 = 0$ ; the Koszul sign cancels the mixed terms. Hence  $d^2 = 0$  on the displayed quotient.

7.4.5. *Root thimbles and exterior faces.* The remaining root-normal collar is explicit. The four radial branches of  $z^4 = t$ ,  $0 \leq t \leq 1$ , descend to a chain  $\mathfrak{T}_4$  with

$$(169) \quad \partial[\mathfrak{T}_4] = [\text{pt}] - 4[B\mu_4].$$

Since  $[B\mu_4]$  pushes forward to  $\frac{1}{4}[\text{pt}]$ , its normalized low endpoint has coefficient one. On the already present closed and continuing annular factors, this gives a thimble with boundary  $\partial_{\text{pt}} - \partial_{\text{oc}}$ .

The wall and joint cylinders are the explicit Cartan homotopies (161)–(163) and their coherent square primitives. A multiple-collision cone is the radial cone over the corresponding Fulton–MacPherson exceptional face; its bottom has smaller dimension and hence zero cellular pushforward.

The named boundary chains are the following oriented sums:

| Chain                    | Defining codimension-one faces                                                                                                      |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| $\partial_{\text{can}}$  | the $u_\psi = 1$ zero chain of $\sigma_\delta^- \oplus s_\delta^+$ , representing descendent degree $\mathbf{d} + \mathbf{e}_{j_1}$ |
| $\partial_{\text{pt}}$   | the $u_\psi = 0$ pointing-section divisor before the radial root collar is added                                                    |
| $\partial_{\text{oc}}$   | the low root-thimble endpoint, multiplied by the compact closed factor and the continuing open factor                               |
| $\partial_{\text{wall}}$ | critical open-moduli boundary graphs carrying the Cartan wall homotopy                                                              |
| $\partial_\psi$          | contraction boundary graphs of the cotangent homotopy, weighted by the one- or two-marking pointing chain                           |
| $\partial_{\text{exch}}$ | the pair of cyclic orders on a two-direction collision face                                                                         |
| $\partial_{\text{deg}}$  | closed-factor faces whose Euler degree is incompatible with the required zero-dimensional pushforward                               |

Metric-zero faces are filled by (142), metric-one faces are glued to cut forests, structural-nerve faces cancel through the coend, wall and joint faces are filled by (163) and its coherent square version, and multiple collisions are filled by their radial cones. Auxiliary caps and root-chart boundaries are disjoint from the zero locus for all sufficiently small perturbations: compactness gives a positive lower bound for  $\|s_\tau\|$  there. These are all facets because (153) is a finite product of the half-space polytope (152), cubes, simplices, Fulton–MacPherson spaces, cotangent and root intervals, and the compact open and closed orbifold factors in (153).

7.4.6. *Assembly and the boundary theorem.* For each resolved type, let  $\alpha_\tau$  be the transported coefficient assembled from its decorated tropical forest by (136). A *compatible moving-circle chain*  $\mathbf{p}$  assigns a signed rational coefficient  $w_\tau(\mathbf{p})$  to every oriented cell  $M_\tau$ , with equal coefficients on identified subdivisions. With product orientation ordered as in (153), define

$$(170) \quad \Psi_{I,\mathbf{d},N}^{\text{raw}}[\mathbf{p}] = \sum_{\tau} w_\tau(\mathbf{p})[Z_\tau^{\text{vir}}, \alpha_\tau].$$

Here the open-bundle component of  $Z_\tau^{\text{vir}}$  is the parameterized zero chain  $\mathcal{Z}_{\delta,w}$  of (151). Hence its named outward boundary is  $\partial_{\text{can}} - \partial_{\text{pt}} + \partial_{\text{wall}} + \partial_\psi + \partial_{\text{exch}} + \partial_{\text{deg}}$ ; its remaining faces are precisely the internal faces listed above. Add to (170) the wall and joint prisms, corolla and refinement primitives, collision cones, and radial thimbles, each with the sign opposite to its incident internal face. Denote the resulting finite sum by  $\Psi_{I,\mathbf{d},N}[\mathbf{p}]$ .

**Theorem 7.4** (Cotangent-homotopy boundary). *For every compatible moving-circle chain  $\mathfrak{p}$ , the chain  $\Psi_{I,\mathbf{d},N}[\mathfrak{p}]$  is well defined in (167) and satisfies*

$$(171) \quad \partial \Psi_{I,\mathbf{d},N}[\mathfrak{p}] = \partial_{\text{can}} - \partial_{\text{oc}} + \partial_{\text{wall}} + \partial_{\psi} + \partial_{\text{exch}} + \partial_{\text{deg}}.$$

*Its relative PL bordism class is independent of all triangulations, multisections, auxiliary primitives, and chosen root arcs, and is natural under window restriction, deformation-order reduction, finite-root refinement, relabelling, target-frame re-expansion, coordinate transposition, and common-enlargement transport by  $K$ .*

*Proof.* Equation (151) supplies  $\partial_{\text{can}} - \partial_{\text{pt}}$  and the open-moduli side faces. The ordinary cellular boundary of  $\mathfrak{p}$  gives their signed pointing and contraction coefficients. The chain-homotopy identity (162) cancels each wall face, and its two-parameter form cancels each joint face. Equations (143) and (148) fill zero-length forest faces and glue length-one faces. The two cyclic orders at a pair collision have opposite outward-normal orientations and form the displayed exchange pair. For a collision of at least three directions, the bottom of the radial cone has smaller dimension and zero cellular pushforward; nested cone sides cancel in pairs. Simplicial boundary-map, affine-overlap, and stabilization faces cancel by the defining coend relation. Coefficient-order terms in  $\mathfrak{m}_S^N$  vanish, while closed degree mismatches are retained in  $\partial_{\text{deg}}$ . The corridor-end faces are relative and are recorded by  $e^\pm$ . Finally, (169) changes the raw boundary  $\partial_{\text{can}} - \partial_{\text{pt}}$  into  $\partial_{\text{can}} - \partial_{\text{oc}}$ , with coefficient  $4 \cdot \frac{1}{4} = 1$ . The product-facet classification preceding (170) proves that no other codimension-one face remains.

For window restriction, deformation reduction, or finite-root refinement, the smaller face diagram is a downward-closed subdiagram of the larger one. Choose the multisections by relative extension along a nested cofinal exhaustion. Relabelling signs cancel between (157) and (133); coordinate transposition is obtained by taking the two-element branch orbit. Transport to another target frame and core transport after common enlargement carry the incidence equations, the open Witten-bundle diagram, the relative contact complex, and every chosen primitive by chain isomorphisms. These maps therefore commute with (168). The parameterized multisection construction gives the claimed compatible cobordisms and independence of the exhaustion.  $\square$

**7.5. Cotangent recursions.** Let  $i^\pm : C^\pm \hookrightarrow C$  be the incoming and outgoing facets of the compact corridor. Restriction of branch polyhedra along  $i^\pm$  and the coefficient pullback (166) define chain maps

$$(172) \quad e^\pm : C_{I,\mathbf{d},N}^{\text{tot}} \longrightarrow C_{I,\mathbf{d},N}^{\text{end},\pm}.$$

Here  $C_{I,\mathbf{d},N}^{\text{end},\pm}$  is the same coend (167) over the full subcategory of cells in  $\partial_{\text{ann}}^\pm$ ; hence its objects, differential, and coefficient maps are explicit restrictions of those already defined. Both targets are transported to the fixed frame  $\Gamma$  using (126).

For any descendent vector  $\mathbf{d}$ , let  $\mathcal{E}_{I,\mathbf{d},N}^\pm$  be the restriction to the corresponding annular end of the canonical Euler zero chain  $Z(\mathbb{E}_\Gamma(\mathbf{d}), \sigma_\Gamma(\mathbf{d}))$ . Relative extension makes these choices compatible in  $\mathbf{d}$  and in the finite window. In particular, the canonical cotangent-homotopy boundary satisfies

$$(173) \quad e^\pm(\partial_{\text{can}} \Psi_{I,\mathbf{d},N}[\mathfrak{p}]) = \mathcal{E}_{I,\mathbf{d}+\mathbf{e}_{j_1},N}^\pm.$$

The moving-circle primitive has zero collar restriction, so the right side does not depend on that primitive. Define the annular moment by

$$(174) \quad \mathcal{A}_{\text{ann}}(I, \mathbf{d}) = \mu_\Gamma(\mathcal{E}_{I,\mathbf{d},N}^-).$$

Wall transport is a chain map,  $\mu_\Gamma$  is a cocycle, and the intervening relative chain has boundary (171); the wall, exchange, and residual-degree terms have zero contact trace as proved below. Thus the greatest-slope endpoint gives the same scalar after transport to the same target frame. The empty endpoint is the unit:  $\mathcal{A}_{\text{ann}}(\emptyset, 0) = 1$ .

For  $A \subseteq I$ , write

$$(175) \quad \sum_{i \in A} a_i = r_A + 4\ell_A, \quad \sum_{i \in A} b_i = s_A + 4m_A, \quad N_A = \ell_A + m_A - |A| + 1 + \sum_{i \in A} d_i,$$

with  $0 \leq r_A, s_A \leq 3$ . When  $N_A = -1$ , let  $z_A$  be the degree-zero open half-node of twist  $(r_A, s_A)$  and set  $Q_A = (I \setminus A) \sqcup \{z_A\}$ . Its compact closed partner has the complementary twist  $(2 - r_A, 2 - s_A)$ , interpreted modulo 4. Denote its Frobenius-normalized Euler pushforward, including the degrees  $\mathbf{d}|_A$ , by

$$(176) \quad \mathcal{C}_A^{\text{ext}} = 16 \int_{[\mathfrak{c}_A]} e(\mathbb{E}_A^{\text{cl}}).$$

The factor (176) is an input from the compact closed quartic  $W$ -spin moduli problem, not a value of an open invariant.

The wall-crossing trace vanishes by the primitive

$$(177) \quad \mathcal{K}_{u,v} = (u+1)X^u Y^{v+1} dX + (v+1)X^{u+1} Y^v dY,$$

whose differential is

$$(178) \quad D_{W_0} \mathcal{K}_{u,v} = 4(v+1)X^{u+4} Y^v \Omega - 4(u+1)X^u Y^{v+4} \Omega.$$

Indeed, the two terms in (178) have traces  $4(v+1)w(u+4, v)$  and  $4(u+1)w(u, v+4)$ , which agree by (123). The complete relative differential also contains its divisor restrictions, so Proposition 7.1 annihilates every wall-crossing face. The exchange trace vanishes because exchange of the two directions reverses the real face orientation and preserves its coefficient.

**7.5.1. Pointing chains.** We next define the two moving-circle chains and compute their contributions. A rooted critical decomposition is

$$Q = ((I_0, u, v); (I_1, p_1, q_1), \dots, (I_h, p_h, q_h)),$$

where  $j_1 \in I_0$ , the root profile  $(u, v)$  is critical, and the other profiles are balanced. Its two quartic phase faces and its  $j$ th contraction face are

$$(179) \quad \begin{aligned} P_X(Q) &= \{(I_0, u+4, v), (I_1, p_1, q_1), \dots, (I_h, p_h, q_h)\}, \\ P_Y(Q) &= \{(I_0, u, v+4), (I_1, p_1, q_1), \dots, (I_h, p_h, q_h)\}, \\ P_j(Q) &= \{(I_0 \sqcup I_j, u+p_j, v+q_j)\} \cup \{(I_i, p_i, q_i) : i \neq 0, j\}. \end{aligned}$$

Put

$$(180) \quad \begin{aligned} \beta_X(j) &= \frac{p_j}{u+p_j}, & \beta_Y(j) &= \frac{q_j}{v+q_j}, \\ A_Q &= \frac{1+u+\sum_j p_j}{u+4}, & B_Q &= \frac{1+v+\sum_j q_j}{v+4}. \end{aligned}$$

A zero numerator gives the value zero in the first line. For a balanced decomposition  $P$ , introduce signed rational cellular coefficients  $(a_P, b_P, c_P, d_P)$ , summing to one, at the  $X$ -flag,  $Y$ -flag, root flag, and second distinguished marking of its moving circle. They are not multisection weights.

The *one-marking pointing chain* uses only the first distinguished marking: set  $d_P = 0$  and define, inductively on the number of components,

$$(181) \quad R_Q^{(1)} = 1 + \sum_{j=1}^h (\beta_X(j)a_{P_j(Q)} + \beta_Y(j)b_{P_j(Q)}).$$

The *two-marking pointing chain* also retains  $j_2 \neq j_1$ : put  $(a_P, b_P, c_P, d_P) = (0, 0, 0, 1)$  when  $j_1$  and  $j_2$  lie in the same block. Otherwise set  $d_P = 0$  and

$$(182) \quad R_Q^{(2)} = 1 + \sum_{\substack{1 \leq j \leq h \\ j_2 \notin I_j}} (\beta_X(j)a_{P_j(Q)} + \beta_Y(j)b_{P_j(Q)}).$$

In either case impose  $A_Q a_{P_X(Q)} + B_Q b_{P_Y(Q)} = R_Q^{(\ell)}$  by the symmetric solution

$$(183) \quad a_{P_X(Q)} = \frac{A_Q R_Q^{(\ell)}}{A_Q^2 + B_Q^2}, \quad b_{P_Y(Q)} = \frac{B_Q R_Q^{(\ell)}}{A_Q^2 + B_Q^2}, \quad c_P = 1 - a_P - b_P.$$

The right sides involve one fewer detached component. The root block determines  $Q$ , so no new variable occurs in two equations at the same level. Thus (183) defines rational, permutation-natural coefficients without using a GKT coefficient. They need not be positive; for example,  $h = u = v = 0$  gives  $(a, b, c) = (2, 2, -3)$ . This is permitted for a rational cellular chain.

For the one-marking chain, the resulting contraction identity is

$$(184) \quad \sum_{j=1}^h (\beta_X(j)a_{P_j(Q)} + \beta_Y(j)b_{P_j(Q)}) - A_Q a_{P_X(Q)} - B_Q b_{P_Y(Q)} = -1.$$

For the two-marking chain, if  $j_2 \notin I_0$ , the same difference omits the unique block containing  $j_2$  and the marked-pointing branch contributes  $+1$ ; the sum is  $-1 + 1 = 0$ . If  $j_2 \in I_0$ , the defining same-component case sets all these coefficients to zero.

We now realize these identities as actual cellular boundaries. List the named points of the oriented moving circle in cyclic order as  $p_0, p_1, \dots, p_{m-1}$ , with  $p_0$  the root direction, and let  $e_i$  be the arc from  $p_i$  to  $p_{i+1}$ , indices modulo  $m$ . The one-marking signed zero-chain is

$$(185) \quad z_Q^{(1)} = p_0 + \sum_{j=1}^h (\beta_X(j)a_{P_j(Q)} + \beta_Y(j)b_{P_j(Q)})p_j - A_Q a_{P_X(Q)}p_X - B_Q b_{P_Y(Q)}p_Y,$$

where coincident named points are combined. Equation (184) says that its total coefficient is zero. If  $j_2 \notin I_0$ , define  $z_Q^{(2)}$  by omitting the terms with  $j_2 \in I_j$  and adding the marked point with coefficient one; if  $j_2 \in I_0$ , put  $z_Q^{(2)} = 0$ . The preceding two-marking calculation again gives total coefficient zero.

For any zero-chain  $z = \sum_i \zeta_i p_i$  of total coefficient zero, set

$$(186) \quad \mathcal{P}_{p_0}(z) = \sum_{j=1}^{m-1} \left( - \sum_{i=1}^j \zeta_i \right) e_j, \quad \partial \mathcal{P}_{p_0}(z) = z.$$

The boundary formula follows from  $\partial e_i = p_{i+1} - p_i$ . It is the unique primitive whose coefficient on the root arc is zero. Compatibility with the complete face diagram is the following direct calculation:

$$(187) \quad F_* z_Q^{(\ell)} = z_{FQ}^{(\ell)}, \quad F_* \mathcal{P}_{p_0}(z_Q^{(\ell)}) = \mathcal{P}_{F(p_0)}(z_{FQ}^{(\ell)}).$$

For a collision, coincident named points are combined and both adjacent arc coefficients are repeated. For a forest contraction, substitution of (181) or (182) is exactly the coefficient of the contracted point. A closed splitting deletes its named block and leaves the lower-support formula occurring on the right side of the same recursion. Wall and affine-overlap maps preserve the pointing flags and act only on the transported coefficient. Stabilization deletes a zero coefficient at an unstable flag, while forgetting and relabelling merely reindex the cyclic list. These are all cases in Proposition 7.3, so they prove (187); it is not inferred from subdivision naturality alone. Summing the primitives over all critical decompositions gives compatible moving-circle chains  $\mathbf{p}^{(1)}$  and  $\mathbf{p}^{(2)}$ . Put

$$(188) \quad \Psi_{I,\mathbf{d},N}^{(\ell)} := \Psi_{I,\mathbf{d},N}[\mathbf{p}^{(\ell)}], \quad \ell = 1, 2.$$

For each named flag  $w$ , the corresponding cell is fibered over the actual cotangent zero cobordism  $\mathcal{Z}_{\delta,w}$  from (151). The coefficients sum to one, so the  $u_\psi = 1$  facets combine to one copy of the degree- $\mathbf{d} + \mathbf{e}_{j_1}$  Euler chain. The positive-weight PL perturbations are the same in both constructions; only the signed rational coefficients on the separate cotangent cobordisms differ.

On a contraction facet, the exact-data bundle map restricts  $E_\delta^-$  to the lower-descendent Euler chain  $\mathcal{E}_{I,\mathbf{d}}$ , while the degree of the pointing map is the corresponding coefficient  $\beta_X(j)$ ,  $\beta_Y(j)$ ,  $A_Q$ , or  $B_Q$  in (180). Applying the boundary pushforward and then  $\mu_\Gamma$  therefore multiplies every term in (184) by the common scalar  $\mathcal{A}_{\text{ann}}(I, \mathbf{d})$ . The resulting contraction moment is  $-\mathcal{A}_{\text{ann}}(I, \mathbf{d})$  for the one-marking chain and zero for the two-marking chain. Outward-normal-first orientation reverses this induced moving-circle boundary. Hence the geometric  $\partial_\psi$  term in (171) has one- and two-marking moments  $+\mathcal{A}_{\text{ann}}(I, \mathbf{d})$  and zero, respectively.

Finally, the normalized radial thimble restricts to a zero-dimensional closed factor exactly when the Euler rank  $\ell_A + m_A + \sum_{i \in A} d_i$  equals the closed dimension  $|A| - 2$ , which is equivalent to  $N_A = -1$ . Equations (169) and (176) then give coefficient  $4 \cdot \frac{1}{4} = 1$  and the product  $\mathcal{C}_A^{\text{ext}} \mathcal{A}_{\text{ann}}(Q_A, \mathbf{d}|_{I \setminus A})$ . If the ranks do not agree, the closed Euler pushforward has nonzero residual degree; coefficient-order terms lie in  $\mathfrak{m}_S^N$ . These are precisely the residual-degree faces, and their degree-zero moment is zero.

**7.5.2. Stokes evaluation and support reduction.** Apply the contact cocycle to the two explicitly constructed chains (186). The degree-zero values of every term in (171) are

| Face                     | one-marking value                                            | two-marking value                            |
|--------------------------|--------------------------------------------------------------|----------------------------------------------|
| $\partial_{\text{can}}$  | $\mathcal{A}_{\text{ann}}(I, \mathbf{d} + \mathbf{e}_{j_1})$ | same                                         |
| $\partial_{\text{oc}}$   | threshold sum over $j_1 \in A$                               | threshold sum over $j_1 \in A, j_2 \notin A$ |
| $\partial_{\text{wall}}$ | 0 by (178)                                                   | 0                                            |
| $\partial_\psi$          | $\mathcal{A}_{\text{ann}}(I, \mathbf{d})$                    | 0                                            |
| $\partial_{\text{exch}}$ | 0 by orientation reversal                                    | 0                                            |
| $\partial_{\text{deg}}$  | 0 by degree or truncation                                    | 0                                            |

Since the contact trace kills a total boundary, this table and the sign  $\partial_{\text{can}} - \partial_{\text{oc}}$  give

$$(189) \quad \mathcal{A}_{\text{ann}}(I, \mathbf{d} + \mathbf{e}_{j_1}) = \sum_{\substack{A \subseteq I, j_1 \in A, |A| \geq 2 \\ N_A = -1}} \mathcal{C}_A^{\text{ext}} \mathcal{A}_{\text{ann}}(Q_A, \mathbf{d}|_{I \setminus A}) - \mathcal{A}_{\text{ann}}(I, \mathbf{d}),$$

and, for  $j_2 \neq j_1$ ,

$$(190) \quad \mathcal{A}_{\text{ann}}(I, \mathbf{d} + \mathbf{e}_{j_1}) = \sum_{\substack{A \subseteq I, j_1 \in A, j_2 \notin A, |A| \geq 2 \\ N_A = -1}} \mathcal{C}_A^{\text{ext}} \mathcal{A}_{\text{ann}}(Q_A, \mathbf{d}|_{I \setminus A}).$$

The vacuum value is the empty broken-line chain. For a singleton at descendent degree zero, the normalized root thimble followed by (124) gives one. Since no subset of a singleton has cardinality at least two, (189) then gives  $\mathcal{A}_{\text{ann}}(\{i\}, d_i + 1) = -\mathcal{A}_{\text{ann}}(\{i\}, d_i)$ . Hence the initial values are

$$(191) \quad \mathcal{A}_{\text{ann}}(\emptyset, 0) = 1, \quad \mathcal{A}_{\text{ann}}(\{i\}, d_i) = (-1)^{d_i}.$$

Subtracting (190) from (189) yields

$$(192) \quad \mathcal{A}_{\text{ann}}(I, \mathbf{d}) = \sum_{\substack{A \subseteq I, \, j_1, j_2 \in A \\ N_A = -1}} \mathcal{C}_A^{\text{ext}} \mathcal{A}_{\text{ann}}(Q_A, \mathbf{d}|_{I \setminus A}).$$

Every open factor on the right has support  $|Q_A| = |I| - |A| + 1 \leq |I| - 1$ . Hence the recursions and (191) determine the moment family uniquely. In particular, no multi-marking primary open invariant is initial data.

## 7.6. GKT comparison and the inverse system.

**7.6.1. Moment comparison and endpoint normal forms.** The two GKT cotangent recursions [GKT22, Thm. 5.10] are (189)–(190) for  $r = s = 4$ , and their singleton condition is (191) [GKT22, Cor. 5.11(1)]. Their closed factors are the same compact closed  $W$ -spin integrals as (176). Uniqueness therefore gives

$$(193) \quad \mathcal{A}_{\text{ann}}(I, \mathbf{d}) = \mathcal{A}_{\text{GKT}}(I, \mathbf{d}).$$

For a nonempty support  $J$ , define  $r_J, s_J, \ell_J, m_J$  by

$$\sum_{i \in J} a_i = r_J + 4\ell_J, \quad \sum_{i \in J} b_i = s_J + 4m_J, \quad 0 \leq r_J, s_J \leq 3,$$

and put

$$(194) \quad N_J = \ell_J + m_J - |J| + 1 + \sum_{i \in J} d_i.$$

The support is active when  $N_J \geq 0$ . GKT Notation 2.40 then lists the balanced graphs  $\Gamma_{J,0}, \dots, \Gamma_{J,N_J}$  with profiles

$$(195) \quad (k_1, k_2)(\Gamma_{J,p}) = (r_J + 4p, s_J + 4(N_J - p)).$$

Write their moment in the triangular form

$$(196) \quad \mathcal{A}_{\text{GKT}}(J, \mathbf{d}; \nu) = \sum_{p=0}^{N_J} q_{J,p} \nu_{\Gamma_{J,p}, \mathbf{d}} + P_{J, \mathbf{d}}(\nu|_{\text{proper supports}}),$$

where

$$(197) \quad q_{J,p} = \frac{\Gamma((r_J + 4p + 1)/4) \Gamma((s_J + 4(N_J - p) + 1)/4)}{\Gamma((r_J + 1)/4) \Gamma((s_J + 1)/4)} \neq 0.$$

Here  $P_{J, \mathbf{d}}$  is the sum over partitions with at least two blocks, and therefore involves only proper supports. Put

$$b_{J, \mathbf{d}} = \begin{cases} (-1)^{d_i}, & J = \{i\}, \\ 0, & |J| \geq 2. \end{cases}$$

Starting with the empty and singleton values, define two labelled systems recursively on  $|J|$  by

$$(198) \quad \begin{aligned} \nu_{\Gamma_{J,p}, \mathbf{d}}^- &= 0 \quad (p > 0), & q_{J,0} \nu_{\Gamma_{J,0}, \mathbf{d}}^- &= b_{J, \mathbf{d}} - P_{J, \mathbf{d}}(\nu^-), \\ \nu_{\Gamma_{J,p}, \mathbf{d}}^+ &= 0 \quad (p < N_J), & q_{J,N_J} \nu_{\Gamma_{J,N_J}, \mathbf{d}}^+ &= b_{J, \mathbf{d}} - P_{J, \mathbf{d}}(\nu^+). \end{aligned}$$

The Gamma factors in (197) are nonzero, and every term in  $P_{J,\mathbf{d}}$  has smaller support. Induction therefore proves existence and uniqueness. Substitution in (196) gives the required singleton moment  $(-1)^{d_i}$  and zero higher-support moment; the construction is twist-preserving permutation invariant and has the prescribed vanishing outside its single extreme graph. These are precisely the chamber-index conditions of [GKT22, Def. 3.47]. GKT Theorem 5.13 and Corollary 5.14 consequently realize the two systems by symmetric canonical multisections.

It remains to show that the annular endpoints give these particular systems. For a fixed active support  $J$ , the balanced exponent line has the integral coordinate

$$(199) \quad \vartheta_J(r_J + 4p, s_J + 4(N_J - p)) = p.$$

The critical wall between consecutive profiles  $p - 1$  and  $p$  is the hyperplane  $\vartheta_J = p - \frac{1}{2}$ ; this is (200) below written in the affine cofactor coordinate. The stops defining the corridor lie in the two unbounded chambers  $\vartheta_J < \frac{1}{2}$  and  $\vartheta_J > N_J - \frac{1}{2}$ . Substituting these inequalities in (152) shows that a one-block cell meeting the incoming facet has  $p = 0$ , while one meeting the outgoing facet has  $p = N_J$ . Thus the chain maps (172) kill all other one-block profiles before the contact moment is taken.

For a fixed active  $J$ , evaluation of the retained connected cell has diagonal weight  $q_{J,0}$  or  $q_{J,N_J}$  by (124); every cut-forest cell gives exactly  $P_{J,\mathbf{d}}$  by (148). The scalar comparison (193) therefore turns the two endpoint equations into (198). Induction on  $|J|$  identifies the endpoint coefficients with  $\nu^-$  and  $\nu^+$ . Since GKT moments are constant across chamber indices [GKT22, Cor. 4.28], this argument cannot and does not identify an unspecified chamber representative.

For reference, the shifted exponent of GKT's critical graph  $\Lambda_{J,p}$  is

$$(200) \quad (k_1 - 1, k_2 - 1)(\Lambda_{J,p}) = (r_J + 4(p - 1), s_J + 4(N_J - p)),$$

which places it precisely on the separating hyperplane used above. Its vector field is the one whose action on  $W_0$  is (178). No equality of a raw annular wall coefficient with a GKT chamber coefficient is used or asserted.

**7.6.2. The inverse-system class and completion of the proof.** We finally define the chain asserted in Theorem 1.7. For  $\delta \in \mathcal{E}$ , let  $I(\delta)$  and  $\mathbf{d}(\delta)$  be its labelled support and descendent vector, and set

$$(201) \quad \mathbf{C}_{S,N}^{\text{lab}} = \bigoplus_{\delta \in \mathcal{E}} \mathbf{C}_{I(\delta), \mathbf{d}(\delta), N}^{\text{tot}}.$$

This direct sum has the literal restriction maps of the finite window.

Let  $\mathcal{A}_S$  be the finite set of twist-descendent label types  $\alpha = (a, b, d)$  appearing in  $S$ , and let  $\Gamma_{\mathbb{Q}}(\mathcal{A}_S)$  be the divided-power algebra with basis  $x_{\alpha}^{[m]}$  and multiplication  $x_{\alpha}^{[m]} x_{\alpha}^{[n]} = \binom{m+n}{m} x_{\alpha}^{[m+n]}$ . For a multiplicity vector  $\mathbf{m} = (m_{\alpha})$ , put  $G_{\mathbf{m}} = \prod_{\alpha} \mathfrak{S}_{m_{\alpha}}$ . Let  $\mathcal{E}[\mathbf{m}] \subset \mathcal{E}$  be the exact data whose labelled supports contain exactly  $m_{\alpha}$  labels of each type, and define

$$(202) \quad \mathbf{C}_{S,N}^{\text{lab}}[\mathbf{m}] = \bigoplus_{\delta \in \mathcal{E}[\mathbf{m}]} \mathbf{C}_{I(\delta), \mathbf{d}(\delta), N}^{\text{tot}}.$$

The group  $G_{\mathbf{m}}$  acts by permuting equal-type leaf labels on the forest, open and closed exact data, moduli factors, determinant lines, pointing sections, and coefficient monomials. The window contains every such relabelling. Each boundary identification and every map  $F_{\Lambda\Gamma}$  and  $\phi_{\Lambda\Gamma}$  is equivariant; therefore the action is by chain automorphisms of (202). Define the coordinate-symmetric target complex

$$(203) \quad \mathbf{C}_{S,N}^{\text{dp}} = \bigoplus_{\mathbf{m}} (\mathbf{C}_{S,N}^{\text{lab}}[\mathbf{m}])_{G_{\mathbf{m}}} \otimes \prod_{\alpha} x_{\alpha}^{[m_{\alpha}]}.$$

On an oriented labelled cell generator  $c$  with stabilizer  $\text{Stab}(c) \subset G_{\mathbf{m}}$ , define

$$(204) \quad p_{\text{dp}}(c) = \sum_{g \in G_{\mathbf{m}} / \text{Stab}(c)} [gc] \otimes \prod_{\alpha} x_{\alpha}^{[m_{\alpha}]},$$

and extend linearly after choosing one generator from each orbit. This is a chain map because the differential is equivariant and preserves  $\mathbf{m}$ . Under the power-series realization  $x_{\alpha}^{[m]} \mapsto x_{\alpha}^m / m!$ , the scalar multiplying the unlabelled orbit is

$$(205) \quad \frac{|G_{\mathbf{m}}| / |\text{Stab}(c)|}{\prod_{\alpha} m_{\alpha}!} = \frac{1}{|\text{Stab}(c)|},$$

which is the required reciprocal automorphism factor. The same computation applies to every degeneration face because its stabilizer is computed after the corresponding labelled boundary map.

Define

$$(206) \quad \text{BL}_{S,N} = p_{\text{dp}}((\Psi_{I(\delta), \mathbf{d}(\delta), N}^{(1)})_{\delta \in \mathcal{E}}) \in \mathbb{C}_{S,N}^{\text{dp}}.$$

Its relative bordism class is the element  $[\text{BL}_{S,N}]_{\text{PL}} \in \Omega_{S,N}^{\text{PL}}$  of Theorem 1.7. Root transport and factorization decorations are retained in (206); the contact moment is applied only afterward.

The naturality in Theorem 7.4 makes (201)–(206) an inverse system over windows, deformation orders, and finite root stages. This is not an extension inferred from the scalar GKT comparison: the multisections, wall cylinders, corolla primitives, and root thimbles on a smaller window are literal restrictions of those on a larger window. Every monomial coefficient of the formal genus-zero descendent potential belongs to some finite divisor- and dependency-closed window, and framed transport and  $K$  act before scalarization. The inverse system therefore is the  $K$ -equivariant inverse system of theta-decorated relative PL bordism classes asserted in Theorem 1.7; applying the contact moment coefficientwise gives the descendent potential. The construction does not identify this bordism module with the theta algebra.

*Completion of the proof of Theorem 1.7.* The boundary theorem, Theorem 7.4, constructs the relative PL chain and identifies all of its exterior faces. Applying the contact cocycle gives (189)–(190); together with (191), these relations determine the moments by induction on the support. Equation (193) identifies them with the GKT moments, while (198) identifies the two annular restrictions. Finally, (206) and the literal restriction maps just established form the asserted inverse system. These statements give every construction, recursion, endpoint identification, and naturality assertion in Theorem 1.7.  $\square$

## 8. CONCLUSION

The principal result is the relative PL broken-line chain over the finite diagrams of quartic open and closed  $W$ -spin moduli spaces. Its exterior boundary realizes the two quartic cotangent recursions by Stokes' formula, and their difference reduces every multi-label contact moment to smaller supports. Vacuum and singleton values, together with the compact closed factors, therefore determine the moment family. Comparison with Gross–Kelly–Tessler identifies these moments and, by triangularity, the two canonical extreme endpoint systems. Literal restriction across finite windows organizes the formal descendent class as a  $K$ -equivariant inverse system in theta-decorated relative PL bordism; its coefficientwise contact moments give the descendent potential.

The annular affine, pro-Kummer, and Jacobi constructions provide the geometry needed for that chain. Descendent exponents force the annular parity cover, transported slab roots force the pro-Kummer system, and the twisted volume form gives a volume line carrying the

Jacobi bracket. In these categories the initial Hamiltonians have a contractibly consistent completion, broken-line products form a framed theta-algebra local system, and logarithmic caps add separated finite-stage boundaries while retaining wall transport at the two ends.

The supporting comparison with open FJRW theory is a factorization statement, not an equality of wall functions. The GKT critical factors embed in the annular Jacobi group, while the residual factors are those required by scattering consistency; the compatible-realization theorem supplies finite GKT homotopies for the enumerative factors. Gross–Kelly–Tessler’s primary mirror theorem supplies the source Saito/FJRW tensor. The contribution here is its transport, including the primitive form, Kodaira–Spencer basis, multiplication, unit, and residue pairing, through every framed annular and finite Kummer chart.

Two limitations remain. The pro-Kummer compactification is not identified with a distinguished toric degeneration of the original quartic pair. The finite-window PL construction also does not produce an analytic infinite-window virtual fundamental class. Moreover, chamber-independent moments cannot determine a raw coefficient in an arbitrary chamber; the coefficient-level statement is deliberately restricted to the two specified extreme normal forms.

#### APPENDIX A. AFFINE AND LOGARITHMIC CALCULATIONS

This appendix records the basis conventions and local algebra used in the main proofs. It is included so that the annular lattice, shear widths, and boundary separation can be checked without reconstructing them from a developed picture.

**A.1. Straight-boundary frames.** At a vertex  $V$  of  $\Delta$ , let  $\ell_V, r_V$  be the primitive boundary tangents toward its two neighbors and  $d_V = -p_V$  the inward radial vector. Both  $(\ell_V, d_V)$  and  $(r_V, d_V)$  are unimodular. Define the left and right straight-boundary frames by

$$F_{V,L}\ell_V = (-1, 0), \quad F_{V,L}d_V = (0, 1), \quad F_{V,R}r_V = (1, 0), \quad F_{V,R}d_V = (0, 1).$$

The resulting matrices in the old integral basis are

$$(207) \quad \begin{array}{c|cc} V & F_{V,L} & F_{V,R} \\ \hline A & \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix} & \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \\ B & \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 1 \\ -1 & -2 \end{pmatrix} \\ C & \begin{pmatrix} -1 & -3 \\ -1 & -2 \end{pmatrix} & \begin{pmatrix} -1 & -3 \\ 0 & 1 \end{pmatrix} \end{array}.$$

The degeneration-side transition is  $\check{M}_V = F_{V,R}^{-1}F_{V,L}$ , and discrete Legendre duality gives  $M_V = \check{M}_V^T$ . Direct multiplication yields precisely the matrices (25). Their invariant primitive directions are  $(1, -1), (1, 1), (1, 3)$  and

$$(208) \quad \begin{aligned} M_A - I &= \begin{pmatrix} -1 \\ 1 \end{pmatrix} (1, 1), \\ M_B - I &= \begin{pmatrix} 1 \\ 1 \end{pmatrix} (2, -2), \\ M_C - I &= \begin{pmatrix} 1 \\ 3 \end{pmatrix} (3, -1). \end{aligned}$$

Thus the disk width profile is  $(1, 2, 1)$ .

The exponent map and inverse are

$$F = \begin{pmatrix} 1/4 & -1/4 \\ 0 & -1/2 \end{pmatrix}, \quad F^{-1} = \begin{pmatrix} 4 & -2 \\ 0 & -2 \end{pmatrix}.$$

Conjugating the disk monodromies gives

$$(209) \quad \begin{aligned} F^{-1}M_A F &= \begin{pmatrix} -1/2 & 9/2 \\ -1/2 & 5/2 \end{pmatrix}, \\ F^{-1}M_B F &= \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}, \\ F^{-1}M_C F &= \begin{pmatrix} -1/2 & 1/2 \\ -9/2 & 5/2 \end{pmatrix}. \end{aligned}$$

This verifies that the trivial cover fails at  $A, C$  and that the parity subgroup is the minimal domain on which  $\widehat{M}$  is a local system.

For completeness, a Reidemeister–Schreier basis of that subgroup is

$$\gamma_B, \quad \gamma_C \gamma_A^{-1}, \quad \gamma_A^2, \quad \gamma_A \gamma_B \gamma_A^{-1}, \quad \gamma_A \gamma_C.$$

Its matrices in the  $F$ -basis are

$$(210) \quad \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}, \quad \begin{pmatrix} -1 & 2 \\ -10 & 19 \end{pmatrix}, \quad \begin{pmatrix} -2 & 9 \\ -1 & 4 \end{pmatrix}, \quad \begin{pmatrix} -14 & 25 \\ -9 & 16 \end{pmatrix}, \quad \begin{pmatrix} -20 & 11 \\ -11 & 6 \end{pmatrix}.$$

All are integral. The second is the core matrix  $K$ ; the third and fourth give the lifted  $A$  and second  $B$  monodromies. Squaring the old  $C$  loop gives the final local matrix in (27). The primitive factorizations (28) follow by subtracting the identity.

The old primitive facet conormals are  $(1, 1), (-1, 1), (-3, 1)$ . Their  $F$ -dual coordinates are

$$(211) \quad F^T(1, 1) = (1/4, -3/4), \quad F^T(-1, 1) = (-1/4, -1/4), \quad F^T(-3, 1) = (-3/4, 1/4).$$

Thus  $p^*\varphi$  is quarter-integral and  $4p^*\varphi$  is the minimal integral multiple.

**A.2. Core and boundary holonomy.** The core matrix satisfies

$$K - I = \begin{pmatrix} -2 & 2 \\ -10 & 18 \end{pmatrix}.$$

The old lattice in the  $F$ -basis is generated by  $g_1 = (4, 0)$  and  $g_2 = (-2, -2)$ . The columns of  $K - I$  are  $2g_1 + 5g_2$  and  $-4g_1 - 9g_2$ , so  $K$  acts trivially on  $D$ . Its characteristic polynomial is  $T^2 - 18T + 1$ , proving the spectral statements in Section 3.

Representatives of the two boundary holonomies are

$$(212) \quad U_0 = \begin{pmatrix} 5 & -4 \\ 4 & -3 \end{pmatrix}, \quad U_1 = \begin{pmatrix} 33 & -64 \\ 16 & -31 \end{pmatrix}.$$

They factor as

$$U_0 - I = 4 \begin{pmatrix} 1 \\ 1 \end{pmatrix} (1, -1), \quad U_1 - I = 16 \begin{pmatrix} 2 \\ 1 \end{pmatrix} (1, -2).$$

Using the invariant tangent–normal bases

$$P_0 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad P_1 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix},$$

one obtains

$$P_0^{-1}U_0P_0 = \begin{pmatrix} 1 & -4 \\ 0 & 1 \end{pmatrix}, \quad P_1^{-1}U_1P_1 = \begin{pmatrix} 1 & -16 \\ 0 & 1 \end{pmatrix}.$$

This derives the two global widths used in the compactification.

The affine translation part of the core holonomy is normalized by placing the branch foci at the common radial fraction  $1/2$ . In the  $F$ -basis the core map then has translation  $(8, 8) = (I - K)(8, 4)$ , so its integral radiance class is zero. The branch segment in old polygon coordinates joins  $(-1/2, -1/2)$  to  $(3/2, -1/2)$  and crosses the negative singleton chords at  $(2/3, -1/2)$  and  $(1/3, -1/2)$ . Crosswise gluing changes only the sheet tag at those intersections.

**A.3. Separation on the two outer flares.** In invariant tangent–normal coordinates, the outer boundary development data are

$$(213) \quad \begin{array}{c|c|c|c} i & \ell_i & w_i & (\phi_i, h_i) \\ \hline 0 & 4 & 4 & (7/2, 1) \\ 1 & 8 & 16 & (15/2, 1/2). \end{array}$$

A support of tangential speed  $\sigma$  is  $x(r) = \phi_i - \sigma(r + h_i)$ , and a deck period at height  $r$  is  $w_i(r + h_i)$ . The exact slope envelopes in the two cofactor representatives are

$$(214) \quad \begin{array}{c|c|c|c} i & \text{chart} & \text{negative slopes} & \text{positive slopes} \\ \hline 0 & + & [1/3, 2/3] & [19/8, 21/8] \\ 0 & - & [0, 3/5] & [9/4, 31/12] \\ 1 & + & [7/3, 11/3] & [21/2, 23/2] \\ 1 & - & [1, 17/5] & [10, 34/3]. \end{array}$$

For  $i = 0$ , every positive-minus-negative difference lies in  $(0, 4)$ ; for  $i = 1$ , it lies in  $(0, 16)$ . Each individual interval also has width smaller than  $w_i$ . Thus no two distinct supports differ by a deck period. This proves Proposition 4.3.

**A.4. Finite root and logarithmic algebra.** For an effective Cartier divisor  $D = (f)$  on  $X = \text{Spec } A$ , our root-stack convention is

$$(215) \quad \sqrt[d]{(X, D)} = \left[ \text{Spec } A[g]/(g^d - f)/\mu_d \right].$$

On  $X \setminus D$  the atlas is a finite étale Kummer torsor; along  $D$  the quotient retains the root automorphism group. If a character algebra and an integral conormal  $N$  are present, the slab map  $z^m \mapsto g^{(N, m)} z^m$  is defined after inverting  $g$ . The divisor  $g = 0$  remains in the root stack and becomes part of the focus chart (62).

Let  $\mathfrak{J}$  be the filtered poset of finite rooted subtrees of the path tree. Compatible affine groupoid presentations of the stacks  $\mathfrak{X}_T$  have coordinate rings whose colimit is the coordinate ring of the pathwise atlas. A finitely presented module, algebra, morphism, or finite collection of descent identities uses finitely many generators, relations, and matrix entries. All therefore occur on one  $T \in \mathfrak{J}$ , after replacing the chosen stages by their rooted union. This proves the finite-presentation assertion in Theorem 2.7.

Finally, let  $R$  be a finite Artin ring,  $P$  an fs toric monoid, and  $S_0$  a finite Kummer algebra over  $R[P]$  given by monic root equations. If  $J_0 = (z^p : \nu(p) > 0)$  is the boundary ideal and  $S_C, J_C$  are transported as in (70), transport gives strict isomorphisms of the prelog charts and

$$S_C/J_C \simeq S_0/J_0.$$

Monicity makes the root atlas finite free before the diagonal quotient, and the finite quotient is an  $R$ -direct summand. Hence the boundary is  $R$ -flat. On a full bridge, the entire chart rings are identified before inverting a boundary parameter; the affine-intersection criterion then gives separatedness. These are exactly the local algebraic statements used in Theorem 4.4.

## APPENDIX B. COMPATIBLE COMPLETION OF GKT HOMOTOPIES

This appendix proves the geometric compatibility used in Proposition 5.2. A coefficient window does not by itself define a restriction map on full GKT canonical families: a multisection on one graph prescribes data on graph boundaries and on the connected components of its base graph. We instead use finite systems closed under these dependencies.

We follow the terminology of [GKT22]. Fix a countable twist-labelled reservoir  $\Omega_\infty$  containing infinitely many labels of each twist. Every construction below uses only a finite subset. The relative analytic extension input is the multisection argument of [BCT24]; the finite slope-ordered realization is the theorem of [Wue26].

**B.1. Finite dependency-closed systems.** For a graded open  $(4, 4)$ -graph  $\Gamma$  with finite internal label set and descendent vector  $\mathbf{d}$ , write

$$(216) \quad \delta = (\Gamma, \mathbf{d}), \quad E_\delta = E_\Gamma(\mathbf{d}) \longrightarrow \mathcal{PM}_\Gamma, \quad \text{supp}(\delta) = u_{I(\Gamma), \mathbf{d}}.$$

Call  $\delta$  an *exact GKT datum*. If  $\Lambda$  is a zero or exchange boundary stratum, canonical assembling requires

$$(217) \quad s_{\Gamma, \mathbf{d}}|_{\mathcal{PM}_\Lambda} = F_\Lambda^* \left( \bigoplus_{\Xi \in \text{Conn}(B\Lambda)} s_{\Xi, \mathbf{d}|_{I(\Xi)}} \right).$$

The immediate dependencies of  $\delta$  are the connected component data of  $B\Gamma$  and of all  $B\Lambda$  occurring here, with restricted descendent vectors.

**Definition B.1.** A finite dependency-closed partial system is a triple  $\mathbb{D} = (I, P, \mathcal{E})$  such that:

- (1)  $I \subset \Omega_\infty$  is finite and twist labelled;  $P$  is a finite divisor-closed set of labelled monomials, containing 1 and invariant under twist-preserving permutations of  $I$ ;
- (2)  $\mathcal{E}$  is a finite set of exact data containing every iterated dependency and connected component datum of each of its elements;
- (3) if a support in  $P$  has a nonempty GKT balanced set,  $\mathcal{E}$  contains its complete chamber block—all balanced and critical graphs and their dependency closures;
- (4)  $\mathcal{E}$  is invariant under the twist-preserving permutation group of  $I$ .

An inclusion means literal inclusion of labels, monomials, and exact data. Restriction is therefore the forgetting map.

**Lemma B.2** (Finite closure). *The dependency closure of finitely many exact data and complete chamber blocks is finite. Moreover, if  $\delta'$  depends on  $\delta$ , then*

$$(218) \quad \text{supp}(\delta') \mid \text{supp}(\delta).$$

*Proof.* A dependency uses a subset of the internal labels and the restricted descendent vector, proving (218). Order connected graphs lexicographically by

$$(|I(\Gamma)|, n(\Gamma), -k_{12}(\Gamma)), \quad n = 4k_1 + 4k_2 + 8k_{12}.$$

Every proper connected base component precedes the original graph. There is no infinite descending chain, and every graph has finitely many boundary types. Thus the finitely branching dependency tree is finite. Chamber blocks and permutation orbits are finite as well.  $\square$

Let  $\mathcal{C}(\mathbb{D})$  be the collections of transverse, strongly positive, automorphism-invariant canonical multisections on  $\mathcal{E}$  satisfying (217). Independent choices are made only on smooth connected graphs; the other data are the pullbacks and external sums prescribed by GKT.

For homotopies, fix a nondecreasing collar function  $\lambda : [0, 1] \rightarrow [0, 1]$ , constant near the endpoints, and distinct times  $t_A \in (0, 1)$  for the subsets  $A \subseteq I$ . Let  $\mathcal{H}(\mathbb{D}; \mathfrak{s}^0, \mathfrak{s}^1; \tau_I)$  denote

the simple canonical homotopies satisfying the GKT normal-form conditions: irrelevant and excess-rank components are nowhere zero; the two critical rank cases are transverse on all graph strata; rank-one-higher zeros occur over  $t_{I(\Gamma)}$ ; and the selected spatial slices are transverse. Endpoint, graph-boundary, and positive-boundary collars are part of the data.

## B.2. Relative extension.

**Lemma B.3** (Equivariant relative multisection extension). *Let  $\delta = (\Gamma, \mathbf{d})$  be free, and let  $Y$  be either  $\mathcal{PM}_\Gamma$  or  $[0, 1] \times \mathcal{PM}_\Gamma$ . Suppose compatible invariant data are prescribed near collared invariant closed sets  $K_0 \subset \text{int } K_1$ , where  $K_0$  contains every protected endpoint, graph-boundary, positive-boundary, prior-stage, and selected-slice collar. If the prescribed data are strongly positive and already satisfy the required transversality or nonvanishing conditions on  $K_0$ , they extend to an invariant global multisection literal near  $K_0$  with all the same open conditions. A nowhere-vanishing critical section may additionally be modified in the smooth interior through a transverse homotopy with prescribed rational signed zero count.*

*Proof.* The extension argument of [BCT24, Lem. 6.5] first extends the assembling data to a neighborhood of the graph and positive boundary; include the endpoint and selected-slice collars in that neighborhood. On the compact complement choose finitely many compactly supported local multisections whose branches span every fiber. Relative parametric transversality, applied to the finitely many graph strata and selected slices, gives a generic linear combination with the required properties. The perturbation vanishes near  $K_0$ , hence fixes all protected collars and preserves strong positivity.

Perform the construction on an injectively labelled cover and take the normalized union of all finite automorphism translates. On  $K_0$  every translate is the same invariant prescribed multisection, so the result is literal there. The generic smooth interior of the injectively labelled cover has trivial isotropy. On this locus, GKT Lemma 5.17 [GKT22] supplies a compactly supported critical modification whose transverse homotopy has any prescribed rational signed zero count. Relative transversality preserves that count and leaves the protected collars fixed.  $\square$

**Theorem B.4** (Relative endpoint lifting). *Let  $\mathbb{D} \subset \mathbb{D}'$  and let  $\mathfrak{s} \in \mathcal{C}(\mathbb{D})$ . Prescribe chamber coordinates on the new blocks, assuming that the old and new coordinates are the restriction of a realizable GKT chamber index. Then there exists  $\mathfrak{s}' \in \mathcal{C}(\mathbb{D}')$  realizing those coordinates and restricting literally to  $\mathfrak{s}$ . If the input and target index are twist symmetric, the extension may be chosen twist symmetric.*

*Proof.* Construct new free data in GKT's lexicographic graph order. Dependency closure and associativity of assembling prescribe compatible boundary values, which Lemma B.3 extends. Define the nonfree entries by their prescribed pullbacks and assemblies.

Process the new supports by increasing cardinality. For one support, temporarily complete the partial family on the finite label set and apply the GKT realization theorem. Proper-divisor coordinates have already been fixed, so the chamber equations reduce to their one weighted linear relation on the present block. The critical sections have rank one greater than the dimension and are nowhere zero. On the injectively labelled smooth-interior cover, GKT Lemma 5.17 supplies compactly supported modifications with prescribed rational signed zero counts. Equations (5.52)–(5.54) of GKT are triangular in these counts: choose them successively to match all but the last balanced coordinate, which is forced by equation (5.50), the weighted chamber relation. Reapply the relative extension lemma and reassemble the affected data.

A later change can affect only the same support or a multiple of it. If such a datum were old, divisor closure and chamber-block completeness would make the present support old, a

contradiction. Hence the original family is fixed literally. For symmetry, make choices orbitwise and take normalized finite orbit-unions; on old orbits these equal the old invariant section.  $\square$

**Theorem B.5** (Relative simple-homotopy lifting). *Let  $\mathbb{D} \subset \mathbb{D}'$ , let  $\mathfrak{s}^0, \mathfrak{s}^1 \in \mathcal{C}(\mathbb{D}')$ , and suppose*

$$H \in \mathcal{H}(\mathbb{D}; \mathfrak{s}^0|_F, \mathfrak{s}^1|_F; \tau_I).$$

*Then the time assignment extends injectively to  $I'$ , and  $H$  extends to a simple canonical homotopy on  $\mathbb{D}'$  which fixes every old homotopy and every protected collar. In particular, all old subset times are retained.*

*Proof.* For an old rank-equals-dimension component, the times at which a spatial slice is not transverse form a measure-zero set by Sard's theorem on the finitely many zero-set strata. Choose the new subset times distinctly, away from the old values and these exceptional sets. No old homotopy must be changed.

Induct on  $\dim \mathcal{PM}_\Gamma$ , choosing only new free connected data. Endpoints are prescribed. On graph boundaries, use the contracted-tail interpolation when a boundary tail is contracted and the simple assembly of component homotopies otherwise; on disconnected faces take external sums. These prescriptions agree at corners by associativity of assembling.

Apply Lemma B.3 with all old and boundary data protected. In the rank-one-higher case, make the total homotopy transverse, separate the finitely many zero projections, and move them by a supported fiber-preserving diffeomorphism to the assigned time. In the rank-equals-dimension case, construct transverse selected slices first, protect collars around them, and perturb the complement. The irrelevant and excess-rank cases are obtained relatively by positivity and transversality. Automorphism orbit-unions preserve the assigned time because it depends only on the invariant label set. Nonfree and disconnected homotopies are then the forced pullbacks and assemblies.  $\square$

Every finite partial family or simple homotopy consequently extends, relative to the partial system, to the full GKT data on the same finite label set: add the missing data support by support, adjoining a complete chamber block before its first datum, and apply the two lifting theorems.

**B.3. The compatible union.** The imported boundary-ray theorem, Theorem 1.1, gives unique extremal chamber indices and a unique increasing-slope factorization of their transition. In every finite divisor-closed quotient it also supplies a finite sequence of simple canonical homotopies realizing precisely the nonidentity reduced factors, one ray at each stage.

*Proof of Theorem 5.5.* Choose increasing finite label sets exhausting  $\Omega_\infty$  and finite divisor-closed permutation-invariant monomial sets exhausting the labelled coefficient ring. At each stage include the complete chamber blocks, the next finitely many exact data, and their finite closures. This gives the systems  $\mathbb{D}_n$ .

The union of the finite sets of active ray factors is countable. Embed its slope order into the complementary intervals of a Cantor set and call the assigned intervals  $J_\rho$ . Inductively apply Theorem B.4 to all intermediate endpoint families and Theorem B.5 to all elementary homotopies, always relative to the previous stage.

Fix  $\delta \in \mathbb{D}_{n_0}$ . Only the finitely many factors visible at stage  $n_0$  can be nonconstant on it. A factor installed later is extended relative to the constant homotopy on the old system. This installation-time finiteness proves Part (2). On an active interval use the elementary homotopy; elsewhere use the chamber value obtained after the finitely many active intervals to the left. Constant collars make these pieces join smoothly. Every assembling identity involves finitely many dependencies and hence holds at some finite stage.

The finite completion statement permits application of the GKT realization theorem at each stage. Exact-support uniqueness identifies the jump with the required ray factor in every finite quotient. Since the transition maps are literal forgetting maps, the compatible restrictions form an inverse limit in sets.  $\square$

Enlargement of the labelled system fixes the old realization literally.

No topology is asserted on equivalence classes of multisections with varying branch presentations. Smoothness means smoothness on each fixed  $[0, 1] \times \mathcal{PM}_\Gamma$ . The residual factors of Section 5 are not GKT jumps; they remain the algebraic comparison required to restore the consistent Jacobi wall word.

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